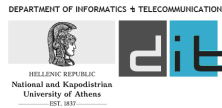


A Prototype Nurse Chatbot for Use in Psychiatric Clinics

A study case



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This dissertation is submitted for the degree of
Master of Science

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I would like to dedicate this dissertation to my loving partner . . .

Declaration

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other university. This dissertation is my own work and contains nothing which is the outcome of work done in collaboration with others, except as specified in the text and Acknowledgements. This dissertation contains fewer than 65,000 words including appendices, bibliography, footnotes, tables and equations and has fewer than 50 figures.

Anastasia Karakitsou
October 2025

Abstract

Background: The prevalence of stress and burnout in healthcare professionals due to excessive administrative work is a well-known problem since the advent of electronic health records (EHR) in 2016. Especially nursing personnel has been found to spend up to 60% of their time for handling documentation, e.g. for compiling progress notes for their patients (Khamisa et al, 2013). In Greece, the problem is even worse, as there still exist handwritten patient records that take up considerable time and energy from nurses to write, read, interpret, and most importantly retrieve on demand. The goal of this project is to “liberate“ nurse staff from time-consuming logistics problems, and offer them the opportunity to focus on patient treatment and care instead. The way to achieve this is by designing a voice chatbot that will automate and expedite the information retrieval process from patient data files.

Objective: This study aims to design, develop, and test an artificial intelligence-powered voice assistant to help psychiatric nurses manage the daily interactions with their patients.

Methods: The voice assistant is being implemented in several steps. First, a comprehensive mock knowledge base is created, which defines real-world user profiles. Second, conversation behavior between users and the voice assistant is defined via a well-designed LLM prompt. Third, advanced features like memory, tools, and an orchestrating “agent“ are implemented to provide continuity, reasoning, and autonomy. Fourth, two separate system evaluations are conducted via use cases (scenarios) using RAGAs framework as well as additional qualitative evaluations.

Results: A prototype voice assistant was evaluated in using various use cases. Preliminary qualitative test results demonstrate reasonable rates of dialogue success and answer correctness.

Conclusions: This study suggests the feasibility of using an intelligent voice assistant to help nurses in managing patients in psychiatric clinics.

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Chapter 1

Introduction

1.1 Background and Context

Burnout among clinicians has become a well-documented challenge in modern medical practice. Nurses and doctors report high levels of exhaustion, impaired attentiveness, and a loss of meaning in their work, symptoms that have been linked to increased use of electronic health records (EHRs), administrative overload, and shift pressures [54, 33, 92, 55]. Psychiatric nurses, the focus of this thesis, combine clinical responsibilities with the emotionally demanding task of supporting individuals living with mental health conditions. Their day-to-day work requires continuous patient observation, medication management, coordination with multidisciplinary teams, and empathetic engagement with vulnerable patients [86]. These competing demands frequently leave little time for documentation or quick access to the comprehensive patient information needed for safe care.

1.2 Problem Statement and Research Gap

Despite the promise of digital systems, most psychiatric clinics continue to rely on fragmented or handwritten patient records. Nurses often depend on memory, the single physical copy of the patient record, or time-consuming manual searches, diverting attention away from patient interaction. Existing conversational assistants primarily target patient-facing use cases or focus on medical specialties other than psychiatry, leaving a gap in tools that support psychiatric nurses in Greek clinical settings. At the same time, large language models (LLMs) and conversational AI have matured to the point where they can summarise and synthesise clinical information, yet their

use raises questions about accuracy, privacy, and hallucinations [59, 76, 57]. There is, therefore, a need to investigate whether an AI-powered assistant can reliably retrieve nurse-specific information while respecting the constraints of mental health care.

1.3 Research Objectives and Questions

The overarching objective of this research is to design, implement, and evaluate a conversational assistant that helps psychiatric nurses retrieve patient information on demand. The study addresses the following research questions:

- RQ1) Is it feasible to develop a conversational assistant in Greek language that helps nurses extract information about patients' profiles on demand?
- RQ2) If feasible, how effective is the assistant in terms of faithfulness, answer relevance and correctness, context precision, and context recall?
- RQ3) Does the assistant improve the daily experience and satisfaction of nurses when interacting with patient information?

1.4 Significance of the Work

Automating access to patient context through an AI assistant promises to reduce cognitive load, mitigate burnout, and allow nurses to focus on therapeutic interactions rather than administrative retrieval. Voice- and text-enabled conversational interfaces can support nurses who need hands-free access during shifts, provide structured summaries for fast decision-making, and serve as a foundation for future integration with electronic systems. By grounding responses in curated patient data and highlighting limitations, the prototype contributes to the growing field of agentic AI in healthcare while foregrounding responsible deployment in high-stakes environments [60].

1.5 Structure of the Thesis

This thesis is organised as follows. Chapter 2 surveys the literature on clinician burnout, healthcare conversational agents, and AI techniques relevant to psychiatric practice. Chapter 3 details the methodology, including data sources, system design, tooling, and evaluation procedures. Chapter 4 presents the experimental results and discusses their implications in relation to the research questions and prior work. Finally, Chapter 5

summarises the key findings, outlines the contributions, and proposes avenues for future research.

Chapter 2

Literature Review

2.1 Overview of the Research Area

2.1.1 Clinicians with burnout

A well-documented current problem in medical practice is healthcare staff work stress that leads to burnout, i.e. physical and mental exhaustion. Burnout is a psychological syndrome whose symptoms comprise feeling emotionally exhausted and ineffective, experiencing impaired memory and attentiveness, finding work no longer meaningful, wanting to quit, and tending to view patients and colleagues as objects [33]. Clinical burnout goes back to at least 2016, since the proliferation of electronic health records (EHRs). Even though EHRs were introduced with the aim to eliminate the drawbacks of an old, paper-based system, they led to increased documentation workload, due to the time and energy placed onto the administrative task of capturing patient information and filling out intricate (i.e. non-intuitive) forms to prove compliance with government rules [54, 33]. For instance, a 2016 study estimated that about 27% of physicians' time is allocated to direct clinical care, with the rest 73% dissipating in administrative work [92]. Other causes of burnout include limited time with patients, shift overload, and taking work at home. The consequences regarding patient care are many: poor communication, non-adherence to guidelines, medical errors, high organizational costs related to absenteeism and turnover, and legal costs due to complaints of negligence [55].

Especially for nurse personnel, a 2023 study showed a high risk of burnout and subsequent mental health concerns (depression, anxiety, suicidality) due to problems with understaffing, immense patient volumes – in particular after the COVID-19 pandemic – and exposure to traumatic events [4].

Turning to the focus of this project, we now shift our attention to the modern psychiatric nurse, whose tasks include caring for, motivating, and supporting individuals suffering from mental health conditions or behavioral problems. The nurse's aim is to assist them patients to adapt their behavior to fit into society, in comparison to the more traditional goal of reactively managing the patients' symptoms [86]. A nurse's typical day-to-day duties include observing patients and evaluating their condition, administering, adjusting, and monitoring medications, educating patients about their illness, assisting with self-care (e.g. bathing, grooming, etc.), daily routines, and general physical health ¹. Psychiatric nurses collaborate with doctors (psychiatrists and psychotherapists) to develop care plans and promote mental healthcare case by case, e.g. they may work with children exposed to trauma, or soldiers returning from the battlefield, among other cases. Psychiatric nurses may work in any setting that provides mental health services, such as correctional and assisted living facilities, rehabilitation centers, hospitals, clinics, and private homes. As the profession of psychiatry by nature deals with the unique psychological and emotional needs of vulnerable individuals – where personalization of the interaction, building a trusting relationship, and showing compassion and empathy ² are necessary skills – the job of a psychiatric nurse can be very stressful [86].

However, due to time pressures, the procedures are not always followed accurately, and information is not always immediately accessible. Of course, nurses have too little time to manually search for patient information on formal documents whenever necessary, and oftentimes they have to rely on their own memory for dozens of patients. As a result, the documentation part of their job diverts nurses' attention from the actual patient care part, leading to both poorer quality of care and increased mental burnout for the nurse staff.

The solution to this may be the automation of information retrieval via an AI assistant. The purpose of a conversational assistant would be to assist nurses to retrieve essential patient information with accuracy and timeliness. This should improve the staff's well-being and job satisfaction, and free up their time to devote to patient care.

¹Psychiatric Nurse: Duties, Pay, and How to Become One'. November 29, 2023. Coursera. Accessed on October 28, 2024. <https://www.coursera.org/articles/psychiatric-nurse>

²“ In client-centered therapy, therapists might convey empathy through techniques such as active listening, reflection, and adopting a nonjudgmental stance and warm tone. Conversational agents might be able to model similar behaviors, at least through speech.” [73]

2.1.2 Automation through AI

In general, the main goal of leveraging AI in the healthcare domain is to improve the quality of care. In this particular case, this goal could be achieved by reducing the burden of documentation for nurses, thus reducing their cognitive load, de-stressing them, and redirecting their time and energy to focus on interactions with their patients. Language models could assist nurses by simplifying or even replacing everyday tasks that require text synthesis or analysis, such as drafting medical notes, reports, and administrative emails or letters, and summarizing dense patient medical records [59]. Ultimately, the goal is to help nurses focus on the outcomes, and not the procedures and protocols. The way this can be achieved is by leveraging the power of LLMs, which can read, summarize, and generate natural text.

Health professionals have already started using such solutions. As we can see from the image below, around two thirds of American doctors reported to have used Augmented Intelligence in their practice in 2024 for various tasks.

Usage of AI use cases nearly doubled

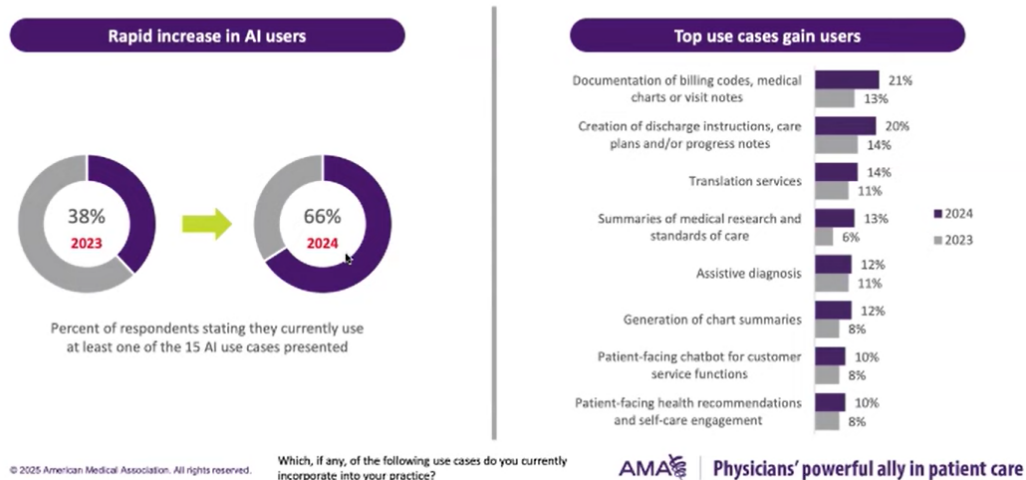


Figure 2.1 AI in Healthcare Series: From Decision Support to Drug Prescriptions, Stanford Online, YouTube (video), Accessed on March 29, 2025. https://www.youtube.com/watch?v=TC64U_rzEuI

2.2 Critical Analysis of Key Prior Studies

2.2.1 Challenges and mitigation strategies

Real-world healthcare data is unstructured

In many healthcare settings, healthcare information is written fast, carelessly, and in an unstructured manner due to time pressures. Often this information is written with poor handwriting (if written manually), or typed without punctuation, and containing spelling errors. To address this challenge, AI-based solutions have emerged. Technologies such as Optical Character Recognition (OCR), e.g. *Amazon Textract* ³, can digitize handwritten medical notes with increased accuracy. These technologies are then combined with natural language processing (NLP) models fine-tuned in the medical domain to correct spelling and punctuation errors, normalize abbreviations, and extract structured information from free-form text. Such systems can transform poorly typed (or written) input into usable, structured digital records, improving data usability.

Hallucinations from LLMs

The unpredictability and non-deterministic nature of LLMs' output is a major red flag for their use in medical practice, where even seemingly small mistakes may prove catastrophic, since they can put patient lives in danger. According to Obradovich et al. "these models can occasionally produce factually incorrect answers, make errors in simple arithmetic or logical problems, or generate responses that are seemingly obvious yet erroneous". Specifically in healthcare, the use of the Automatic Speech Recognition (ASR) system *Whisper* that powers *Nabla* – an assistant for automated medical documentation used for millions of visits in the US, and by 30,000 clinicians – has been recently reported to have introduced fabricated information into medical transcriptions, with even higher error rates for multilingual transcription and speech by aphasia patients ⁴.

To ensure AI safety, a solution is to employ professionals for AI red-teaming efforts, where the vulnerabilities of the system are challenged and exposed, with the aim of improving its weakest spots [76].

³"Amazon Textract", AWS (website), Accessed July 17, 2024. <https://aws.amazon.com/textract/>

⁴"RWS and BLEND Appoint New CEOs, Microsoft and Amazon Speech AI", Slator (video), December 6, 2024. Accessed on December 7, 2024. <https://www.youtube.com/watch?v=GvhflrVZNGo>

Another solution is to use retrieval augmented generation (RAG), which is a method to take advantage of the parametric (embedded) memory and generative capabilities of an already trained LLM, while also greatly reducing its notorious hallucinations. The RAG architecture works by augmenting – and also “grounding” – the LLM via connecting it to an external corpus of information, i.e. a non-parametric (separate) memory storage. This corpus may be any type of data (PDF, spreadsheet, etc.), even proprietary or confidential. The way RAG works is that the LLM combines the user input (query) together with relevant documents, passages or facts retrieved from the corpus, in order to generate an accurate answer to the query [57]. In this way, the generated responses are based on factual context, bypassing the need to fine-tune and retrain the whole language model on numerous extra examples of domain-specific knowledge in order to achieve similar informed and current responses.

Privacy of data and patient confidentiality

Any information that can be used to trace an individual’s identity is called “personally identifiable information” (PII). The type of such information can be virtually anything, such as one’s name, date of birth, passport number, driver’s license number, physical address, email address, photos, etc. Protected health information (PHI) is a subset of PII, but it specifically refers to health information acquired or generated during a health service, for example medical records, lab reports, discharge notes, and hospital bills.

Various problems have arisen concerning medical data use for LLM training or deployment. For instance, the kind of patient data that will be uploaded and made available to an external LLM server to train an AI model, or the fact that many of the health data that might be used to adapt LLMs to medical contexts may fall under formal privacy protections (e.g. the Health Insurance Portability and Accountability Act, or HIPAA). Another open question is what part of that training data would probably leak into the model’s responses [76]. Apart from the ethical and social aspect of disclosing confidential data about patients’ mental and physical health (i.e. the individual may feel emotionally distressed and embarrassed, or they may face discrimination or harassment after their private records have been exposed publicly), security violations of private and sensitive patient records can be detrimental for both individuals and organizations from an economic and legal perspective, too. At the individual level, the damage may include identity theft, loss of money, extortion, or even physical harm. At the organizational level, the damage may include a loss of

public trust, deteriorating reputation, criminal or civil penalties, or remediation costs to compensate the compromised individuals [37].

To protect sensitive health information, the U.S. Department of Health and Human Services has enacted the Health Insurance Portability and Accountability Act Privacy Rule, which is a document that sets standards for the proper use and disclosure of PHI, how to store such information securely, and what to do if there is a breach. The Rule defines 18 items that mark health information as PHI ⁵:

1. Names
2. Physical addresses (street address, city, county, zip code)
3. Dates (including birth date, admission date, discharge date, and date of death)
4. Phone numbers
5. Fax numbers
6. Email addresses
7. Social Security numbers
8. Medical record numbers
9. Health plan beneficiary numbers
10. Account numbers
11. Certificate/license numbers
12. Vehicle identification numbers (including license plate numbers)
13. Device identifiers and serial numbers
14. Web Universal Resource Locators (URLs)
15. Internet Protocol (IP) address numbers
16. Biometric identifiers (including fingerprints and voice)
17. Full face photographic images and any comparable images

⁵“HIPAA Administrative Simplification Regulation Text“, March 26, 2013, p. 97, Accessed January 28, 2024. <https://www.hhs.gov/sites/default/files/hipaa-simplification-201303.pdf>

18. Any other unique identifying number, characteristic, or code

In order to align with ethical standards and societal norms, the solution is to follow specific guidelines and principles from the emerging field of Responsible AI for an ethical use of AI systems, including significant human oversight and impact assessments [76]. Regarding training data specifically, a more practical approach is to de-identify, i.e. anonymize, PHI in order to prevent any potential linking of that data with any particular person. De-identification can be done via performing named entity recognition (NER) on clinical notes, in order to locate and then obscure the actual PHI information, e.g. names, dates, etc., by replacing it with a generic label, e.g. <NAME>, <DATE>, etc. An alternative for obscuring real PHI data is by replacing it with pseudo-random data, e.g. pseudonyms.

(Mis)trust in AI and ‘alert fatigue’

There is still a substantial number of people who are more or less skeptical about the use of AI in healthcare. For example, there were relatively few applications of AI directly used in widespread clinical care as of the beginning of 2023, with some rare exceptions, even in medical imaging [52]. “One of the major challenges for technology in healthcare is that it needs to work well enough to drive widespread end-user adoption, demonstrate measurable time savings, and create a clear financial return on investment.”⁶ Therefore, facing resistance from some nurse staff to full-fledged AI adoption in their work pipeline is to be expected. However, even tech-savvy nurses who otherwise trust AI in general, may resist the adoption of a conversational chatbot in their work. The reason may be attributed to the fact that healthcare employees are overworked already (see Section 2.1.1), and incorporating new ideas into existing workflows can be daunting and stress-inducing. A solution could be to make the use of the AI tool optional, and not mandatory, allowing nurses themselves to have agency on its use. With time, confidence in its use will grow, as with any other new tool.

2.2.2 Conversational AI applications in healthcare

Conversational assistants based on generative AI are being used in healthcare as goal-oriented dialogue systems with a wide range of functionalities. As will be explained

⁶“Ambience Healthcare Raises \$70M to Scale the Most Comprehensive AI Operating System for Healthcare Organizations“, PR Newswire (press release), Accessed January 29, 2025. <https://www.prnewswire.com/news-releases/ambience-healthcare-raises-70m-to-scale-the-most-comprehensive-ai-operating-system-for-healthcare.html>

below, their aim is to assist professionals in the delivery of care by supporting them in their diagnoses, treatment, administrative and communication duties, by freeing up time and energy. Patients benefit, too, as the access to expert care becomes immediate, consistent, as well as location- and time-independent. There are at least eight areas where such assistants have proven useful through research or launched products:

Informing patients about their condition and developing medical literacy

Comprehension of their condition has been found to empower patients, boost their satisfaction in regards to the medical care they are given, and cultivate a collaborative and trusting patient-clinician relationship [3, 26]. Communicating the patient's health condition to them and to their family in a clear, understandable, non-jargon manner can help enhance the delivery of individualized and compassionate medical care [52]. Clear communication has also been reported to improve compliance with medication regimens and treatment therapies, lessen patient anxiety, and improve clinical outcomes and quality of life, while decreasing treatment times and hospitalization costs [35, 63]. However, due to administrative and other time-consuming demands, physicians lack the time, the concentration or the willpower to engage in a genuine, detailed, and terminology-free conversation during a face-to-face patient visit [63]. So, due to the clinical setting's pressures, physicians tend to intentionally refrain from explaining medical tests, results, or the pros and cons of a therapy, and even patients themselves hesitate to ask for information to be repeated [7]. There have been attempts to bridge this gap, for example via the well-known and successful *OpenNotes* initiative, through which clinicians share medical notes with their patients ⁷.

Nowadays patients have the means to know more about their condition, without being dependent on their physician's availability. It is now possible to simply input their doctors' notes or medical tests into any commercial, general-purpose AI chatbot and begin a conversation by "Explain this to me in plain English", or any other language [59]. The assistant can provide medical summarization, explanation of terminology, procedures, and medications, as well as general (i.e. not personalized) information on medical conditions, symptoms, causes, and treatments, based on publicly available reputable sources, such as MayoClinic or WebMD. For instance, some popular voice assistants are Google Assistant, Amazon Alexa, Apple Siri, and Samsung Bixby, and

⁷A 2020 study enhanced this process with external AI software that converted medical notes into plain language of 8th grade reading level [3]. With this tool, clinicians did not need to change their note-taking style – filled with abbreviations and medical jargon – nor spend extra time to manually simplify or clarify their notes. Ultimately, patients judged the simplified notes to be desirable, useful, and empowering.

Table 2.1 Functionalities of voice assistants in healthcare

Functionalities of AI Assistants in Healthcare	Example Sources
Informational chatbots: Informing patients about their condition and developing medical literacy	Study on the performance of ChatGPT on US Medical Licensing Examination [52]
Health coach chatbots: Promoting public health and behavior change	Physical activity intervention [10, 51] education on responsible alcohol use [30] asthma education [50] cancer education [19] sexual health [15] sleep hygiene [88] eating disorders [20]
Diagnostic chatbots: Checking personalized symptoms and offering a diagnosis	Google’s <i>AMIE</i> [97] excessive daytime sleepiness [83] depression and occupational burnout [84, 1] Post-Traumatic Stress Disorder (PTSD) and mental distress [66, 62] alcoholism [30] diagnostic assistant for cancer referrals [5]
Medical Research Assistant chatbots: Empowering medical students	Study on the performance of ChatGPT on US Medical Licensing Examination [52]
Treatment assistant chatbots: Monitoring post-treatment plans and supporting self-management for chronic diseases	Diabetes [21, 36] cancer [19] asthma [85, 50] chronic pain [38] chronic stress [41]
Mental health chatbots: Managing mental health conditions and providing behavioral counselling	Depression [32, 44, 91] panic disorder [77] Attention Deficit Disorder (ADD) [43] schizophrenia [8] Post-Traumatic Stress Disorder (PTSD) [66, 62] Substance Use Disorder (SUD) [30, 28, 16, 14, 75]
Patient engagement chatbots: Providing communication therapy	Parkinson’s and dementia [42, 24, 93] Autism Spectrum Disorders (ASD) [96]
Administrative task chatbots: Automating communication and documentation tasks for medical staff	Study on the use of AI assistants for patient questions [2] virtual nurse for discharge education [9] support for non-professional caregivers on diet management for patients with dementia [58] support for clinicians on local guidelines and recommendations for treatment options [48] documenting patients’ progress (e.g. <i>Suki</i> , <i>Nursebrain</i> , <i>Rover</i> , <i>Know My Patient</i> , <i>ChatEHR</i>)

a common text-based assistant is ChatGPT. Such conversational AI systems, having absorbed a great amount of medical knowledge during their training, can perform tasks such as clinical reasoning, and answering medical questions quite accurately, even without fine-tuning [52, 46]. Since the early 2022 it has been established that language-based AI systems (e.g. ChatGPT running on GPT-3) can perform at such a level that they reach or exceed the 60% passing grade for the US Medical Licensing Examination (i.e. the standardized exam used to license doctors), even with no specialized training input or prompting [52]. Later in the same year it was found that GPT-4's performance had matured and was comparable to the median physician in all core medical disciplines of the Israeli board medical examinations (i.e. internal medicine, general surgery, pediatrics, psychiatry, and obstetrics and gynecology), with the exception of psychiatry, where it ranked *higher* than the majority of physicians [46]. GPT-4's performance reached or exceeded the passing rate levels in all five core medical domains, a fact which gives confidence in using the technology for clinical matters.

However, recent studies show that commercial, general-purpose, voice-based agents are neither always complete, safe nor communicatively suitable in their advice. In 2018, 30 health and lifestyle prompts were posed to *Siri*, *Google Assistant*, *Alexa*, *Bixby*, and *Cortana*, which collectively reached only 41.0% of appropriate responses to safety-critical prompts that needed a referral to a professional (e.g. depression, heart attack), and 39% to non-safety critical ones (i.e. physical health symptoms and lifestyle questions) [47]. In this study, there were cases where the agents provided differing responses to the same prompts, or cases where they exhibited no sensitivity or empathy in their responses. Moreover, Palanica et al. have examined the speech recognition and comprehension performance of the top 50 US – most of them complex – medication names by popular conversational agents *Siri*, *Google Assistant*, and *Alexa*, also from users with foreign accents of English. Comprehension accuracy for *Google Assistant* was 91.8%, for *Siri* 58.5%, and for *Alexa* just 54.6%, and notably the last 2 agents' performance deteriorated by around 10% with a non-native English accent. The study raises concerns about the trustworthiness of responses based on misrecognized medication names. Further worrying results were reported by another 2018 study, which concluded that to rely on the aforementioned speech-based conversational assistants for actionable medical information, such as medication recommendations or emergency tasks, poses a safety risk for patients [11]. The study advises users to always consult health care professionals before following an AI response on any medical question. It is remarkable that, as mentioned before, the study participants were chosen to be only

native users of English, because conversations with non-native users in a earlier pilot resulted in extremely high misrecognition rates.

Promoting public health and behavior change

Targeted conversational agents can play a decisive part in improving health literacy, health behaviors, and decision-making in the general population. They can be used as tools to encourage adults to increase and maintain their physical activity [10, 51, 90], follow sun protection guidelines [90], manage eating disorders [20], and improve contraceptive knowledge and behavior [68], or inform young people about alcohol, sex, and drug use [23], and teach them healthy lifestyle strategies to ultimately build resilient minds [82]. Even though they cannot replace human medical practitioners, they can act as a useful first point of contact, e.g. by doing initial screening and referring to a professional [30] or as a virtual companion for positive behavior change [10]. The upsides include confidentiality [62], a wide range of trusted medical information, and the delivery of care irrespective of the user's location or time availability [10, 90, 30]. Typically, they feature a friendly, mentor-style personality that establishes a close relation with the user [10, 30].

On physical activity interventions for public health, *Ally* (*Assistant to Lift your Level of activity*) is a chat-based smartphone app used during a 2017 1.5-month study that promotes walking through delivering personalized planning and self-monitoring prompts [51]. The app tracks the user's steps and provides daily and weekly overviews of steps, distance, and energy consumed, offers a weekly planning of date, time and location of upcoming walks, and also sends reminders on the planned dates via the chatbot interface initiating a conversation with the user; the reminders include the daily step goal, the difference between the user's current step count and the goal, an estimation of minutes needed to reach the goal, and an actionable tip on how to increase daily steps. The way *Ally* sets the daily personalized step goal is based on the user's past activity data and current average step count. Interestingly, the component that was linked with an 8.1% greater probability of reaching daily step goals was the financial incentive, i.e. earning US\$ 1 per day for reaching the personalized step goal. A similar agent, *FitTrack*, was designed in a 2003 study as an exercise advisor to help users achieve the recommended minimum level of daily physical activity of 10,000 steps or 30 minutes of activity while maintaining a social relationship with them

[10]. The embodied ⁸ computer-based agent used synthesized speech and synchronized nonverbal behavior (hand gestures, body posture shifts, gazing at and away from the user, raising and lowering eyebrows, head nods, and walking on and off the screen) to build emotional connection. Users interacted in a constrained manner, i.e. by selecting items from multiple-choice menus of text phrases, which were dynamically updated based on the conversational context. They received standard behavioral interventions, such as self-monitoring (progress charts showing their activity levels over time), and educational content on the topic of walking for exercise. They also held a daily conversation with the agent about their progress and any obstacles they had regarding exercising, and were offered tips to overcome them. A similarly informative, interactive, and engaging agent was developed in 2012 for a 12-month clinical trial to promote and maintain positive behavior change regarding physical activity and ultraviolet light exposure protection, in order to preempt certain types of cancer, e.g. skin cancer [90]. While conversing with the participants, the agent provided answers, support, and encouragement, and appeared empathetic and attentive to them thanks to its non-verbal (intonation, human-like mannerisms, etc.), and social dialogue skills (humor, storytelling, small talk, etc.). The agent also emailed periodic reminders to motivate consistent physical exercise and sun protection behaviors, and it eventually achieved an 88% user satisfaction, and a 77% behavior change.

A few text-based assistants have also been developed to manage distorted weight and shape concerns of people who suffer from or are at high risk of eating disorders (EDs) [20]. One of them, called *Tessa*, was designed in 2022 to target young adult women at risk of developing EDs by delivering “Body Positive“, an EDs prevention program. The program comprised the following sessions: challenging the thin-body ideal, media literacy, healthy eating, critical comments, exercise, bingeing, and maintenance, all through synchronous free-form conversation.

Chatbots have also been developed to aid family planning by providing fertility as well as contraceptive information, and linkages to products and services to adults of reproductive age [68]. However, there are conflicting results on their impact, i.e. on the increase in contraceptive knowledge and on the intention of users to actually use contraception.

As we will see in a subsequent Section (2.2.2), there exist conversational agents that promote understanding of asthma [50], and various types of cancer [19], which is fundamental to self-management and better disease control.

⁸“Embodied conversational agents are animated humanoid computer-based characters that use speech, eye gaze, hand gesture, facial expression, and other nonverbal modalities to emulate the experience of human face-to-face conversation with their user“. [17]

On alcohol abuse, a 2016 study on young adults examined a text-based chatbot intervention that provided education on responsible alcohol use, and performed a standardized risk assessment of the user's drinking habits [30]. The chatbot gave relevant feedback, advice, and links to government agencies and private organizations for more information on how to seek medical assistance, if necessary.

On sexual education, many young people refrain from getting medical advice, and therefore a diagnosis and medication, for fear of stigma, discrimination and homophobic attitudes, thus remaining unaware of their status and endangering their partners' health [15]. *SHIHbot* was developed to educate on HIV and AIDS disseminating reliable, accurate information. An earlier chatbot was developed in 2011 by [23] to give answers to questions about sex, alcohol, and drug use to high school students, who do not often get adequate information from their parents or school. The goal was to help adolescents make informed decisions before experiencing these risky behaviors themselves. The chatbot was used extensively, and was perceived to be faster, more concise, and more anonymous than typical search engines.

Keeping to young people, a recent 2020-2021 intervention of 12 weeks was designed to decrease sedentary behaviors, and increase physical activity, sleep quality, and breakfast frequency for 12-15 y.o. school students with the aim of promoting general well-being and mental resilience [82]. The mobile-based health app, called *#LIFEGOALS*, had several components, as it comprised an activity and sleep tracker, self-regulation techniques (goal-setting, coping strategies, self-monitoring, etc.), gamification elements (earning virtual coins after completing goals to personalize the app's avatar), weekly health-related narrative video episodes, and a chatbot for answering questions on lifestyle behaviors, and sending encouraging messages and timely reminders for engaging with the app. The intervention had positive outcomes on learning about and adopting good lifestyle habits; however, the COVID-19 pandemic "shelter-at-home" restriction measures moderated some of these.

On sleep, a 2021 research study developed a dialogue system that acts as a sleep coach, aiming to provide relevant information, and make a diagnosis on all potential behavioral causes of sleep issues. Finally, the system recommends possible next steps, directing users to helpful lifestyle changes, such as limiting caffeine and alcohol consumption before bedtime.

Checking personalized symptoms and offering a diagnosis

Diagnosing patients means classifying them into the most fitting class of disease, via examining their clinical status. This is why for a proper diagnosis a clinician needs to

gather accurate symptoms. In order to so, face-to-face, meaningful, and lengthy clinical interviews are very important for communicating empathy and eliciting patient trust. However, problems such as the limited visit time, the patient’s geographical location, their hesitancy to share intimate information with another human [84, 61, 69, 62], their fear of stigmatization [62], or diverging from the standardized procedure on the side of the physician can harm a diagnosis, especially of mental health disorders [84]. Therefore, the benefits of safe, ethical, and robust diagnostic conversational agents can be many: access to primary care even to under-served populations [83], fewer misdiagnoses as well as earlier diagnoses that lead to better treatment of diseases and recovery rates [84], “humanized” anonymity and higher patient satisfaction [84, 62, 1], and cost reduction [84, 97]. Such agents could act as a decision aid to both patients and physicians: “AI systems capable of such diagnostic dialogues could increase availability, accessibility, quality and consistency of care by being useful conversational partners to clinicians and patients alike.”⁹

Google’s *AMIE* (*Articulate Medical Intelligent Explorer*) is an AI agent based on a medical LLM that enables natural, synchronous, and text-based conversation with patients with the aim of offering an accurate diagnosis. *AMIE* is an impressive tool for the field of clinical reasoning, not only because it can maintain fluent and empathetic conversations, but also because it actively elicits missing information through the conversation. Its training is novel: through self-play in the roles of doctor and patient, chain-of-reasoning, and automated feedback mechanisms it has learnt to evaluate medical history and symptoms with high precision. *AMIE* has been trained on datasets of real-world medical reasoning, summaries, and clinical conversations, but also on simulated conversations. *AMIE* was tested on text-based dialogue by simulating consultations with trained actors, and was evaluated on areas of consultation quality, such as history-taking completeness, diagnostic accuracy, management plan, and communication skills. The system was rated superior to human primary care physicians in 28 out of 32 areas from the perspective of specialist physicians, and in 24 out of 26 areas from the perspective of patient actors [97]. As for its conversational performance, *AMIE* scored higher than human physicians in communication and empathy, and was reported to have longer and more structured replies. This conclusion is consistent with a prior study where chatbot (ChatGPT) responses to health questions posted by patients on a social media forum (cf. Section above) were rated as more

⁹Alan Karthikesalingam and Vivek Natarajan, ‘AMIE: A research AI system for diagnostic medical reasoning and conversations’, Google Research (blog), January 12, 2024. Accessed on September 3, 2024, <https://research.google/blog/amie-a-research-ai-system-for-diagnostic-medical-reasoning-and-conversations/>

preferable, and significantly more empathetic, more detailed, and of higher quality than the responses from actual, verified clinicians [2].

Older initiatives include “virtual humans” programmed to follow standardized diagnostic procedures to help the professional clinician with automating the diagnosis procedure. For sleep disorders, and especially the common condition of excessive daytime sleepiness, a voice-based, embodied conversational agent was developed for a 2014 study to screen patients and offer an accurate diagnosis, with the aim of helping physicians. Indeed, very few are trained to adequately differentiate sleeping disorders from fatigue or even psychiatric conditions, resulting in prescribing the wrong treatment [83]. The agent was programmed to conduct a simple interview based on the widely used Epworth Sleepiness Scale, which is a self-administered, standardized questionnaire. After scoring a 80% for diagnosing alert patients, and 98% on diagnosing the sleepiest ones, it showed that a virtual physician can assist real ones in diagnosing patients with sleeping disorders; it was also well received by patients.

Similarly, an embodied conversational agent was developed in 2015 to identify specific symptoms and diagnose major depressive disorder in the general population via a face-to-face interview, following a list of criteria of a highly structured, reliable, and validated procedure [84]. Patients rated the agent as trustworthy, commended on its ability to communicate empathy, and also noted the agent helped reduce their feeling of being judged by a human interviewer. The agent efficiently identified patients without depression and also those with severe depression. Another chatbot, named *Vickybot*, was developed in a 2022 study to self-assess, monitor, and deliver psychological treatment modules according to severity (e.g. mindfulness) for depression, anxiety, and work-related burnout, and for detecting suicidal risk. The smartphone-based app was useful in diagnosing and significantly reducing occupational burnout (but not depressive and anxiety symptoms) and in intervening in emergency situations (e.g. suicidal thoughts and behaviors) by immediately providing emergency resources and alerting the research team [1].

On PTSD screening, two interventions have been studied specifically for military service members, who are much more likely to have such disorder compared to civilians. First, Lucas et al. developed a 2017 virtual human interviewer that establishes an anonymous as well as rapport-building interaction via verbal and non-verbal channels, e.g. utterances of agreements, head nods, etc. The virtual interviewer was highly accepted, and was found to increase accurate disclosure of PTSD and mental distress symptoms, compared to a traditional assessment. Also, *SimCoach*, another virtual human that uses conversational speech and gestures, was designed in 2015

to encourage military service members, especially those with signs or symptoms of PTSD or depression, to seek help to improve their psychological health [66]. The intervention offered voice-based screening for assessing psychological health through a symptom checklist, and conversational dialogue through which users' entries signifying distress would trigger personalized recommendations, e.g. direction to web pages that mention treatment or help-seeking services. However, there was no benefit regarding the participants' intent to seek help, according to the study's evaluation.

On alcoholism, as described in Section 2.2.2, Elmasri and Maeder have studied a text-based chatbot intervention that performed a screening of risky drinking habits, and gave relevant advice and links to government agencies and private organizations for more information on how to seek medical assistance, if necessary.

A much earlier mixed-initiative and voice-based conversational agent, *ERA*, was used in a 2006 study for decision support to general practitioners and nurse practitioners to help determine whether a patient should be referred urgently to a cancer specialist for suspected cancer [5]. The system was designed to be used over the phone to acquire data regarding the patient's signs and symptoms, while also re-phrasing and elaborating a question, asking clarification questions on the user's responses, and explaining its reasoning; the final decision is made by the user.

In the domain of clinical decision support, treatment selection after a diagnosis is also a field that has been examined for conversational agents. A preliminary study of 2023 examined medication recommendations by ChatGPT-plus (GPT-4 model) regarding prescriptions of anti-depressant drugs for major depression after initial antidepressant nonresponse [79]. The model identified optimal treatment for 76% of test cases, but also advised less optimal or contraindicated approaches (e.g. prescribing medication associated with weight gain to obese patients) for almost half of them. Interestingly, the model provided as rationale for treatment selection a number of simple heuristics commonly applied by human physicians in psychopharmacologic decision-making to better match patients with optimal treatments, such as avoidance of prior non-responsive medications and avoidance of medications with adverse effects based on medical comorbidities. A follow-up study of 2024 examined next-step treatment options for bipolar disorder, where the GPT-4 model was augmented with recent bipolar treatment guidelines within each prompt [80]. The model was asked to identify 5 optimal next-step pharmacotherapies and 5 poor or contraindicated choices, and explain its rationale. For 94.9% of test cases the optimal treatment was among the top 5 selected by the model, and for 84.4% the optimal treatment was among the top 3. Only for 12% of cases did the model select a medication considered by the experts

to be a poor choice or contraindicated. Both studies are encouraging observations for future applications of such models as interpretable tools for guiding prescriptions in other clinical domains, too.

An important, recent study from October 2024 by Goh et al. indicated that a commercially available chatbot (ChatGPT Plus, GPT-4) could outperform generalist physicians with a range of experience levels in making accurate clinical diagnoses. Interestingly, the chatbot alone was found to perform significantly better in diagnosing the given – albeit curated and summarized – patient cases in comparison to the two human groups: physicians with access to the same LLM, and physicians with access to conventional resources. The study demonstrates that, for human physicians, access to generative LLMs will not automatically improve diagnostic reasoning; they will need targeted training to integrate these resources appropriately into their workflow (e.g. in prompt engineering techniques).

Empowering medical students

Using generative AI in healthcare can have a lot of advantages in regards to the clinical, but also the educational setting, especially the domain of medical education. AI-generated answers have been found to bring along significant and novel insights, coupled with a deductive reasoning process [52]. What is more, conversational agents like *AMIE* can act as an educational tool for medical students, and embodied conversational agents can be used as virtual patients for training in detecting certain conditions, e.g. suicidal behavior [64].

Monitoring post-treatment plans and supporting self-management for chronic diseases

People with chronic diseases, like cancer, hypertension, diabetes, chronic pain, etc., can benefit from AI companions in keeping track of their ongoing medication routines and monitoring their symptoms [72, 19, 85, 69, 50, 21], gaining more knowledge and self-confidence in managing their disease [85, 87], and even getting personalized feedback from their physicians [50]. Specifically, self-management of a condition comprises adherence to a certain medication, activity and diet plan, as well as lifestyle changes to eliminate long-held dysfunctional behaviors and prevent relapses [41, 87, 38]. Also, using AI agents instead of paper records to check with chronic patients can promote more accurate and complete recordings, while saving time for health care providers [13, 49, 29] and reducing the health systems burden in general [36]. Scheduling and managing medical appointments is another feature of medical agents that helps towards

reducing missed appointments, and prevents patients from forgetting to bring their record sheets to their health centre, as they are carried within the app [29]. From a social perspective, conversational agents can coordinate the communication among health care professionals, family members, and patients, and achieve a holistic support network, while saving patients from visiting hospitals for on-site consultations [49, 50]. Specifically for chronic pain (e.g. back pain due to accident, surgery, physical decline, etc.), where cognitive behavioral therapy (CBT) is the standard therapy, there exist significant barriers to its management, e.g. a lack of adequately qualified providers, an inconvenient distance regarding the location of the few qualified ones, and a lack of time and money for on-site proper consultations [38]; there are similar challenges connected to chronic stress management [41, 87]. For all the above, interventions via conversational agents can be a cost-effective, accessible and flexible solution.

A 2019 study demonstrated that conversational agents can improve medication adherence rates of patients with breast cancer as well as their quality of life in general [18]. *Vik*, a text-based chatbot, was designed to empower patients with breast cancer and their relatives, by responding to their queries and concerns with quality-checked medical information, and reminding them of their prescription. The study analyzed the usage data of patients conversing with *Vik* for a year, showing that the more the patients used the chatbot, the more adherent they were. Further, *Vik*'s answers to common questions asked by breast cancer patients about their therapy management were found to be non-inferior and as satisfactory as information by a multidisciplinary committee (medical, radiation, and surgical oncologists) [6]. Actually, there exist many *Viks*, each designed to provide quality-checked information about the treatment, side-effects, quality of life, reimbursement and patients rights for other cancers and chronic diseases, such as headaches, asthma, etc. [19].

Turning to hypertension, however, two clinical trials have shown no improvement in home management compared to a purely blood pressure tracking app [81] or compared to pen-and-paper [29]. Specifically, the text-based smartphone app *HPCP* (Hypertension Personal Control Program) app was developed to support users on hypertension self-management and coach them on healthy behaviors including medication adherence, blood pressure (BP) measurement, diet, physical activity, sleep, and stress management. For example, to promote home monitoring *HPCP* encouraged, reminded, and tracked blood pressure measurements regularly, and recommended retaking readings or contacting a medical professional in case of outlier or extreme values. Similarly, the text-based *TensioBot* was a chatbot operating on Telegram and designed for assisting patients and physicians with blood pressure home measurements. Its features include alerts

for measurements, notifications of data recording errors, video- and text-based advice on good BP measurement practices, reminders and rescheduling of upcoming medical appointments. The trial confirmed that chatbot reminders do not affect treatment adherence (i.e. BP check at a fixed schedule) or number of BP checkings. However, it is important to note that chatbot users scored higher in knowledge and skills regarding home BP measurement technique, and that the bot assisted doctors and nurses by informing them accurately about BP values, thanks to a downloadable report sheet.

Turning to asthma, improved self-management and treatment adherence was reported in two studies targeting adolescent patients [85, 50], where parents greatly appreciated the deeper understanding and empathy towards the previously unknown degree of afflictions their children were facing. Despite being one of the earliest chatbots (2014), the text-based *mASMAA* (mobile phone-based Asthma Self-Management Aid for Adolescents) was used to interact with teenagers in free-form text and facilitate symptom monitoring, increase awareness of symptoms and triggers, support treatment adherence, give a sense of control over the disease, and promote adolescent–parent partnership via emailing daily reports to parents. Conversations between patients and *mASMAA* included questions and answers about asthma and its triggers, interpretation of the patients’ medical records, system-initiated messages to help them monitor symptoms (through automated daily asthma-monitoring inquiries) as well as activity levels, and reminders to take medications according to the patients’ individual schedule. A similar study of 2021 showed improved cognitive and behavioral skills after the agentic intervention of *MAX*, a text-based conversational agent aiming to help asthmatic young patients to adopt healthy behaviors (i.e., inhalation technique) as well as gain knowledge about their condition [50]. *MAX* took the concept of virtual asthma coach one step further by engaging the complete triangle of stakeholders: health care professionals (via email), patients (via the app), and family members (via SMS text messaging), thereby establishing a robust working alliance acting as a mediator. For instance, parents would record a video of their children’s inhalation procedure, upload it on a secure server, and then *MAX* would notify the physician for review and feedback, which would then be sent to the child through the app. What is more, it also introduced gamification for patients, so as to increase their perceived enjoyment and understanding, by introducing the point system after successful asthma quizzes, and for each finished coaching session.

About monitoring chronic pain, the Pain Monitoring Voice Diary was implemented in 2006 for real-time assessment of experience and behavior in the patient’s natural environment [56]. This electronic data collection technique allowed for real-time data

collection of information from patients over the phone, as they were responding to spoken dialogue system prompts using unconstrained speech. A more recent implementation is *SELMA*, a text-based smartphone chatbot designed and implemented in a 2018 clinical trial to deliver psychoeducation (via cognitive behavior therapy) for pain management [38]. Psychoeducation enhances peoples' understanding of their disease and their self-help motivation, while reducing their insecurity. The pain self-management coach guides patients through CBT modules that addresses dysfunctional cognitions and emotions, and teaches coping strategies with the aim to control pain. What is more, it provides relevant exercises, such as on stress release and mindfulness, and follow-up questions for the participants. Nevertheless, comparison of the intervention and control groups showed similar effects in terms of pain-related impairment, general well-being, and pain intensity.

On chronic stress management, a 2017 clinical trial showed that a conversational agent in combination with medical group visits was able to reduce long-term stress – the primary symptom for patients with chronic pain, depression, and anxiety – by motivating adherence to positive behaviors and providing personalized guidance [87]. *Gabby*, the voice-based animated conversational agent, was designed for patients to use at home for reviewing the educational material covered in more detail, as well as for practising self-care skills, such as meditation, self-massage, and yoga based on the widely-used training program 'Mindfulness Based Stress Reduction' (MBSR). *Gabby* could express herself with eyebrow movements, hand gestures, and gaze and posture shifts, apart from having a natural, calming speech, and received user input via multiple choice selection of utterances on the screen. An earlier intervention, the embodied *Virtual Mindfulness Coach* prototype, was developed in 2013 to provide meditation coaching via mixed initiative, text-based, free-form conversation enhanced with multiple-choice based input for reduction of stress, pain, and anxiety disorder's symptoms as well as for prevention of relapse into depression [41]. The coach, a female character named *Chris*, showed a variety of facial expressions, gestures, and gazes, which she adjusted reacting with empathy to the user's input and emotional state. *Chris* offered customized guidance on the lesson materials of MBSR curriculum, providing encouragement, advice and additional information, correcting misconceptions, and asking follow-up questions on specific content based on the previous dialog and the user's concerns (e.g., "Is mindfulness good for stress?" "Will meditating help me feel less stressed?"). She was demonstrated to help users establish a more frequent and longer mindfulness practice, compared to a self-administered program via written and audio materials.

On diabetes self-management, the current state of the art mostly assists patients via mobile applications, like *Care4Life Diabetes*¹⁰ or *MyDiabetesConnect*¹¹. These applications offer a dashboard for diabetic-related data points, recipes for diabetics, exercise and weight monitoring, customizable reminders for checking glucose levels as well as for taking medications and not missing doctor appointments, blood glucose and blood pressure tracking and summary reports, and educational messages [45]. Some apps include encouraging messages¹², a comprehensive diabetes management education plan¹³, the ability to connect the glucose meter with the app for error-free logging¹⁴, downloadable statistics and charts¹⁵, the ability to upload photos of meals to log nutrition information¹⁶, pairing with a smartwatch to display glucose levels¹⁷, sharing real-time data to the whole family's smartphones, and also a synced web platform for healthcare professionals to view their patients' logs¹⁸. However, inputting information on a screen can be difficult and even frustrating for elderly patients. *Healthy Coping* is a voice-based application that assists the older generation with self-management of type 2 diabetes by combining the Google Home's device voice interface for hosting the conversational agent, and a paired web interface for a visual display of the patient's monitoring results, intended for both the patients themselves and their physicians. It focuses on monitoring blood glucose levels, taking medication, being active, eating a healthy diabetic diet, and mental health coping (via offering a depression screening survey). It also offers personalized advice for maintaining target glucose levels, and reminders for taking medication, all through intuitive voice commands or questions expressed in natural language (e.g. "I want to take the coping survey.", "I want to monitor my health", "What should I eat?") [21]. It is notable that this dialogue system design is not innovative compared to much earlier telemedicine and phone monitoring solutions, such as *DI@L-log* from 2005 [12]. These solutions were used for data collection and for offering encouraging feedback and advice to patients through a voice user interface, and also supplied physicians with progress charts of weight, blood

¹⁰Care4Life, Southern Nevada Health District, YouTube (video), Accessed on September 25, 2024. <https://www.youtube.com/watch?v=y1VxFZkajGg>

¹¹Diabetes Connect LTD, Google Play, YouTube (video), Accessed on September 25, 2024. <https://play.google.com/store/apps/details?id=com.mydiabetesconnect&hl=en>

¹²BEAT Diabetes, Accessed on September 25, 2024. <https://www.beatdiabetes.org/>

¹³Glucose Buddy, Accessed on September 25, 2024. <https://www.glucosebuddy.com/>

¹⁴mySugr, Accessed on September 25, 2024. <https://www.mysugr.com/en-us>

¹⁵Diabetes:M, Accessed on September 25, 2024. <https://diabetes-m.com/features/>

¹⁶Undermyfork, Accessed on September 25, 2024. <https://www.undermyfork.com/>

¹⁷Gluroo, Accessed on September 25, 2024. <https://gluroo.com/>

¹⁸Health 2 Sync, Accessed on September 25, 2024. <https://www.health2sync.com/careproviders>

glucose values, etc. as well as alerts for deviant readings through a graphical user interface [13].

On childhood obesity, a 2017 5-month study implemented a text-based smartphone app that monitored and supported young patients during their treatment, and assisted their health carers too [49]. For rapport building there were a female or male persona to select, use of the teenager's first name, use of informal language and emoticons, engaging in small-talk, and expressing empathy. This assistant was programmed to offer a challenge a day, e.g. a relaxation technique for stress management, a step counter for physical activity, photographing meals for diet adherence, and answering entertaining quiz questions for promoting health literacy. The results of the challenges were reviewed in the next consultation hour.

Interestingly, attitudes of long-term patients on healthcare chatbots vary. A survey study conducted between January and March 2023 reported AI hesitancy among such population: they were skeptical about the use of AI chatbots for health care purposes, and less than 30% were likely to use a health chatbot in the following 12 months due to mistrust in its accuracy [100]. In the survey, chronic patients seemed more receptive to using voice-based chatbots for nutrition, sleep, and mental well-being, i.e. stress and emotional state [100]. Contrary results have been reported for the cancer agent *Vik*, whose users expressed an overall satisfaction at a 94% and even recommended the chatbot to their friends [18]. Also, a large majority of experiment participants have commended on the chatbot's ease of use and usefulness for the asthma agents *mASMAA* and *MAX*, and the hypertension agent *TensioBot* [85, 50, 38]. Specifically for *TensioBot*, most patients indicated they prefer to register their BP at home using *TensioBot* instead of manually; additionally, 85% continued its use even after the experiment was over.

Managing mental health conditions and providing behavioral counseling

Mental illnesses are very common: it has been estimated that 46% Europeans experienced an emotional or psychosocial problem during 2022 ¹⁹. Also 1 in 5 adults in the U.S. live with mental health [74], and 1 in 8 people worldwide ²⁰. Some mental health issues are chronic, non-curable, and can only be managed by medication, therapy, and lifestyle changes throughout the patient's life, in the same way other chronic health conditions described in Section 2.2.2 are managed. Such mental illnesses include

¹⁹European Commission. Mental health. Accessed on October 8, 2024. https://health.ec.europa.eu/non-communicable-diseases/mental-health_en

²⁰Mental health worldwide - Statistics & Facts. Statista. Accessed on October 8, 2024. <https://www.statista.com/topics/8066/mental-health-worldwide/>

schizophrenia, bipolar disorder, certain types of depression, Post-Traumatic Stress Disorder (PTSD), and Substance Use Disorders (SUDs). Some others are non-chronic and treatable with medication and therapy, e.g. mild-to-moderate depression, Acute Stress Disorder (ASD), panic disorder (in some cases), and phobias. Unfortunately, less than 1 in 3 people get mental health care worldwide ²¹. Untreated mental health conditions lead to high healthcare costs, impaired functioning and loss of autonomy of the patient in everyday life and in the professional domain, and social isolation [8, 93, 77]. Access to therapies that can manage or even cure a mental health disorder is often hindered by shortages in mental healthcare professionals [93, 71, 44, 91, 94], high costs for the individual [44, 69] and lack of time for frequent on-site consultations on the patients' side [9, 93], as well as avoidance of seeking support in-person through formal mental health services, or of giving an honest reporting of symptoms during official screenings, due to a variety of reasons: fear of social stigma [8, 25, 30, 32, 62, 9, 70, 75], judgment and discrimination [62], risk of blocking career advancement [66], reluctance to disclose intense or negative emotions and behaviors to other humans [9, 61, 69, 84], and impairments in emotions and communication due the illness itself [8]. Thankfully, conversational agents can be an affordable, accessible, acceptable by users, and reliable solution in providing 24/7 remote psychoeducation ²² [8, 77, 43], therapy [93, 32, 69, 44, 77, 43] and mental health assessment to those who need it [66, 62, 84], while simulating the interpersonal relationship of a mental health care provider [8, 62], securing anonymity [62, 91], and providing unbiased recommendations [30] as well as psychological support and information immediately, anytime, and anywhere [8, 39, 32, 69, 75, 44, 28, 16]. In this way, medication and exercise adherence can be increased, and hospital re-admissions as well as symptoms can be reduced [8]. A systematic review on the feasibility and effectiveness of interventions based on one-on-one, synchronous, written conversations has shown significant and sustained improvements in mental health outcomes, which are even equivalent (not superior, though) to face-to-face therapy and telephone counseling [39]. Last, let us not forget that the first well-known conversational chatbot, *ELIZA*, was

²¹WHO's Science in 5: Mental Health - 1 July 2022, Accessed on October 8, 2024. <https://www.youtube.com/watch?v=IOEXhd6c5k0>

²²Psychoeducation, the goal of which is to help people better understand (and become accustomed to living with) mental health conditions, is considered to be an essential aspect of all therapy programs. It is generally known that those who have a thorough understanding of the challenges they are facing as well as knowledge of personal coping ability, internal and external resources, and their own areas of strength are often better able to address difficulties, feel more in control of the condition(s), and have a greater internal capacity to work toward mental and emotional well-being.", GoodTherapy (blog). Accessed on October 19, 2024. <https://www.goodtherapy.org/blog/psychpedia/psychoeducation>

programmed back in 1966 to simulate a text-based conversation with a psychotherapist [98].

On depression and anxiety, the use of generative AI chatbots has recently increased, and research has indicated promising results in their use for meaningful mental health support. Two recent 2024 pilot studies used ChatGPT as a support tool in psychiatric care, albeit with a small sample size (12 and 19 patients, respectively) [67, 89]. The chatbot offered effective and accessible mental health care in the form of self-help advice, relationship coaching, psychoeducation, and emotional support, and scored high in both patient engagement and satisfaction. It also showed a substantial improvement in self-perceived quality of life, such as better interpersonal relationships and mood, and healing from trauma and loss. However, despite the overall positive feedback, a few participants complained about the irrelevant or overly long responses, and the restrictive safety guardrails that often urged users to contact a human professional, discontinuing the conversation. Moving to rule-based bots, in a 2023 pilot trial a virtual, voice-based coach, *Lumen*, offered problem-solving treatment (PST) therapy to adults with mild-to-moderate depression and/or anxiety [44]. The PST coach acts as a guide to identify a problem, set a goal, brainstorm solutions, choose a solution, develop an action plan, and to implement and evaluate the plan. After 4 months with *Lumen*, self-reported rational problem-solving, impulsivity scores as well as total psychological distress showed meaningful improvements. Six years earlier, a text-based conversational agent, named *Woebot*, was developed in study to deliver a 2-week self-help CBT program for young adults with symptoms of anxiety and depression with great success in reducing these symptoms and in keeping the users engaged nearly every day [32]. The reasons for its success were attributed to *Woebot's* interactional skills: use of emojis to express affect, showing empathy on users' responses, checking in and following up on users' mood, and sending a personalized message every day to initiate a conversation. The bot also offered weekly charts depicting each participant's mood over time, accompanied by a brief summary to facilitate reflection. Another text-based intervention was implemented in 2015 with *TeenChat*, designed to listen, comfort, and encourage stressed out young adults to practise coping strategies, as a "virtual friend" would, with the aim of releasing stress and avoiding deterioration inside the young and immature minds (e.g. depression, suicide) [40]. Turning to launched products, a popular mental health app is *Wysa*²³, which is a text-based coach designed to help its users deal with anxiety, worry, and low mood by teaching skills to manage their emotional resilience, while ensuring their anonymity. *Wysa* uses

²³Wysa (website). Accessed on October 13, 2024. <https://www.wysa.com/>

CBT, meditation, breathing, yoga, and motivational interviewing to help build mental skills in a self-help context; it also offers direct access to one-on-one sessions with mental health professionals. Also, a 2020-2021 study on users' engagement with *Wysa* during the COVID-19 pandemic confirmed that conversing with the app can result in a significant reduction in symptoms of anxiety and depression, further suggesting that digital health interventions can provide large populations with efficient support for mental health care [91]. A similar, text-based conversational assistant that offers mental health assessment and one-on-one counseling on a wide range of mental health and well-being issues, such as depression, anxiety, work stress, grief, eating disorders, bullying, addiction, etc., is *Noni* ²⁴. *Noni* also offers the opportunity to chat live with a volunteer listener, take part in guided chat-based group therapy with other users, and connect with human therapists via videocalls.

Panic disorder, which falls under anxiety disorders, is characterized by recurrent and unexpected panic attacks, i.e. sudden surges of intense fear or discomfort. A 2020 clinical trial developed a mobile, text-based app, named *Todaki*, to relieve panic symptoms, such as social anxiety, rapid heart rate, sweating, trembling, shortness of breath, or a feeling of losing control [77]. *Todaki* offers CBT training to correct distorted thinking, as well as reliable information on the disorder which it tests with quizzes in every conversation, it describes the effects and side-effects of medication, it shows videos of psychiatrists for breathing and muscle relaxation training, it suggests coping skills for real-time symptom management, and last provides a “daily panic report” for the users to record and assess their panic episodes, depressive moods, and dysfunctional thoughts, which it then displays as a graph. The results of the one-month trial showed that the severity of panic disorder symptoms were significantly decreased, and that interactive CBT using the chatbot is both feasible and effective for patients with panic disorder. *Todaki* was later repurposed for Attention Deficit Disorder (ADD) – also known as Attention Deficit Hyperactivity Disorder (ADHD) – which is very common to depression and anxiety patients. Similarly, psychoeducation sessions and CBT were used to mitigate associated psychiatric symptoms, such as inattention, impulsivity, and hyperactivity through behavioral change (time, space, and work management), emotional control, and daily conversation checks on concentration and anxiety [43]. *Todaki*'s addition was the use of social dialogue skills, such as empathy and reflection, to create a therapeutic bond with the users. The study's results showed a decrease in inattention/memory problems, and hyperactivity/restlessness, and verified the effective use of interactive CBT via a chatbot.

²⁴7cups (website). Accessed on October 19, 2024. <https://www.7cups.com/>

On schizophrenia management, an early 2010 voice-based relational agent named *Laura* was used in a month-long pilot study to promote long-term anti-psychotic medication adherence and walking as a physical activity. Medication non-adherence in schizophrenia approaches 50% the first year, and 75% the second year with ensuing relapse estimated at an annual rate of 46%. Interestingly, *Laura* was designed to establish a warm interpersonal relationship with the patient by projecting emotions and using supporting language, also known as “therapeutic alliance” [8]. According to Bickmore et al. relational agents are “animated conversational agents designed to establish trust and therapeutic alliance with users over time [...] through displays of empathy, close proximity, more frequent eye gaze and attentiveness, social dialogue and humor”. This empathetic relationship has been shown to improve symptoms during psychotherapy [71]. *Laura* interacted with patients once a day to check and remind adherence to their medication and exercise regimen, set new exercise goals for the patient to work towards before the next conversation, displayed the self-reported adherence charts, and also offered tips and education, e.g. on managing medication side effects, strategies for preempting forgetting, or overcoming common barriers to physical activity. *Laura* kept track of medication refills and reminded patients of pharmacy visits, and also prompted the patient to call their nurse in critical circumstances, e.g. when the patient had reported self-harming thoughts. Towards the end of the intervention she taught techniques for self monitoring. *Laura* exhibited empathy and trust through verbal (calling user by name, using first person plural grammar types, empathetic exchanges, social chat, humor, and reciprocal self-disclosure) and non-verbal behavior (hand gestures, head nods, eye gaze movements, eye brow raises, posture shifts, proxemics, and facial displays of emotion), but without the stigma and anxiety of a face-to-face human encounter. Interaction with users was simplified: they responded not with free-form speech, but by selecting an utterance from a dynamically-updated multiple-choice input menu. This design choice was made to accommodate individuals with a wide range of computer literacy levels and voice qualities, and also to safeguard against potentially dangerous effects of misunderstanding ambiguous utterances about medical topics. *Laura* scored high on acceptance, usability, and self-reported adherence.

On PTSD, as mentioned earlier in Section 2.2.2, two voice-based interventions for user self-assessment have been studied so far for military service members. Both included embodied virtual human interviewers that used anonymity as well as rapport-building behaviors, and have resulted in higher and more accurate disclosure of symptoms of PTSD [66, 62].

Substance Use Disorders (SUDs) are considered a common form of mental health disorder [30]. On alcohol abuse, as already mentioned in Section 2.2.2, a text-based chatbot was studied in 2016 that provides education on drinking responsibly, performs a standardized screening, and offers expert advice and links to organizations for seeking medical assistance [30]. On smoking addiction, an intervention was studied in 2016, in which a text-based prototype chatbot was developed to divert smokers from smoke cravings during the critical first three weeks of a cessation program [28]. The bot offered immediate motivating comments, multimedia content, games, and connection with the other participants, thus helping to keep the around 7,000 participants from relapsing. In a subsequent edition of the program, the chatbot’s successor tracked the users’ smoking behavior (location, time, mood, desire intensity, etc.), and offered the statistics to raise the patients’ awareness on their smoking patterns; then, it provided more personalized support with possible substitution and avoidance strategies [16].

However, even though clinical AI has shown great potential in performing tasks related to psychotherapy (i.e. assessment and delivery of therapy), launched products and prototypes are not yet sophisticated enough to fully replace psychotherapists²⁵. Moreover, concerns about patient safety have already been raised. Major voice-based conversational agents have been shown to provide no information on SUD treatment according to a 2019 study [75]. The study examined responses to people seeking help to quit an addiction (alcohol, opioids, marijuana, tobacco) by 5 popular virtual assistants: Amazon’s *Alexa*, Apple’s *Siri*, *Google Assistant*, Microsoft’s *Cortana*, and Samsung’s *Bixby*. Almost no agent gave a singular and coherent response (with the exception of *Google Assistant* that directed to a mobile smoking cessation app), and their responses often led to confusion or even promoted substance retailers! A similar study conducted in October 2017 concluded that there is still “room for improvement” for smartphone digital assistants on the topic of smoking cessation advice [14]. The researchers compared *Google Assistant* and *Siri* on quality of information, that is information taken from health agencies with medical expertise, such as hospitals, specialist clinics, and universities, versus low-quality sources, such as magazines or individual blogs. *Google Assistant* returned expert content only about half the time, whereas *Siri* just one in four times. A 2015-2016 research examined the responses on mental health issues and interpersonal violence of 4 commercial, general-purpose conversational assistants: Apple’s *Siri*, Google’s *Google Now*, Samsung’s *S Voice*, and

²⁵Toward Responsible Development and Evaluation of LLMs in Psychotherapy’ (Policy Brief). Human-Centered Artificial Intelligence (HAI), Stanford University. June 13, 2024. Accessed on October 30, 2024. <https://hai.stanford.edu/toward-responsible-development-and-evaluation-llms-psychotherapy>

Microsoft’s *Cortana*. Victims hesitate to report such sensitive problems for fear of stigmatization and retaliation, and this is why effective interventions are sometimes implemented too late or not at all. The agents were evaluated on their ability to flag a human utterance as concerning (e.g. “I want to commit suicide.”, “I was beaten up by my husband.”) and refer them to the appropriate helpline, for instance for suicide prevention, depression, domestic violence, or sexual abuse. The responses, though, were assessed as inconsistent and incomplete regarding recognition of a personal crisis, referrals to resources, or even showing respect [70]. Last, the PTSD agent was found to be lacking in recognizing trigger-worthy user responses, therefore missing cases that need referral to emergency services [66]. Therefore, for conversational agents to be able to offer guidance during personal crises, more collaboration among software developers and health authorities is necessary.

Providing communication therapy

Neurological conditions, such as autism, Parkinson’s and Alzheimer’s disease, stroke, and dementia, lead to difficulties in cognitive and motor skills, social interaction, speech, and behavior. Conversational agents can help these patients to increase their socialization skills by providing cognitive stimulation and repetitive practise in social dialogue.

Harlie (Human And Robot Language Interaction Experiment) was a 2016 smartphone voice- and text-based application designed to hold a large variety of conversations with people suffering from neurological conditions, such as Parkinson’s disease and dementia [42]. These populations have difficulty initiating and managing a conversation, which leads to distress and gradual social exclusion. *Harlie*’s job is to place a random call daily to monitor for challenges these people may be facing while analyzing their language articulation, vocabulary range, and the length of mid-sentence pauses. The system logs this information and then passes it, with consent, to health and research teams to aid further research and provide more targeted therapy [22]. Ultimately, *Harlie* provides conversational practise, feedback as well as non-judging companionship.

For Alzheimer’s disease (AD) – the most common form of dementia – the standard form of therapy is regular (even daily) reminiscence therapy, which involves conversations between patient and caregiver that evoke the patient’s old past experiences. The goal is to focus on the patient’s long-term memory, which is easily accessible during the moderate AD phase compared to their short-term memory, in order to stabilize the patient’s psychological state, stimulate their memory and cognitive functions, and increase their self-confidence and sociability [93]. To this end, a voice-based conversa-

tional agent has been proposed to provide daily, personalized dialogs for each patient based on their life history and lifestyle information [24]. Also, as recently as in 2025 a purely voice-based chatbot was presented, *Remi*, which combined a continuously growing memory context through summarizing past sessions, as well as empathetic engagement and emotional support ²⁶. However, concerns have been raised since 2009 regarding the interaction behavior of older users, who typically treat spoken dialog systems like real humans [99]. Thus, they become very sociable and chatty, putting a strain onto the speech recognition and natural language understanding components of a voice companion.

For patients of Autism Spectrum Disorders (ASD), who lack communication skills, social interaction difficulties and even isolation are common problems [96]. Social skills training (SST), which is the recommended treatment, is typically carried out by a therapist, but it can be automated by a virtual agent. The benefits of using such agents include the augmentation of social interactions, through the establishment of a relationship between the user and the infinitely patient SST coach [10]. In a 2017 study, a voice-based embodied agent was developed to coach the narrative skill, explaining its goals and demonstrating it for the patients to imitate afterwards in a role-play [96]. Interestingly, the system recorded an audiovisual video of the patient and could then analyze the speech (i.e. F0 variation, amplitude, voice quality, pauses, words per minute, words over six letters, and fillers) and facial expressions (i.e. horizontal and vertical head pose, smiling ratio) to provide feedback on strengths and weaknesses. Also, the agent set minor homework challenges for users to finish over the week, and report on the experience in the following session. The results showed improvement and maintenance of narrative skills, and demonstrate that virtual SST coaches can be used either as an automated social skills trainer or as a supplement to human-based SST.

Automating communication and documentation tasks for medical staff

Dialogue systems can assist clinical staff by saving them time and energy from executing certain repetitive and monotonous tasks in the inpatient setting, such as answering common patient questions, remembering and following lengthy protocols, and documenting the patients' health status, among other standardized tasks of the clinical and nursing workflow. By digitally supporting these tasks via a mobile device (e.g. a smartphone), medical staff can save time, increase their productivity, reduce the risk for

²⁶Chatbot: Empowering Medical Communication using AI, NLP Healthcare Summit 2025, Accessed on April 6, 2025. <https://app.swapcard.com/event/healthcare-nlp-summit-2025/planning/UGxhbm5pbmdfMjU3Njg5Mg==>

errors, and ultimately focus on physically interacting with their patients. Also, doctors' and nurses' job satisfaction can be enhanced, and the so-called "clinician burnout", already mentioned in Section 2.1.1, can be mitigated. Since patient satisfaction and treatment outcomes are connected to the medical staff's workplace contentment [27], improving the latter should improve the former, too.

On patient question answering, generative AI chatbots, like ChatGPT, are already being used by doctors for drafting medical reports or sending letters and emails ²⁷, thus saving time and effort. Also, since conversational agents can empower patients, as explained earlier, less strain and fewer demands are placed on physicians. AI-powered patient portals may soon become the "primary contact point" for health information, as AI-curated answers of patients' medical questions can save doctors time, provide quick and accurate answers, and free up resources, eliminating unnecessary clinical visits [2, 59]. Specifically, the commercial ChatGPT chatbot (version GPT-3.5) was used in a 2023 study to produce answers to patients' health questions publicly posted in a social media forum [2]. Its responses were afterwards compared with the actual responses given by verified health care professionals on that forum. The evaluators – who were themselves licensed health care professionals – rated both the quality of information and the empathy provided to be considerably higher in chatbot responses than physician responses, and ultimately preferred the AI-generated responses 4 to 1. This study suggests that AI chatbots could help physicians in messaging patients by providing a first draft that a professional or even a support member of staff would then proofread and edit ([2]). A practical example is Mayo Clinic's nurse staff who have been using generative AI-drafted responses to patient messages since early 2023. This initiative saves the staff around 30 seconds of time spent per message (1 min. and 20 sec. compared to 1 min. and 50 sec. when writing from scratch) with the added advantage that the final messages receive the patients' appreciation, as they are more conversational, friendly, and detailed than the ones that nurses used to write on their own ²⁸. However, a 2024 study conducted by Nigam Shah, Professor of Medicine, Stanford School of Medicine, and UC San Diego, examining LLMs answering patients' questions in text format, showed that the time spent by doctors reviewing and editing the AI-drafted responses has remained the same (i.e. there were no productivity gains),

²⁷Stermer C., 'AI ChatGPT for Medicine', TikTok (video), December 12, 2022. Accessed on September 7, 2024. <https://www.tiktok.com/@tiktokrheumdok/video/7176340747170467114>

²⁸"Gen AI Saves Nurses Time by Drafting Responses to Patient Messages", Epic (blog), 04 March 2024, Accessed July 29, 2025. <https://www.epicshare.org/share-and-learn/mayo-ai-message-responses>

and only their cognitive load has reduced (i.e. only efficiency has increased) ²⁹. What is more, another 2024 study published by the medical schools at Harvard, Yale, and Wisconsin indicated an alarming 7% possibility of significant patient safety risks when AI-generated responses were left unedited or not reviewed critically by clinicians ³⁰.

On memorizing and following lengthy protocols, regarding the care of patients with major depression, a relational embodied nurse agent, *Elizabeth*, was developed in a 2010 study to orally provide information and counseling in an empathetic fashion to hospitalized patients about their discharge plan, following a standardized hospital discharge protocol designed to reduce readmissions [9]. The protocol involves a human nurse spending half an hour with each patient, ensuring they understand their post-discharge self-care regimen: medications, follow-up appointments, exercise and diet regimens, and pending lab tests. While this protocol is effective, the additional nursing time required makes it impractical to implement, motivating the development of the relational nurse agent to perform this task. The nurse agent produces a personalized “After Hospital Care Plan” and reviews it linearly with the patient, while also engaging in social dialogue and giving comprehension tests. Interaction with the patient was chosen to take place through selecting options on a touch screen, due to the noisy environment of the hospital room, as well as non-English accents. At the end of the interaction, any unresolved patient questions are displayed for a human nurse to follow up with the patient. All participants rated the agent very high on satisfaction and ease of use and, notably, most preferred receiving their discharge information from the agent compared to the hospital doctors or nurses. It can be concluded that agents with relational behavior could provide an effective medium for delivering discharge education as well as comfort to patients with depression.

On following guidelines, a recent example is the project by “Penda Health” company in Kenya in February 2024, where an application was introduced to offer advice on diagnosis and treatment to nurses as well as doctors handling a wide range of primary care cases [48]. The way it works is that clinicians enter the patient’s diagnosis, and then receive some local guidelines and recommendations for treatment as options to either accept or reject. Notably, the people responsible for implementing the solution kept in mind the ‘alert fatigue’ described in the Subsection 2.2.1, and made the tool completely optional. By not forcing its use for every patient visit, ultimately clinicians’

²⁹AI Transforms Health Care | Artificial Intelligence: The Future of Medicine & Health Care Is Here, Stanford Health Care, YouTube (video), Accessed on September 30, 2024. <https://www.youtube.com/watch?v=wD1qn2i3Wb4>

³⁰“Patients Not Told AI Drafted Messages From Their Doctors“, Douglas Mark, April 07 2025, Accessed July 29, 2025.<https://www.mhaonline.com/blog/ai-messages-from-doctors>

rating for the AI consultant was 99.6% positive. On AD and related dementias (ADRD), apart from therapy already mentioned in Section 2.2.2, having proper eating behavior and nutrition can also slow the illness’s progression and cognitive decline [58]. In 2020 the *ADRD-DietAssistant* was developed to educate caregivers – who are typically non-professionals, e.g. spouses, family members or friends of the patient – on healthy diet (i.e. which food items it should include and how to cook meals), guidelines, and constraints for people with ADRD, interacting in free-form, natural language. The assistant provided personalized answers and recommendations based on the caregiver’s and patient’s profile, such as on the caregiver’s time limitations, financial ability, and education level as well as on the patient’s age, stage, food preferences and allergies.

On documenting patients’ progress either in real-time or post-hoc, a variety of products that assist with note-taking and entering structured information directly onto Electronic Health Records (EHRs) by voice alone are already in use by clinicians to streamline administrative processes. All of them promise to decrease the “documentation burden“, as discussed earlier in Section 2.1.1. For instance, *Suki* is an AI-powered voice assistant that allows physicians to dictate clinical notes in free-form speech, suggesting notes after a patient visit, and recommending ICD codes (International Classification of Diseases) for diagnoses. *Suki* achieves high accuracy, is available for 99 medical specialties and 80 languages, and is HIPAA compliant ³¹. Specifically for nurses, the *NurseBrain* smartphone and web app assists nurses with real-time documentation and note generation, AI suggestions for patient management during bedside care (i.e. actions to take and observations to make), general organization of the nursing workflow, and facilitation of handover communication. The app features voice dictation for recording information about patients on the fly, which it then shows in one place (e.g. name, age, sex, room, current situation, background, assessment, recommendation, etc.). *Nursebrain* keeps the nurse organized with to-do checklists and customized reminders of tasks, their priority, due dates and times, and also provides mood check-ins for nurses. Last, it easily hands off the patients’ information to the next nurse in shift by a simple QR scan ³². Another popular mobile app for nurses is *Rover*, which assists with secure barcoded medication administration (BCMA), that is the right dosages, dates, and times for the right patient. The app features the interesting functionality of taking advantage of the in-built scanner and camera of the device, with which the nurse can scan the barcode on the patient’s bracelet for verification and for pulling up patient lists and charts, or scan the barcode of the medication for confir-

³¹Suki (website). Accessed on November 5, 2024. <https://www.suki.ai/>

³²Nursebrain (website). Accessed on November 7, 2024. <https://nursebrain.com/>

mation, or photograph a wound for documentation ³³. *Know My Patient* is another nurse-facing mobile app used for gathering and accessing clinical (i.e. physiological and psychosocial status) and personal (e.g. the patient's behaviors and responses to treatments, schedules at home, preferences for care during hospitalization) information about multiple patients, thus providing instant, and consistent, individualized care ³⁴. More recently, *ChatEHR* was released in 2025 by Stanford Medicine to allow clinicians to interact with their patients' actual medical records, boasting accurate and secure responses ³⁵. Even though still in pilot stage, the chatbot assists 33 Stanford Hospital clinicians in their medical practice by answering questions about a patient's medical history as well as summarizing patient's charts.

Limitations of conversational assistants in healthcare

Using AI technology in healthcare – a domain with very high stakes – brings forth a range of concerns about its design and deployment, mainly falling into two categories: safety and effectiveness, and privacy. Especially the use of AI in mental healthcare, must balance not only these, but also sensitivity to patients' psychology, cultural adaptability, and avoidance of reinforcement of stigmas [76].

The safety and effectiveness category comprises two subcategories: being safe as well as effective. Being safe means that the technology should work to prevent, and should not actively promote, dangerous or unhealthy human behaviors, e.g. self-harm or substance abuse. Being effective means that it should decrease the symptoms of the disorder or the clinical issue being targeted, and increase well-being and quality of life [95]. Human supervision of AI-generated content is necessary to ensure accuracy, and avoid false, biased, or harmful information [65, 71]. A systematic literature review carried out in February 2018 on general-purpose conversational agents that accept any unconstrained natural language input (i.e. spoken and typed) for health-related issues, concluded that the studies screened lacked adequate evidence of patient safety and effectiveness [53]. Regarding the same assistants, as analyzed in Section 2.2.2, other studies have reported inappropriate responses to both critical and non-critical medical questions [47], as well as significant NLU misrecognitions of drug names and even of

³³How Can the Epic Rover App Transform Your Clinical Workflows?'. SuretySystems. February 2, 2023. Accessed on November 7, 2024. <https://www.suretysystems.com/insights/how-can-epic-rover-app-transform-clinical-workflows/>

³⁴Know My Patient, Nightingale Apps (website). Accessed on November 7, 2024. <https://nightingaleapps.com/know-my-patient/>

³⁵Clinicians can 'chat' with medical records through new AI software, ChatEHR", June 05, 2025, Stanford Medicine News Center (blog). Accessed on June 15, 2025. <https://med.stanford.edu/news/all-news/2025/06/chatehr.html>

whole health-related queries due to non-native English accents [78, 11]. Specifically for mental health, as mentioned in Section 2.2.2, these commercial agents have been found to offer no coherent or helpful advice to patients with SUDs [75] and to patients seeking smoking cessation guidance [14]. They have also been found to respond incompletely and disrespectfully to mental health as well as physical health emergencies [70, 66]. As expressed concisely by Miner et al.: “If conversational AI isn’t safe it should not be used, and if it isn’t trusted, it won’t be.”

Regarding privacy, the blurring of the laws, norms, and ethics of confidentiality and data sharing is a great challenge [71]. We know that it is forbidden for human clinicians to disclose identifiable patient information as well as a patient’s sensitive – and potentially stigmatizing – information, e.g. sexual history, trauma, substance abuse, suicidal ideation, etc., otherwise the clinician will face legal consequences. Therefore, similar accountability must be applied to clinical AI agents involved in interactions with patients, and other related third-parties, especially since such agents have an almost infinite capacity to remember and share conversations.

As a last note, it is crucial to responsibly develop and continuously evaluate clinical conversational agents to ensure that their large-scale deployment is guided by the principles of clinical effectiveness, safety, and explainability [52, 94].

2.3 Identification of Gaps or Unresolved Issues

The existing work demonstrates the breadth of conversational assistants deployed across public health, chronic disease management, and mental healthcare focusing mainly on patient-facing solutions, but few studies focus on tools that serve clinical staff directly. Nurse-facing systems are typically limited to discharge education or administrative reminders, with little emphasis on psychiatric settings, where documentation is still handwritten and fragmented [9]. Moreover, most evaluations concentrate on patient outcomes, leaving questions about workflow integration, data source, and sustained adoption among nurses unresolved. Finally, very limited literature addresses Greek-language assistants or solutions designed around Greek regulation constraints, despite the documented burnout and workload pressures experienced by the local psychiatric nurses.

2.4 Theoretical and Conceptual Framework

This thesis is grounded in the intersection of responsible AI, retrieval-augmented generation (RAG), and agentic workflows. RAG offers a principled mechanism for grounding generative models in a curated document database, thereby reducing hallucinations while also preserving the expressive power of LLMs [57]. Agentic orchestration layers provide decision-making heuristics for when and how to invoke tools, enabling the assistant to combine search, summarization, and dialog generation [60]. These capabilities are aligned with responsible AI practices that emphasize privacy protection, transparency, and human oversight in high-stakes environments [76]. Together, they form the conceptual scaffolding for a system that can supply accurate, contextual responses without compromising patient confidentiality.

2.5 Positioning of the Present Research

Building on the identified gaps and theoretical grounding, this research positions itself as a bridge between nurse workflow needs and recent advances in conversational AI. This prototype focuses on psychiatric nurses operating in Greek clinics, addressing language localization, privacy-preserving data curation, and the requirement for fast, hands-free information retrieval. By focusing on faithfulness, relevance, and context precision, the work contributes empirical evidence about the feasibility of deploying agentic, RAG-enabled assistants in frontline mental healthcare. It also complements the existing literature by articulating design considerations, evaluation protocols, and human feedback loops that are specific to nurse-facing applications.

Chapter 3

Methodology

3.1 Methodology Overview

This chapter outlines the design, implementation, and preliminary testing of an AI-powered conversational assistant (CA) prototype, whose goal is to assist Greek psychiatric nurses in their day-to-day tasks. In particular, the assistant's role is to support nurses in performing the concrete task of accessing important pieces of information about a patient's profile (e.g. medications prescribed, etc.) accurately, on the spot, and in spoken natural language. The project followed a ten-month (November 2024 - August 2025) cycle that combined requirements elicitation, data curation, prototype implementation, and quantitative and qualitative evaluation.

The overarching methodological goal was to determine whether a retrieval-augmented conversational workflow could deliver timely, accurate information while aligning with clinical expectations.

For this prototype, hypothetical data were used and protected health information (PHI) was not included.

This work aims at answering the following research questions :

- RQ1) Is it feasible to develop a CA that helps nurses to extract information about a patients' profiles on demand?
- RQ2) If so, how effective is the CA in terms of faithfulness, answer relevancy and answer correctness, context precision and context recall ?
- RQ3) Is it effective? Does the CA improve the daily experience and satisfaction for the job of the nurse?

3.2 Data Sources and Collection Methods

Typically, the design and development of dialog systems begins with the collection and analysis of sample dialogs between real people (e.g. between health providers and patients) [7]. The recordings are then transcribed and analyzed carefully to understand the range of topics, concepts, terms, and general structure typically used in patient–physician communication. However, for the particular project – as well as for most industry-led ones – there were no real template-dialogs to be followed, due to GDPR regulations, and the ethical principle of medical confidentiality, both of which prohibit the dissemination of medical conversations and data. The requirements gathering phase was implemented through interviews with professional psychiatric nurses.

The design process, from project start to completion of the user studies, lasted three months (November 2024 to January 2025). During this period, I interviewed three nurses who worked in psychiatric clinics. Their experience and insights helped me learn about the stakeholders and the basic workflow requirements of the system. The interview material can be found in Appendix C.

Three psychiatric nurses were interviewed for 2 hours each, over Zoom during the month of November, 2024. The interviewed nurses gave information voluntarily about what psychiatric care in Greek psychiatric clinics involves. They explained the different psychiatric facilities, the daily tasks of a nurse during a typical shift, what a nurse should pay attention to when dealing with patients, and how they collaborate with the rest of the medical team, i.e. doctors, psychologists, and social workers. From the discussions, certain “pain points” emerged.

First, the daily job of a psychiatric nurse is critical and very demanding, since nurses are responsible for the following tasks:

1. observe the patients at all times, so they do not harm themselves or others
2. make sure the patients do not flee the clinic
3. get to know the patients well, mainly in order to understand who can pose a threat to themselves or others
4. administer medication and make sure the patients actually swallow it
5. convince the patients to follow their medication treatment, and set goals about their health
6. monitor the patients and change their medication or their doses

7. liaise with other medical specialists (e.g. cardiologist, ophthalmologist, etc.) on behalf of the patients, and accompany them during visits

Notably, there is no specific patient assigned to each nurse; rather, all nurses are responsible for all patients, and any nurse can observe and treat any patient.

Second, there are no comprehensive electronic patient records. A full record for each patient exists as hand-written notes from the relevant medical professionals, called “Patient Record“, and includes information on the patient’s demographic status (sex, age, etc.), social life (if they have a family, a social network, a job, etc.), medical history (their symptoms and diagnosis, hospitalization dates, other medical conditions, etc.), and psychological profile (their desires, dislikes, fears, etc.). Interestingly, there exists only one record for each patient (without extra copies), and it is updated manually in the event of a patient’s readmission. There are, however, a few pieces of information that can also be retrieved electronically via the patient’s Social Security Number (AMKA), that is the patient’s medication regimen (i.e. dosage, schedule, duration of treatment), the doctor who prescribed it, and their admission and discharge dates.

All three nurses expressed their excitement for a phone application that would aid them in their job. As a suggestion, they offered the following: having an assistant that knows pieces of information about each patient, which are personal to each nurse and customized by her/him. This customized patient record would help nurses in accessing important data about a specific patient on the spot, without the need to refer to the unique copy of the official, hand-written Patient Record, or rely on memory.

The resulting requirements and expectations were the following:

1. The assistant will not take the initiative to speak, but will answer questions asked by the nurse fast and reliably, in both voice and text format
2. Voice-enabled for fast note recording and for hearing brief pieces of information on demand (e.g. - “Who prescribed medication A? “, - “Doctor X“)
3. Text-enabled for reading longer notes (e.g. a summary of the medical history) and committing to memory
4. Data that would be helpful are:
 - Medical Record Number (AMKA)
 - Cellphone of patient or of a contact
 - Diagnosis (name and date)

- Admission and discharge dates, if any
- Prescribed treatment plan (doses and schedule)
- Past side-effects of drugs
- Last injection and next injection
- Past blood tests and next blood tests
- Evaluation of patient’s adherence to the medical plan
- Evaluation of patient’s mental and physical health
- Brief social profile
- Important advice or warning from the doctor

Symptoms are well-known to the nurses, so they are not included in the data.

In terms of data sources, actual, proprietary data from a real-world clinic were not made available for this project due to privacy concerns. This is why mock – yet realistic – data were used instead. Fifty patient records were created based on the insights from the three interviews. The first ten patient records were manually written to mirror authentic documentation styles, while the rest were generated by *ChatGPT* following the same structure. The dataset features structured lists and free-form narratives to represent the variability of clinical notes. No personally identifiable or protected health information was included, ensuring compliance with privacy requirements. The dataset can be found and downloaded freely from Kaggle here: <https://www.kaggle.com/datasets/anastasiakarakitsou/mockpatientdata/settings>.

3.3 Instruments, Tools, and Software

Development and experimentation were conducted in Kaggle notebooks equipped with dual Tesla T4 GPUs (15 GB each). I will describe five experiments overall: two early experiments, and then the three latest systems: the Baseline, System A, and System B.

The retrieval component I experimented with FAISS and Chroma for vector search, coupled with the multilingual sentence-transformer `paraphrase-multilingual-MiniLM-L12-v2` to embed both queries and patient note chunks.

Three large language models (LLMs) were explored: the 2 billion parameter “Gemma_trained_for_Greek” model, the 8 billion parameter `ilsp/Llama-Krikri-8B-Instruct`, and Gemini 2.5 Flash.

LangChain provided orchestration utilities, including `ConversationalRetrievalChain` and `ConversationBufferMemory`, to inject retrieved context and maintain dialog state.

Custom functions were introduced to resolve patient names and pronouns when native memory capabilities were insufficient.

3.4 Step by Step Workflow

The methodological workflow progressed through the following stages:

1. **Requirement elicitation:** synthesized interview findings into functional expectations, such as voice and text interaction, and rapid and accurate retrieval.
2. **Data curation:** transformed interview themes into realistic mock patient data, covering diagnoses, treatments, exam schedules, and social histories.
3. **Prototype implementation:** configured the RAG pipeline, and integrated an agent with memory and tool calling for follow-up queries.
4. **Evaluation design:** defined multi-turn scenarios in Greek that reflect daily nurse interactions. The scenarios included happy paths (e.g. medication recall), error handling (unknown patients or missing information), disambiguation cases, free-form summaries, and full-record overviews. Golden responses were added for each scenario to mark faithfulness, relevance, and correctness.
5. **Iterative refinement:** analyzed system outputs, adjusted chunking strategies to improve patient-specific retrieval, and updated prompt instructions or memory handling when hallucinations appeared.

3.5 Justification of Methodological Choices

For my experiments, I chose to use the RAG approach, a hybrid technique of combining an external and up-to-date database for information retrieval, and an LLM for the answer generation part. The RAG technique was selected as it is suitable for knowledge-intensive tasks (tasks that humans cannot be expected to execute without accessing an external knowledge source), and it is also well-known for the control and interpretability it offers during answer generation [57]. More importantly, RAG architecture offers traceability and transparency, as it can surface the answers' sources,

while also mitigating the notorious LLM hallucinations (see Section 2.2.1). Additionally, I chose to apply an agent as a dialog orchestrator, as it enabled tool switching and chat history management, thus supporting more natural multi-turn conversations. In terms of the synthetic data, they allowed controlled experimentation without breaching confidentiality, while interviews ensured that the mock patient records reflected the linguistic and informational patterns of Greek psychiatric practice. Lastly, evaluating through scenario-based testing provided a transparent link between user needs and the system's behavior. Below follows a simple image depicting the RAG functionality: ¹

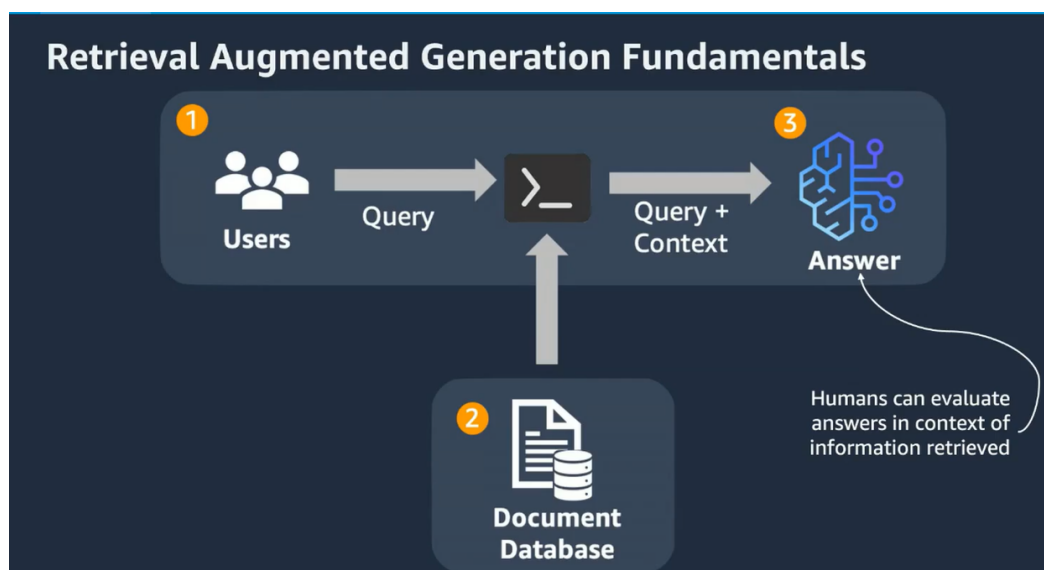


Figure 3.1 Productizing Healthcare ChatBots with John Snow Labs' Medical LLM-as-a-Judge

3.6 Limitations of the Chosen Approach

Relying on synthetic patient records limits external validity until the assistant is used with real patient data under clinical supervision. GPU memory ceilings on Kaggle constrained the length of conversations and the size of models that could be deployed, occasionally causing inference crashes after multiple queries. Custom memory solutions depended on accurate pronoun resolution, and therefore struggled when multiple patients were referenced in quick succession. Finally, the absence of practicing nurses in the initial evaluation means that perceived usefulness and workflow fit must be validated in future in-clinic studies.

¹Healthcare NLP Summit 2025, John Snow Labs. <https://app.swapcard.com/event/healthcare-nlp-summit-2025/planning/UGxhbm5pbmdfMjU3Njg5Ng==>

Chapter 4

Results and Discussion

4.1 Scenario-based Evaluation

The assistant was evaluated across thirteen multi-turn scenarios designed to represent the workflows uncovered during requirements elicitation. Each scenario compared the assistant's responses against golden references to assess faithfulness, relevance, and correctness. The evaluation covered medication recall, reference resolution, abbreviation expansion, error handling, disambiguation, free-form reasoning, and patient summarization tasks, all in Greek language.

4.1.1 Scenarios of multi-turn interactions in Greek

Happy Paths

Scenario 1: Memory

Nurse	Η Μαρία Παπαδοπούλου τι αγωγή παίρνει στην παρούσα φάση;
System - golden	Η Μαρία Παπαδοπούλου παίρνει αλπραζολάμη για τις κρίσεις πανικού με δοσολογία 0,5 μγ και πρόγραμμα 2 φορές την ημέρα.
System - actual	Η Μαρία Παπαδοπούλου λαμβάνει αλπραζολάμη (0.5 μg) 2 φορές την ημέρα για τις κρίσεις πανικού, σύμφωνα με συνταγή του ιατρού Γιάννη Γαστουνιώτη.
Nurse	Και σε ποια ουσία έχει παρενέργειες;
System - golden	Στο λίθιο.
System - actual	Στα έγγραφα της Μαρίας Παπαδοπούλου αναφέρονται παρενέργειες για το λίθιο, το οποίο η ίδια διέκοψε την 1η Οκτωβρίου 2023. Οι παρενέργειες του λιθίου είναι αυξημένη ορμόνη TSH (υποθυρεοειδισμός), καταστολή και υπνηλία και τρέμουλο στα χέρια. Δεν αναφέρονται παρενέργειες για την αλπραζολάμη.

Table 4.1 Scenario 1 (audios 1, 2): Memory - MARIA

Scenario 2: Memory and reference resolution

Nurse	Πότε έγιναν οι προηγούμενες αιματολογικές εξετάσεις του Κώστα Καρά;
System - golden	Στις 24/11/2024.
System - actual	Οι προηγούμενες αιματολογικές εξετάσεις του Κώστα Καρά έγιναν στις 10/02/2024 και 24/11/2024.
Nurse	Και οι επόμενες εξετάσεις του;
System - golden	Στις 15/5/2025.
System - actual	Οι επόμενες αιματολογικές εξετάσεις του Κώστα Καρά είναι προγραμματισμένες για τις 15/5/2025.
Nurse	Τι συστάσεις υπάρχουν;
System - golden	Προώθηση ψυχοθεραπείας για διαχείριση άγχους και καθοδήγηση για ενίσχυση κοινωνικών δεξιοτήτων.
System - actual	Οι τρέχουσες ιατρικές συστάσεις για τον Κώστα Καρά περιλαμβάνουν την προώθηση ψυχοθεραπείας για διαχείριση άγχους και καθοδήγηση για ενίσχυση κοινωνικών δεξιοτήτων.

Table 4.2 Scenario 2 (audios 3, 4, 5): Memory and reference resolution - KOSTAS

Scenario 3: Abbreviations and Acronyms

Nurse	Η Δήμητρα Τσακίρη τι διάγνωση έχει;
System - golden	ΓΑΔ (διάγνωση το 16/04/2004) και κρίσεις πανικού (διάγνωση το 23/01/2008)
System - actual	Η Δήμητρα Τσακίρη έχει διαγνωστεί με ΓΑΔ (Γενικευμένη Αγχώδης Διαταραχή) στις 16/04/2004 και με κρίσεις πανικού στις 23/01/2008.
Nurse	Πες μου τη δοσολογία της αγωγής για την Ιωάννα Λυμπερίου.
System - golden	Δραστική ουσία: κουετιαπίνη Δοσολογία: 300 mg, 2 φ/ημ.
System - actual	Η Ιωάννα Λυμπερίου λαμβάνει κουετιαπίνη σε δοσολογία 300 mg, 2 φορές την ημέρα. Η έναρξη της αγωγής ήταν στις 30/08/2018 και συνταγογραφήθηκε από τον καρδιολόγο Ιωάννη Μακρή.
Nurse	Δώσε μου το τηλέφωνο και το ΑΜΚΑ του Γιώργου Γεωργίου.
System - golden	Τηλ.: 6974513248 (μητέρα, κα. Αθανασία) και ΑΜΚΑ: 67980831881
System - actual	Το ΑΜΚΑ του Γιώργου Γεωργίου είναι 67980831881 και το τηλέφωνο είναι 6974513248 (της μητέρας του, κα. Αθανασίας).
Nurse	Γιώργος Μακρής. Τι διαγνώσεις έχει;
System - golden	- Κρίσεις πανικού (διάγνωση το 12/11/2002) - Διπολική διατ. (διάγνωση το 15/01/2021) - Ψύχωση (διάγνωση το 25/04/2012)
System - actual	Ο Γιώργος Μακρής έχει διαγνωστεί με τα εξής: Κρίσεις πανικού (12/11/2002), Διπολική διαταραχή (15/01/2021) και Ψύχωση (25/04/2012).

Table 4.3 Scenario 3 (audios 6, 7, 8, 9): Abbreviations and Acronyms - PATIENT 01 and PATIENT 03 and PATIENT 04 and PATIENT 05

Error Scenarios**Scenario 4: Non-existent patient**

Nurse	Σε ποια ουσία έχει παρενέργειες η Δήμητρα Καραχίτσου;
System - golden	Δεν ξέρω.
System - actual	Δεν βρέθηκαν πληροφορίες σχετικά με το σε ποια ουσία έχει παρενέργειες η Δήμητρα Καραχίτσου.

Table 4.4 Scenario 4 (audio 10): Non-existent patient

Scenario 5: Non-existent patient who shares the same last name with a known patient

Nurse	Πες μου τις συστάσεις για την ασθενή Θάλεια Παπαδοπούλου.
System - golden	Δεν ξέρω.
System - actual	Δεν βρέθηκαν πληροφορίες για την ασθενή Θάλεια Παπαδοπούλου.

Table 4.5 Scenario 5 (audio 11): Non-existent patient who shares the same last name with a known patient

Scenario 6: Known patient with non-existent info

Nurse	Πότε είναι η επόμενη ενέσιμη αγωγή του Κώστα Καρά;
System - golden	Δεν έχει ενέσιμη αγωγή.
System - actual	Ο Κώστας Καράς δεν έχει ενέσιμη αγωγή. Έχει όμως προγραμματισμένες αιματολογικές εξετάσεις στις 15/5/2025.

Table 4.6 Scenario 6 (audio 12): Known patient with non-existent info - KOSTAS

Scenario 7: Doctor's name

Nurse	Πες μου το AMKA της Κωνσταντίνας Λυμπερίου.
System - golden	Δεν ξέρω.
System - actual	Δεν βρέθηκαν δεδομένα για την Κωνσταντίνα Λυμπερίου. Δεν μπορώ να σου δώσω το AMKA της.

Table 4.7 Scenario 7 (audio 13): Doctor's name - GEORGIA M

Disambiguation Scenarios

Scenario 8: Disambiguation of identical patient name

Nurse	Θύμισέ μου το κοινωνικό ιστορικό της Ελένης Κανελλοπούλου.
System	Έχω βρει πληροφορίες για δύο ασθενείς με το όνομα Ελένη Κανελλοπούλου. Η μία είναι 54 ετών (ΑΜΚΑ: 63681990357) και η άλλη είναι 41 ετών (ΑΜΚΑ: 65357602736). Για ποια από τις δύο Ελένες Κανελλοπούλου θέλετε το κοινωνικό ιστορικό;
Nurse	Πες μου για τη μεγαλύτερη.
System - golden	PATIENT 15.txt
System - actual	OK. Το κοινωνικό ιστορικό της Ελένης Κανελλοπούλου (ΑΜΚΑ: 63681990357) είναι: * Μένει μόνη σε ενοικιαζόμενο διαμέρισμα * Έχει ελάχιστη οικογενειακή υποστήριξη * Δεν εργάζεται επί του παρόντος * Πρώην υπάλληλος γραφείου, αποχώρησε λόγω επιδείνωσης της ψυχικής υγείας * Διατηρεί περιορισμένη επαφή με μία εξαδέλφη. Όσον αφορά τις συνήθειες και την καθημερινότητά της: * Καπνίζει περίπου 10 τσιγάρα την ημέρα * Δεν καταναλώνει αλκοόλ * Παρουσιάζει διαταραχές ύπνου (δυσκολία έναρξης και διατήρησης ύπνου, συχνές διακοπές για ούρηση) * Βλέπει τηλεόραση και ράβει για να ηρεμεί. Οι συστάσεις για την Ελένη Κανελλοπούλου είναι: * Συστηματική ψυχοθεραπεία και παρακολούθηση * Ενίσχυση κοινωνικών δεξιοτήτων μέσω ομάδων υποστήριξης * Εκπαίδευση στη διαχείριση στρες και τεχνικές αναπνοής * Συμμετοχή σε πρόγραμμα επανένταξης στην εργασία (ΔΥΠΑ) * Προγραμματισμός αξιολόγησης από Κοινωνική Υπηρεσία για πιθανή παροχή βοήθειας κατ' οίκον.

Table 4.8 Scenario 8 (audio 14, 15): Disambiguation of identical patient name - PATIENT 13 and PATIENT 15

Scenario 9: Disambiguation of identical patient name and double intent

Nurse	Πότε νοσηλεύτηκε τελευταία φορά η Δήμητρα Μακρή και τι αγωγή παίρνει;
System	Υπάρχουν δύο ασθενείς με το όνομα Δήμητρα Μακρή. Για να σας δώσω τις σωστές πληροφορίες, θα μπορούσατε να επιβεβαιώσετε τον AMKA ή την ηλικία της ασθενούς για την οποία ενδιαφέρεστε;
Nurse	Δε θυμάμαι το AMKA της. Πες μου για την πιο μικρή.
System - golden	Στις 20/03/2013: Επιθετικό-μανιακό πεισόδιο με παραμονή 2 ημέρες, και στις 27/09/2017: Επεισόδιο σχιζοφρένειας με παραμονή 1 ημέρες. Παίρνει ρισπεριδόνη, 3 μγ, 2 φορές την ημέρα, και ζιπραζιδόνη, 40 μγ, 2 φορές την ημέρα μαζί με φαγητό.
System - actual	Η Δήμητρα Μακρή, 32 ετών (AMKA 41661927592), νοσηλεύτηκε τελευταία φορά στις 27/09/2017 και λαμβάνει ρισπεριδόνη (3 μγ, 2 φ. την ημέρα) και ζιπραζιδόνη (40 μγ, 2 φ. την ημέρα μαζί με φαγητό).

Table 4.9 Scenario 9 (audio 16, 17): Disambiguation of identical patient name and double intent - PATIENT 10 and PATIENT 16

Scenario 10: Free-form text

Nurse	Θύμισέ μου αν ακολουθεί τη θεραπεία της η Βασιλική Αντωνίου.
System - golden	Λαμβάνει τα φάρμακά της με σχετική συνέπεια, αν και παρατηρούνται περιστασιακές παραλείψεις, κυρίως κατά τις περιόδους ύφεσης. Παρουσιάζει μειωμένη αυτογνωσία σχετικά με την ανάγκη για συνεχή θεραπεία, γεγονός που αποδίδεται εν μέρει σε μερική ανοσογνωσία.
System - actual	Η Βασιλική Αντωνίου λαμβάνει τα φάρμακά της με σχετική συνέπεια, αν και παρατηρούνται περιστασιακές παραλείψεις, κυρίως κατά τις περιόδους ύφεσης. Παρουσιάζει μειωμένη αυτογνωσία σχετικά με την ανάγκη για συνεχή θεραπεία, γεγονός που αποδίδεται εν μέρει σε μερική ανοσογνωσία.
Nurse	Έχει κάποιες συστάσεις στο φάκελό της;
System - golden	Οι θεραπευτικές κατευθύνσεις περιλαμβάνουν τη συνέχιση της φαρμακευτικής αγωγής υπό τακτική παρακολούθηση, την επανένταξή της σε ομάδα υποστηρικτικής ψυχοθεραπείας σε εβδομαδιαία βάση, την καθιέρωση ενός σταθερού προγράμματος δραστηριοτήτων που να περιλαμβάνει βόλτες και ελαφριές οικιακές εργασίες, καθώς και τη συμμετοχή σε πρόγραμμα κοινωνικοποίησης στο κέντρο ημέρας. Επιπλέον, προτείνεται παραπομπή σε ψυχολόγο για γνωσιακή-συμπεριφορική θεραπεία, παροχή συμβουλευτικής υποστήριξης προς την οικογένειά της, ενίσχυση της αίσθησης αυτοαποτελεσματικότητας και ενδυνάμωσης, και μηνιαία παρακολούθηση από κοινωνικό λειτουργό.
System - actual	Ναι, υπάρχουν κάποιες συστάσεις στο φάκελο της Βασιλικής Αντωνίου. Προτείνεται παραπομπή σε ψυχολόγο για γνωσιακή-συμπεριφορική θεραπεία, παροχή συμβουλευτικής υποστήριξης προς την οικογένειά της, ενίσχυση της αίσθησης αυτοαποτελεσματικότητας και ενδυνάμωσης, και μηνιαία παρακολούθηση από κοινωνικό λειτουργό. Επιπλέον, οι θεραπευτικές κατευθύνσεις περιλαμβάνουν τη συνέχιση της φαρμακευτικής αγωγής υπό τακτική παρακολούθηση, την επανένταξή της σε ομάδα υποστηρικτικής ψυχοθεραπείας σε εβδομαδιαία βάση, την καθιέρωση ενός σταθερού προγράμματος δραστηριοτήτων που να περιλαμβάνει βόλτες και ελαφριές οικιακές εργασίες, καθώς και τη συμμετοχή σε πρόγραμμα κοινωνικοποίησης στο κέντρο ημέρας.

Table 4.10 Scenario 10 (audio 18, 19): Free-form text - PATIENT 40

Scenario 11: Free-form text

Nurse	Ποια είναι η τρέχουσα ψυχολογική κατάσταση του Νίκου Τρανού;
System - golden	Κατά την τελευταία ψυχολογική εκτίμηση, αναφέρθηκαν εναλλαγές διάθεσης με περιόδους απογοήτευσης και μειωμένο ενδιαφέρον για δραστηριότητες, χωρίς ωστόσο ενδείξεις ενεργού αυτοκτονικού ιδεασμού.
System - actual	Η τρέχουσα ψυχολογική κατάσταση του Νίκου Τρανού (AMKA: 71462734135), σύμφωνα με την ψυχολογική αξιολόγηση, παρουσιάζει εναλλαγές στη διάθεση και περιστασιακή απογοήτευση. Η ενέργειά του είναι μέτρια, με μειωμένη όρεξη για δραστηριότητες, ενώ η κοινωνική του συμπεριφορά είναι περιορισμένη. Δεν έχουν αναφερθεί ενδείξεις αυτοκτονικού ιδεασμού.
Nurse	Ακολουθεί την αγωγή του;
System - golden	Η συμμόρφωση του ασθενούς στη θεραπεία είναι μέτρια. Ο ίδιος αναφέρει ότι λαμβάνει τα φάρμακά του με συνέπεια, ωστόσο παραδέχεται πως τα παραλείπει περιστασιακά, ιδίως όταν αισθάνεται πως «είναι καλά» — φαινόμενο συχνό σε χρόνιους ψυχιατρικούς ασθενείς.
System - actual	Ο Νίκος Τρανός (AMKA: 71462734135) παίρνει τα φάρμακά του με σχετική συνέπεια, αλλά αναφέρει περιστασιακές παραλείψεις όταν νιώθει καλά.

Table 4.11 Scenario 11 (audio 20, 21): Free-form text - PATIENT 38

Nurse	Δείχνει επιθυμία για απομόνωση; Πώς είναι κοινωνικά;
System - golden	Η κοινωνική του αλληλεπίδραση παραμένει περιορισμένη, ενώ φαίνεται να βιώνει συναισθήματα απομόνωσης, κάτι που επιτείνεται από την ελλιπή οικογενειακή στήριξη. Διαμένει μόνος του σε διαμέρισμα που του παραχωρήθηκε από την κοινότητα, ενώ η επικοινωνία με την οικογένειά του είναι περιορισμένη — διατηρεί επαφή μόνο με έναν αδερφό του, τον οποίο βλέπει περιστασιακά. Δεν έχει δημιουργήσει σταθερό οικογενειακό κύκλο ή στενές φιλίες, αν και τον τελευταίο χρόνο συμμετέχει σποραδικά σε δραστηριότητες Κέντρου Ημέρας για άτομα με ψυχικές διαταραχές, όπου φαίνεται να δείχνει διστακτικό αλλά θετικό ενδιαφέρον για κοινωνική αλληλεπίδραση. Ανέφερε επίσης ότι κάποιες Κυριακές πηγαίνει σε καφενείο της γειτονιάς, κυρίως για να «παρακολουθεί τους άλλους» και όχι για να συμμετάσχει ενεργά στη συζήτηση.
System - actual	Ο Νίκος Τρανός έχει περιορισμένη κοινωνική αλληλεπίδραση, γεγονός που υποδηλώνει επιθυμία για απομόνωση.
Nurse	Απαιτείται ενημέρωση ιατρού ή παρέμβαση ψυχολόγου ή ψυχιάτρου;
System - golden	Ο συνολικός ψυχιατρικός του φάκελος υποδεικνύει την ανάγκη για πιο στενή ψυχιατρική παρακολούθηση, ενίσχυση της ψυχοκοινωνικής υποστήριξης, και πιθανή ένταξή του σε προγράμματα ψυχοεκπαίδευσης και ενδυνάμωσης, ώστε να βελτιωθεί η ποιότητα ζωής και να περιοριστούν οι υποτροπές.
System - actual	Με βάση τα έγγραφα του ασθενούς Νίκου Τρανού, λαμβάνοντας υπόψη τις διαγνώσεις του (ψύχωση, μανιοκατάθλιψη), το ιστορικό επεισοδίων, την τρέχουσα φαρμακευτική αγωγή (ολανζαπίνη), τις παρενέργειες που είχε από προηγούμενη φαρμακευτική αγωγή (φλουοξετίνη), την περιορισμένη κοινωνική συμπεριφορά και τις εναλλαγές στη διάθεση, είναι απαραίτητη η ενημέρωση του θεράποντος ιατρού του, Ιωάννη Μακρή, καθώς και η ψυχολογική υποστήριξη από ψυχολόγο ή ψυχίατρο.

Table 4.12 Scenario 11 (audio 22, 23): Free-form text - PATIENT 38

Summarization Scenarios

Scenario 12: Summary of a patient's record from structured text

Nurse	Δώσε μου τη γενική εικόνα για τον Θανάση Πράσινο.
System - golden	PATIENT 29.txt
System - actual	Η γενική εικόνα για τον Θανάση Πράσινο είναι η εξής: Ο Θανάσης Πράσιнос, 24 ετών, με AMKA 54565609013, έχει διαγνωστεί με μανιοκατάθλιψη (από 30/04/2014) και σχιζοφρένεια (από 11/01/2005). Είχε ένα επεισόδιο στις 25/03/2024 που χρειάστηκε νοσηλεία για 1 ημέρα. Αυτή τη στιγμή λαμβάνει κουετιαπίνη 300μγ, δύο φορές την ημέρα, για την αντιμετώπιση των συμπτωμάτων του, με συνταγογράφο τον ψυχολόγο Δημήτριο Κατσαρό (έναρξη 25/03/2024). Στο παρελθόν είχε παρενέργειες (ναυτία και μείωση λίμπιντο) από την ολανζαπίνη, η οποία διακόπηκε στις 23/05/2024. Η διάθεσή του παρουσιάζει απότομες εναλλαγές και έχει μέτρια ενέργεια, με μειωμένη όρεξη για δραστηριότητες. Έχει παρατηρηθεί περίεργη/βίαιη συμπεριφορά στα social media (TikTok, Facebook) σε περιόδους έξαρσης της ασθένειας. Δεν έχουν αναφερθεί ενδείξεις αυτοκτονικού ιδεασμού. Η οικογένειά του αποφεύγει να συζητάει για το πρόβλημά του. Εργάζεται ως γυμναστής στα Alter Life. Συνιστάται η συνέχιση της συστηματικής ψυχοθεραπείας και παρακολούθησης. Παίρνει τα φάρμακά του με σχετική συνέπεια, αλλά αναφέρει περιστασιακές παραλείψεις όταν νιώθει καλά. Έχει κάνει αιματολογικές εξετάσεις στις 31/05/2024 και 23/11/2024, και οι επόμενες είναι προγραμματισμένες για τις 06/10/2025.

Table 4.13 Scenario 12 (audio 24): Summary of a patient's record from structured text - PATIENT 29

Scenario 13: Summary of a patient's record from free-form text

Nurse	Πες μου κάποια σημαντικά πράγματα που πρέπει να ξέρω για τη Χριστίνα Παγούτση.
System - golden	PATIENT 41.txt
System - actual	Εδώ είναι μερικά σημαντικά πράγματα που πρέπει να γνωρίζετε για την Χριστίνα Παγούτση: **Ιστορικό:** * Διαγνώστηκε με ιδεοψυχαναγκαστική διαταραχή (από 11 Ιουνίου 2010), μείζονα καταθλιπτική διαταραχή (από 25 Ιανουαρίου 2006), παραληρητική ψύχωση (από 1 Δεκεμβρίου 2005) και αϋπνία σχετιζόμενη με αγχώδη διαταραχή (από το 2021). * Νοσηλεύτηκε στις 4 Οκτωβρίου 2008 για τέσσερις ημέρες λόγω έντονων εμμονικών σκέψεων με αυτοτραυματικές τάσεις. * Στις 12 Σεπτεμβρίου 2020 υπήρξε παρέμβαση κινητής μονάδας, εξαιτίας επιδείνωσης των συμπτωμάτων. * Είχε ένα ψυχωσικό επεισόδιο στις 21 Απριλίου 2013, που οδήγησε σε νοσηλεία δύο ημερών. ***Φαρμακευτική αγωγή:** * Λαμβάνει σερτραλίνη 100 μγ ημερησίως, ρισπεριδόνη 2 μγ ημερησίως, τραζοδόνη 50 μγ πριν τον ύπνο και λοραζεπάμη 1 μγ κατά περίπτωση. * Στο παρελθόν λάμβανε αλπραζολάμη, η οποία διακόπηκε στις 6 Μαρτίου 2023. * Έχει μέτρια συμμόρφωση στη φαρμακευτική αγωγή και ιστορικό διακοπής της αγωγής χωρίς συνεννόηση με το θεραπευτικό προσωπικό. ***Εργαστηριακές εξετάσεις:** * Πρόσφατες αιματολογικές εξετάσεις (17 Αυγούστου 2024) έδειξαν μέτρια αυξημένα τριγλυκερίδια. * Στις 8 Ιουλίου 2023 είχε εμφανίσει ελαφρά αναιμία και χαμηλή φερριτίνη. Η επόμενη εξέταση έχει προγραμματιστεί για τις 29 Δεκεμβρίου 2025. ***Κοινωνικό προφίλ:** * Διαμένει μόνη της σε ενοικιαζόμενο διαμέρισμα με περιορισμένους οικονομικούς πόρους. ***Δημογραφικά στοιχεία:** * Είναι 59 ετών. AMKA 93599718720.

Table 4.14 Scenario 13 (audio 25): Summary of a patient's record from free-form text - PATIENT 41

4.2 Presentation of Findings (organized by Experiments)

4.2.1 First experiment with LLM: Gemma_trained_for_Greek

I first started experimenting with “Gemma_trained_for_Greek” of 2 B parameters, which I downloaded from Kaggle ¹, the sentence transformer embedding model "sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2", as it is multilingual and includes Greek, and the FAISS vector database.

To add a memory component during inference, I used two custom functions: one to extract the current patient’s name from the query, and the second to resolve personal pronouns such as του, της, αυτήν, etc. and replace the pronoun with the latest patient’s name. Memory updates were fully manual and did not hold any conversational steps (i.e. past question-answer pairs). Only later did I learn that *LangChain* had a built-in memory module, so I used it in my next experiment (see Section 4.2.2).

The constraints of this approach were not unimportant: first, the query had to have a pronoun in order for the custom memory to work, and second, the system crashed after 3 queries due to the available Kaggle GPU (VRAM) memory threshold of 30 GB. I experimented with one (1), and then with two (2) patients. As it is obvious from the examples below, the results with one patient were mostly fine, but with two patients they were disappointing, as the answers were either incorrect or mixed-up, or contained English words.

One patient (MARIA.txt)

Experimental Query 1: Η Μαρία Παπαδοπούλου τι αγωγή παίρνει στην παρούσα φάση;
System - golden: Η Μαρία Παπαδοπούλου παίρνει αλπραζολάμη για τις κρίσεις πανικού με δοσολογία 0,5 μγ και πρόγραμμα 2 φορές την ημέρα.

System - actual: Η Μαρία Παπαδοπούλου παίρνει αλπραζολάμη με δοσολογία (0.5 μg) και πρόγραμμα 2 φορές την ημέρα.

Experimental Query 2: Και σε ποια δραστική ουσία έχει παρενέργειες αυτή;

Resolved Query 2: Και σε ποια δραστική ουσία έχει παρενέργειες Μαρία Παπαδοπούλου;
System - golden: Στο λίθιο.

System - actual: Η Μαρία Παπαδοπούλου έχει παρενέργειες από το λίθιο.

¹Gemma_trained_for_Greek, Accessed May 2025. https://www.kaggle.com/models/soloupis/gemma_trained_for_greek

Experimental Query 3: Πότε είναι οι επόμενες αιματολογικές της εξετάσεις;

Resolved Query 3: Πότε είναι οι επόμενες αιματολογικές Μαρία Παπαδοπούλου εξετάσεις;

System - golden: Στις 15 Μαΐου 2025.

System - actual: Οι επόμενες αιματολογικές εξετάσεις της Μαρίας Παπαδοπούλου είναι στις 15 Μαΐου 2025.

Experimental Query 4: Πες μου την αγωγή του Κώστα Καρά.

Resolved Query 4: Πες μου Κώστα Καρά αγωγή Κώστα Καρά Κώστα Καρά.

System - golden: Παίρνει σερτραλίνη των 50 mg, 1 φορά το πρωί και ραμιπρίλη των 5 mg 1 φορά το πρωί.

System - actual: Δεν έχω αρκετές πληροφορίες για να απαντήσω στην ερώτηση. Θα μπορούσατε να παρέχετε περισσότερα πλαίσια ή πληροφορίες για να μπορέσω να απαντήσω στην ερώτηση;

Two patients (MARIA.txt, KOSTAS.txt)

Experimental Query 1: Η Μαρία Παπαδοπούλου τι αγωγή παίρνει στην παρούσα φάση;

System - golden: Η Μαρία Παπαδοπούλου παίρνει αλπραζολάμη για τις κρίσεις πανικού με δοσολογία 0,5 μγ και πρόγραμμα 2 φορές την ημέρα.

System - actual: Δεδομένων των απαντημένων πληροφοριών, δεν είναι γνωστό ποια φάρμακα παίρνει η Μαρία Παπαδοπούλου. Θα μπορούσατε να παρέχετε περισσότερα στοιχεία ή πληροφορίες για να μπορέσετε να απαντήσετε στην ερώτηση;

Experimental Query 2: Και σε ποια δραστική ουσία έχει παρενέργειες αυτή;

Resolved Query 2: Και σε ποια δραστική ουσία έχει παρενέργειες Μαρία Παπαδοπούλου;

System - golden: Στο λίθιο.

System - actual: Η Μαρία Παπαδοπούλου έχει παρενέργειες από τη δραστική ουσία ολανζαπίνη.

Experimental Query 3: Πότε είναι οι επόμενες αιματολογικές της εξετάσεις;

Resolved Query 3: Πότε είναι οι επόμενες αιματολογικές Μαρία Παπαδοπούλου εξετάσεις;

System - golden: Στις 15 Μαΐου 2025.

System - actual: Οι επόμενες αιματολογικές εξετάσεις της Μαρίας Παπαδοπούλου είναι

scheduled για την 15/5/2025

Experimental Query 4: Πες μου την αγωγή του Κώστα Καρά.

Resolved Query 4: Πες μου Κώστα Καρά αγωγή Κώστα Καρά Κώστα Καρά.

System - golden: Παίρνει σετράλίνη των 50 mg, 1 φορά το πρωί και ραμπεπρίλη των 5 mg 1 φορά το πρωί.

System - actual: Δεν υπάρχουν αρκετές πληροφορίες για να καθοριστεί η αγωγή για τον Κώστα Καρά. Θα μπορούσατε να παρέχετε περισσότερα δεδομένα ή συγκεκριμένες παραγγελίες για να μπορέσετε να καθορίσετε την αγωγή του Κώστα Καρά;

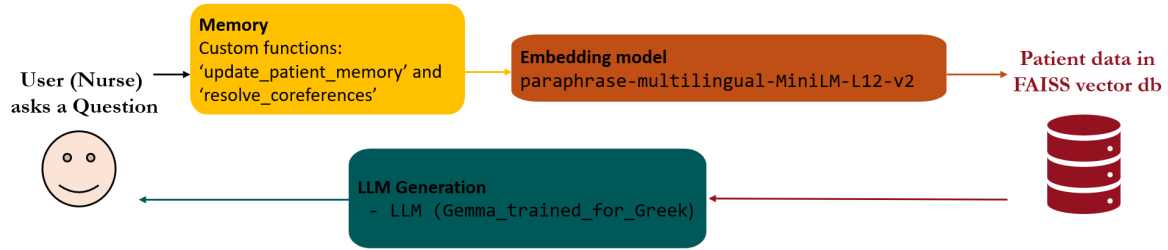


Figure 4.1 **Memory:** Custom memory. Stores current patient’s name. Replaces pronouns like “της”, “αυτό”, “η ασθενής” with the stored patient’s name. **Encoder:** The embedding model paraphrase-multilingual-MiniLM-L12-v2 encodes the question. **Retriever:** Finds relevant context passages from patient documents using the embedding model. **LLM:** At the chain’s core lies the Gemma LLM with its default prompt.

4.2.2 Second experiment with LLM: KriKri

Then, I experimented with the quantized version of the LLM "ilsp/Llama-Krikri-8B-Instruct", because it was one of the newest models at that time (in May 2025) promising advanced language capabilities in both Greek and English - my two languages of interest - and an expanded context window of 128k tokens². For this instruction-focused LLM I added a custom prompt to minimize hallucinations, which can be found in Appendix A. As an improvement from the previous custom memory, I added LangChain’s “ConversationBufferWindowMemory”. I also used LangChain’s “ConversationalRetrievalChain” to connect the memory to the LLM’s prompt as “chat history” together with the query and the context from the FAISS database. The system is outlined in Figure 4.2.

Despite the memory boost, due to the available Kaggle GPU (VRAM) memory threshold of 30 GB, the system was still crashing after the third query, as the LLM

²Llama-Krikri-8B, 9 Feb 2025, <https://www.ilsp.gr/en/news/krikri/>

model occupied almost half of the available memory (16 GB). I started experimenting with 2 patients reaching up to 8 patients, past which the results deteriorated, and the answers were mixed-up or plainly wrong, e.g.:

Eight patients

Experimental Query 1: Η Μαρία Παπαδοπούλου τι αγωγή παίρνει στην παρούσα φάση;

System - golden: Η Μαρία Παπαδοπούλου παίρνει αλπραζολάμη για τις κρίσεις πανικού με δοσολογία 0,5 μγ και πρόγραμμα 2 φορές την ημέρα.

System - actual: Η πληροφορία που παρέχεται δεν περιέχει δεδομένα για ασθενή με όνομα Μαρία Παπαδοπούλου. Οι περιπτώσεις αφορούν τις ασθενείς Γεωργία Αναστασιάδη, Γεωργία Μαντά και Κώστα Καρά.

Resolved Query 2: Και σε ποια δραστική ουσία έχει παρενέργειες Μαρία Παπαδοπούλου;

System - golden: Στο λίθιο.

System - actual: Η δραστική ουσία που προκαλεί παρενέργειες στην παρούσα φάση για τη Γεωργία Αναστασιάδη είναι η βουπροπιόνη, με αναφερόμενες παρενέργειες την ταχυκαρδία και την αϋπνία.

Experimental Query 3: Πότε είναι οι επόμενες αιματολογικές της εξετάσεις;

System - golden: Στις 15 Μαΐου 2025.

System - actual: Δεν υπάρχουν πληροφορίες για ασθενή με όνομα Μαρία Παπαδοπούλου στο παρεχόμενο ιατρικό ιστορικό.

Experimental Query 4: Πες μου την αγωγή του Κώστα Καρά.

System - golden: Παίρνει σερτραλίνη των 50 mg, 1 φορά το πρωί και ραμιπρίλη των 5 mg 1 φορά το πρωί.

System - actual: Δεν υπάρχουν πληροφορίες για ασθενή με όνομα Κώστας Καρά στο παρεχόμενο ιατρικό ιστορικό.

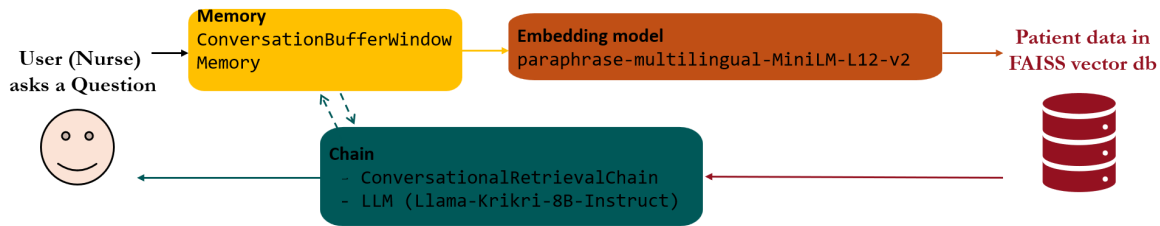


Figure 4.2 **Memory:** Retrieves the last 3 exchanges and passes these as chat history to the prompt. **Encoder:** The embedding model paraphrase-multilingual-MiniLM-L12-v2 encodes the question. **Retriever:** Finds relevant context passages from patient documents using the embedding model. **Chain:** At the chain’s core lies the Llama-Krikri-8B-Instruct LLM. In its prompt it receives the retrieved context, the chat history, the nurse’s original question, and custom instructions on how to answer it.

4.2.3 Third experiment: The Baseline System

After experimenting with the two LLMs described above, I built upon both of them onto a less memory-intensive solution using Gemini Flash 2.0 via an API call, as it allowed for more sequential queries. This evolved into my baseline system.

Chunking

In the baseline system, each patient’s text file was first loaded and then annotated with the patient’s name as helpful metadata to enable filtered, and more accurate, retrieved results. The documents were then segmented into smaller, overlapping chunks using the “Recursive Character Text Splitter“. This splitter was chosen as it maintains semantic coherence while respecting character boundaries and hierarchical structures (i.e. first sentences, then paragraphs, and then words) within a document ³. Each chunk was limited to 800 characters with an overlap of 100 characters, ensuring continuity of context between adjacent segments while avoiding excessive fragmentation.

Embedding Model

The chunks were then converted into numerical representations. The embedding model that was used was the multilingual “sentence-transformers/paraphrase-multilingual-

³How to Use RecursiveCharacterTextSplitter in LangChain". Gary Svenson. Medium (blog), Last accessed July 6, 2025. <https://medium.com/towards-agi/how-to-use-recursivecharactertextsplitter-in-langchain-23bcb0448fca>

MiniLM-L12-v2“ because Greek language was included in its training data and because of its suitability in semantic search ⁴.

Vectorstore

The resulting embeddings of the chunks, enriched with attached metadata including the patient’s name, were later stored in a *FAISS* vector index for filtered similarity-based retrieval. For retrieval, the system was configured with $k = 3$, meaning that the three most semantically similar chunks were returned for each query. This relatively small value of k was chosen to reduce noise and ensure that only the most relevant context was passed to the LLM, thereby improving answer precision.

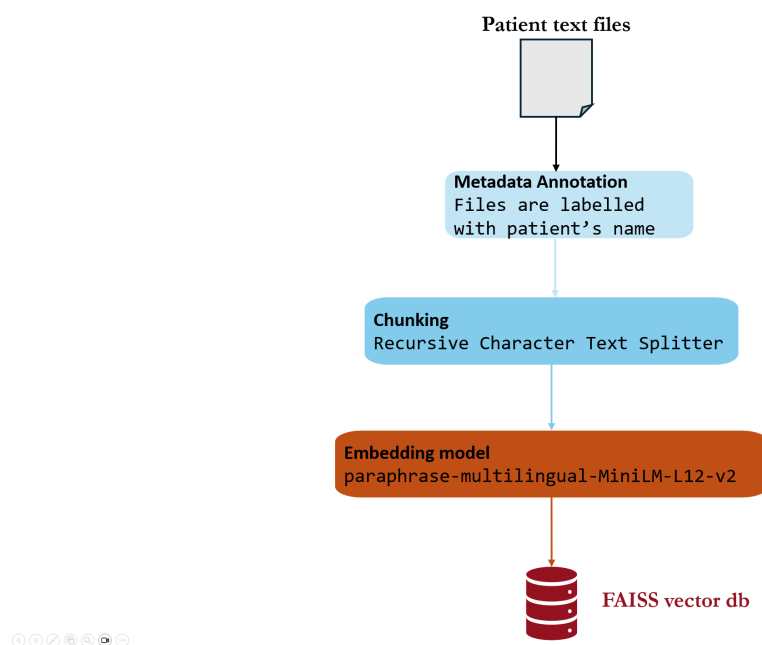


Figure 4.3 **Metadata Annotation:** Each patient file is labelled with the patient’s name. **Chunking:** “Recursive Character Text Splitter“ splits the text into chunks of 800 characters with 100-character overlap. **Embedding model:** The embedding model “paraphrase-multilingual-MiniLM-L12-v2“ encodes each chunk and stores it into the FAISS vector database, together with the patient’s name annotation as its metadata.

⁴HuggingFace, ModelCard for sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2, Accessed July 5, 2025. <https://huggingface.co/sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2>

LLM

The baseline system was based on Google’s LLM “gemini-2.0-flash” model, which was integrated through the *LangChain* framework. This model was chosen because of its very good performance and instruction-following capability, support of the Greek language, and generally better results than other Gemini models, even “gemini-2.5-flash”⁵. The model was configured with a low temperature (set to 0.2) for consistent and deterministic responses, which is especially important in the clinical domain. A customized system prompt written in Greek defined the model’s behavior, which can be found in Appendix A. In addition, a structured RAG prompt template was added to inject the necessary context into each query. This template incorporated the conversation history, the retrieved document chunks, and the user’s query, ensuring that the model grounded its responses in the available patient data rather than hallucinating unsupported information.

Memory

To support conversational continuity, the system employed “ConversationBufferWindowMemory” from *LangChain*. This component maintained a history of the most recent dialogue turns, storing both user inputs and model outputs. The window size was set to 3, as the expected interactions by the nurses are supposed to be brief and straightforward. In this way, the memory preserved the three latest conversations without overwhelming the LLM model with the entire dialogue history. This enabled the system to correctly interpret follow-up questions, resolve references to previous queries, and sustain natural multi-turn interactions.

Chain

The language model, the retriever, the custom prompt, and the memory component were integrated into a “ConversationalRetrievalChain”. At each query, the chain retrieved the most relevant document chunks from the vector store, combined them with the current conversation history stored in memory, and passed them into the custom RAG prompt template. The Gemini 2.0 Flash model, then, generated a response conditioned on both the retrieved knowledge and the conversational flow. The chain was further configured to return the source documents used in each response, ensuring transparency and enabling later evaluation of grounding accuracy.

⁵Google Cloud (blog). Gemini 2.0 Flash. Accessed on July 5, 2025. <https://cloud.google.com/vertex-ai/generative-ai/docs/models/gemini/2-0-flash>

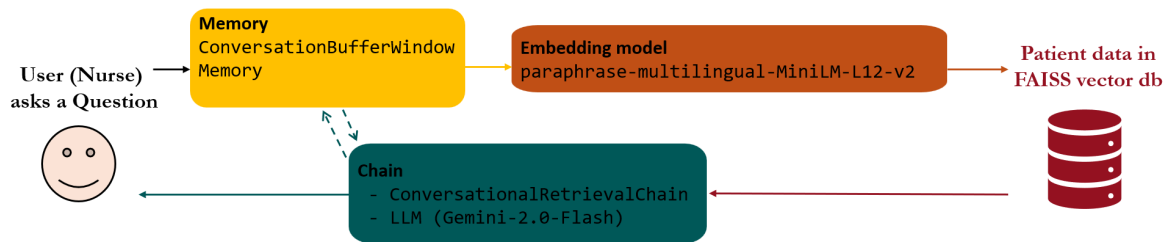


Figure 4.4 **Memory:** Retrieves the last 3 exchanges and passes these as chat history to the prompt. **Encoder:** The embedding model paraphrase-multilingual-MiniLM-L12-v2 encodes the question. **Retriever:** Finds relevant context passages from patient documents using the embedding model. **Chain:** At the chain's core lies the Gemini Flash 2.0 LLM. In its prompt it receives the retrieved context, the chat history, the nurse's original question, and custom instructions on how to answer it.

Below follows a table with the features of my first three experiments:

Feature	Gemini	Krikri	Gemma
LLM	Gemini-2.0-Flash (Google)	Llama-Krikri-8B- Instruct (Hugging- Face)	Gemma (Local)
Framework	Full LangChain RAG pipeline	Full LangChain RAG pipeline	Manual FAISS + SentenceTrans- former pipeline
Memory	ConversationBuffer- WindowMemory (window = 3)	ConversationBuffer- WindowMemory (window = 3)	Custom dict-based patient memory (coreference only)
Embedding Imple- menta- tion	HuggingFace- Embeddings (LangChain)	HuggingFace- Embeddings (LangChain)	SentenceTransform- er.encode() (man- ual)
Embedding Model Used	paraphrase- multilingual- MiniLM-L12-v2	paraphrase- multilingual- MiniLM-L12-v2	paraphrase- multilingual- MiniLM-L12-v2
Vector Store	FAISS (LangChain- managed)	FAISS (LangChain- managed)	FAISS (manually constructed)
Distinctive Feature	Long context win- dow (1M tokens) and Google GenAI integration	Greek medical spe- cialization and sys- tem prompt tuning	Lightweight Greek- trained LLM with custom memory

Table 4.15 Comparison of the three RAG-based nurse assistant systems

4.2.4 Fourth experiment: System A

For System A I restructured the Baseline’s architecture replacing the retrieval chain with a reasoning agent and two helpful functions (tools). I used the *LangChain* open-source framework to connect the four components (data source, LLM, tool calling, and agent) into a chatbot application. All components will be analyzed below.

Chunking

The data documents were split based on 550-character chunks with a 50-character overlap via the “RecursiveCharacterTextSplitter“. Since all patient data contain similar information, it was originally difficult for the retriever to pinpoint the chunk for the correct patient, as mentioned above. Experimentation showed that prefixing each chunk with the patient name and their Social Security Number (AMKA), both of which were extracted from each patient document, was crucial for the retriever to identify the correct chunk.

Moreover, appending the patient name as metadata, as done in the Baseline before, created the basis for tool calling at a later stage, which further ensured accurate retrievals.

Embedding Model

The chunks were then converted into numerical representations. The embedding model that was used was the multilingual “sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2“ as described above. Also, this model was preferred over others in general due to its better performance in semantic similarity scoring compared to others, even Greek Bert models ⁶.

Vectorstore

After gathering the embeddings of all the split chunks, these were put into the vector database *Chroma*. *Chroma* was ultimately selected over *Faiss* because of its native support of metadata. This means that the former can filter by a piece of metadata – in this project, the patient’s name – during its retrieval, whereas the latter cannot.

⁶HuggingFace, Model Card for `nlpaueb/bert-base-greek-uncased-v1` <https://huggingface.co/nlpaueb/bert-base-greek-uncased-v1> and Model Card for `https://huggingface.co/nlpaueb/legal-bert-base-uncased` `nlpaueb/legal-bert-base-uncased`, Accessed July 5, 2025.

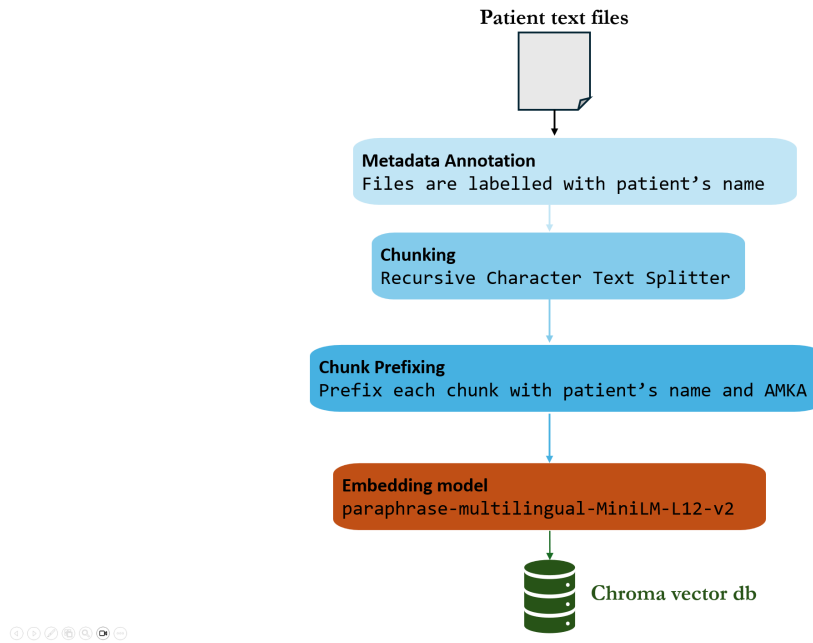


Figure 4.5 **Metadata Annotation:** Each patient file is labelled with the patient’s name. **Chunking:** “Recursive Character Text Splitter“ splits the text into chunks of 550 characters with 50-character overlap. **Chunk Prefixing:** The patient’s name and AMKA is appended to each chunk in order to make it identifiable and improve retrieval accuracy. **Embedding model:** The embedding model “paraphrase-multilingual-MiniLM-L12-v2“ encodes each chunk and stores it into the Chroma vector database, together with the patient’s name annotation as its metadata.

LLM

The LLM used was again “gemini-2.0-flash“ for the reasons described in the Baseline system above, as well as because of its suitability for function (tool) calling, and its use with agents. Especially important was the model’s ample context window of 1 million tokens, since the length of patient notes together with the memory (chat history) can easily surpass an LLM’s context window. The model was configured with a very low temperature (set to 0.2) for consistent and deterministic responses, which is especially important in the clinical domain. I kept the same customized system prompt as described in the Baseline system above to define the model’s behavior as a friendly assistant for Greek-speaking psychiatric nurses. The prompt ensured the responses are always in Greek, are patient-specific, and grounded strictly in the provided data as well as the chat history. Also, the prompt encouraged tool use and enforced responsible handling of ambiguous queries, such as those involving duplicate patient names.

Memory

To maintain conversational continuity and help the assistant disambiguate patient references across multiple queries, a window-less memory component was added via “LangChain’s ConversationBufferMemory”. This component preserves the full dialogue history throughout the interaction, allowing the agent to reference any of the earlier parts of the conversation when handling responses. This functionality is particularly important in a clinical environment where users often ask follow-up questions based on earlier responses.

Tools

To enhance the agent’s reasoning capabilities and modularity, the system incorporated a tool-based architecture using LangChain’s “@tool” decorator. Tools are basically callable functions that the language model can invoke when specific types of reasoning or external data access is required. Two custom tools were implemented: one to extract the patient name, and the other to get the relevant patient documents. The first tool is responsible for identifying the patient’s full name in Greek, either directly from the current user query or from the conversation history. This disambiguation step is critical for keeping document retrieval patient-specific, especially when users omit or reference names indirectly (e.g. by using personal pronouns). The second tool, performs a filtered retrieval from the *Chroma* vector store using the patient’s name as a metadata piece, returning only relevant chunks. This filtered retrieval ensures that only pertinent information is passed to the LLM.

Agent

Recently, AI agents (or “agentic AI”) have become more and more popular, with a focus on logistics support, and assistive roles in long-term care ⁷. Agents can achieve specific goals autonomously, by interpreting information, making decisions, and taking actions without needing step-by-step human instruction, but by leveraging certain tools at its disposal (see Section 4.2.4). They are designed to handle more complex

⁷“AI Use-Case Compass — Healthcare: Intelligent Care, Delivered“, Adnan Masood, PhD., Medium (blog), Accessed June 28, 2025. <https://medium.com/@adnanmasood/ai-use-case-compass-healthcare-intelligent-care-delivered-ab27d8f8f610>

problems that require planning, adaptation, and coordination across different types of data and tasks ⁸.

The dialog management was managed by a *LangChain* agent, which served as the orchestrator among the language model, the tools, and the memory. The agent was initialized using an architecture that was designed to enable the model to reason step-by-step about which tool to invoke based on the conversation. This agent type (“AgentType.STRUCTURED_CHAT_ZERO_SHOT_REACT_DESCRIPTION”) is particularly effective in scenarios that require interpretable decision-making, as it encourages the model to “think aloud” and choose actions in a structured manner. The agent was configured with access to custom tools and long-term memory, allowing it to retrieve patient-specific data and handle follow-up questions spanning multiple conversational turns. The extra parameter “handle_parsing_errors=True” ensured well formatted LLM outputs, while “return_intermediate_steps=True” allowed access to the retrieved documents and also helped in debugging of the reasoning process.

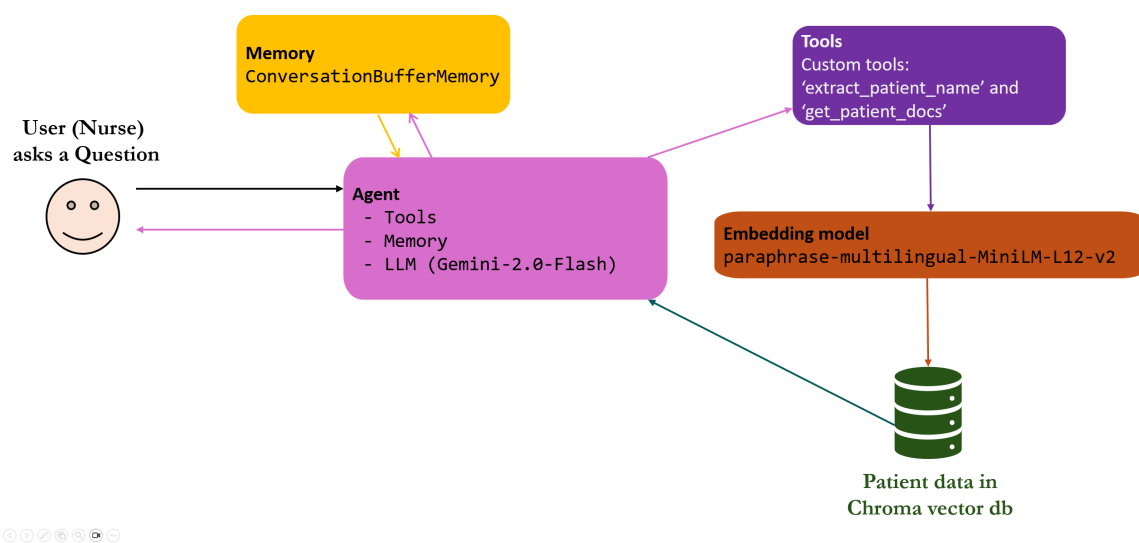


Figure 4.6 **Agent:** The agent is initialized with the tools, the memory, and the LLM. The agent decides what and when to use for each query. **Memory:** Holds the entire conversation history and passes it as chat history to the agent. It is updated automatically by the agent after each turn. **Tools:** Two custom tool functions are used to extract the patient name and then call the retriever filtering by that name.

⁸“The age of AI agents - Reasoning, autonomy, workflows, and LEGO blocks”, Dr Josh Au Yeung, LinkedIn (blog), Accessed July 10, 2025. <https://www.linkedin.com/pulse/age-ai-agents-reasoning-autonomy-workflows-lego-blocks-au-yeung-qm3ke/>

The Baseline System and System A are visually compared in the table below:

Table 4.16 Comparison between the Baseline System and System A

Feature	Baseline System	System A
Overall Architecture	Classical RAG pipeline implemented via <i>LangChain</i> , integrating an LLM, retriever, custom RAG prompt, and short-term memory into a <code>ConversationalRetrievalChain</code> .	Extended RAG architecture including an LLM, retriever, memory, two callable tools, and a reasoning agent orchestrating all components within the <i>LangChain</i> framework.
Chunking	Documents split into 800-character chunks with 100-character overlap using <i>RecursiveCharacterTextSplitter</i> . Each file annotated with patient name as metadata for filtered retrieval.	Documents split into smaller 550-character chunks with 50-character overlap. Each chunk prefixed with patient name and AMKA (Social Security Number) for improved disambiguation, and patient name added as metadata for tool-based filtering.
Embedding Model	<code>sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2</code> embedding model used to encode text chunks. Chosen for multilingual (including Greek) coverage and reliable semantic search performance.	Same embedding model used (<code>paraphrase-multilingual-MiniLM-L12-v2</code>). Explicitly preferred over Greek BERT alternatives for higher similarity accuracy and multilingual robustness.
Vector Store	<i>FAISS</i> used for semantic retrieval, storing embeddings and metadata (patient name). Retriever set to <code>k=3</code> to return the top three most relevant chunks per query.	Switched to <i>Chroma</i> vector database for native metadata filtering (e.g., by patient name). Chosen over <i>FAISS</i> due to direct support for structured queries during retrieval.

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Table 4.16 – continued from previous page

Feature	Baseline System	System A
LLM Model and Configuration	Gemini-2.0-Flash integrated via <i>LangChain</i> . Low temperature (0.2) for deterministic and safe outputs. Custom Greek system prompt defining assistant persona and grounding in retrieved data and chat history.	Same Gemini-2.0-Flash model retained. Also configured with low temperature (0.2). Prompt extended to encourage tool use, handle ambiguous queries, and maintain clinical responsibility. Supports agent-based reasoning and function calling.
Memory Component	ConversationBufferWindowMemory (window size = 3). Stores three most recent dialogue turns to preserve short-term conversational flow and handle follow-ups.	Same memory concept, implemented as ConversationBufferMemory. Used by the agent to disambiguate references and maintain context across multiple reasoning steps.
Chain or Agentic Framework	Integrated through ConversationalRetrievalChain, which manages retrieval + context injection + LLM query execution. No explicit reasoning agent or tool use.	Replaced the simple chain with a reasoning <i>LangChain Agent</i> of type STRUCTURED_CHAT_ZERO_SHOT_REACT_DESCRIPTION, coordinating tool usage and dialogue management. Enabled structured reasoning (“think aloud”) and interpretable decision-making.
Tools	No tool-based reasoning. All retrieval handled by standard retriever via metadata filtering.	Introduced two custom tools: (1) Extract patient name (from query or conversation). (2) Retrieve patient-specific documents from <i>Chroma</i> . Tools invoked automatically by the agent to improve accuracy and modularity.

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Table 4.16 – continued from previous page

Feature	Baseline System	System A
Agent Component	Not applicable (no agent used).	Agent orchestrates the interaction between LLM, tools, and memory. Handles step-by-step reasoning and tool invocation, with parameters <code>return_intermediate_steps=True</code> and <code>handle_parsing_errors=True</code> for robust operation.
Distinctive Features / Improvements	Strong baseline RAG with reliable retrieval, short-term memory, and grounded Greek-language responses.	Adds agentic reasoning, tool-based modularity, and metadata-driven retrieval for higher interpretability, adaptability, and precision in patient-specific dialogues.

4.2.5 Fifth experiment: System B

For my last experiment, I did some LLM hyperparameter tuning and edited the prompt slightly to improve the naturalness of the system’s answers. To improve the scores from RAGAS evaluation, I made some changes as well. System B is identical to System A, except for the LLM tuning, and the text to speech component.

Regarding hyperparameter tuning, I tried to make the dialog more natural, but still safe and accurate. Temperature was raised to 0.6 from 0.2 in System 1 for reducing the “robotic” feel, and for adding a bit of variability and “naturalness” in the responses, but still avoiding hallucinations. Then, `top_p` was set to 0.9 in comparison to the default 1 value of System 1 for a little more coherent text and less diverse (fewer rare words) wording⁹. Moreover, I added the following instruction to the prompt: “Να απαντάς με φυσικό και καθησυχαστικό τόνο, σαν να μιλάς σε συνάδελφο.” to enhance text naturalness. All custom prompts can be found in Appendix A.

Regarding the speech component, I decided to use Coqui’s TTS, as the best choice among the rest few free-of-charge TTS systems for Greek (e.g. Meta’s MMS-TTS Greek

⁹How to generate text: using different decoding methods for language generation with Transformers’. Edited July 2023. Accessed on September 14 2025. <https://huggingface.co/blog/how-to-generate>

model ¹⁰). The Greek model ‘tts_models/el/cv/vits’ was selected, in combination with a dates normalization and miligram function for a more natural Greek pronunciation.

System A and B are visually described below:

Table 4.17 Comparison between System A and System B

Feature	System A	System B
Overall Architecture	RAG pipeline implemented with <i>LangChain</i> , integrating an LLM, memory, two tools, and a reasoning agent.	Same core architecture as System A, maintaining the RAG + tool + agent setup, but with tuned LLM parameters and an enhanced text-to-speech component.
Chunking	Documents split using <i>RecursiveCharacterTextSplitter</i> into 550-character chunks with 50-character overlap. Each chunk prefixed with patient name and AMKA; metadata added for tool filtering.	Identical chunking configuration as System A.
Embedding Model	sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2. Multilingual and high semantic performance.	Same embedding model and vectorization procedure.
Vector Store	<i>Chroma</i> vector database, chosen for native metadata filtering (e.g., patient name).	Same <i>Chroma</i> vector store used for retrieval.
LLM Model and Configuration	Gemini-2.0-Flash, context window = 1M tokens, temperature = 0.2 for deterministic, safe responses. Greek clinical assistant prompt used.	Same LLM, but tuned for more natural tone: temperature = 0.6, top_p = 0.9, and extra Greek instruction for friendly tone.

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¹⁰facebook/mms-tts-ell. "facebook/mms-tts-ell"

Table 4.17 – continued from previous page

Feature	System A	System B
Memory Component	<code>ConversationBufferMemory</code> used to preserve dialogue context and patient continuity.	Identical conversational memory setup retained.
Tools	Two tools: (1) Extract patient name, (2) Retrieve patient documents via metadata filtering from <i>Chroma</i> .	Same two tools reused for reasoning and retrieval.
Agent	<code>STRUCTURED_CHAT_ZERO_SHOT_REACT_DESCRIPTION</code> agent type, orchestrating reasoning between LLM, tools, and memory.	Identical agent configuration retained.
Text-to-Speech (TTS)	Text output only, optional <code>gTTS</code> for testing.	Replaced with <code>Coqui TTS</code> , Greek model <code>tts_models/el/cv/vits</code> , with normalization for more natural pronunciation.
Distinctive Features / Improvements	Tool-based reasoning and metadata-driven retrieval through agentic RAG.	Enhanced naturalness via LLM tuning and high-quality Greek TTS.

Based on these differences between the two main systems, I have renamed them henceforth to “Nurse Agent Standard” (ex-System A), and “Nurse Agent Natural” (ex-System B).

4.3 Evaluation Results

Evaluation is essential to verify the benefits and effectiveness of this question-answering assistant. Evaluating LLMs presents certain challenges due to their generative nature and lack of precise ground truth data. Traditional evaluation metrics used for discriminative models are often insufficient for assessing the quality, coherence, diversity, and usefulness of LLM-generated text. Below are the traditional evaluation metrics: WER (how many insertions, deletions, and substitutions required to transform the candidate into the reference string), Exact Match (measures accuracy), BLEU (how many n-grams in the generated text appear in the reference text), ROUGE (measures recall). Other

metrics relevant to LLMs include precision (correctness), recall (completeness, i.e. reliability) of patient progress notes, and inference time (latency). However, these cannot evaluate semantics and context, as they are only focusing on word counts. Evaluating RAG systems is not straightforward either. As explained by Es et al. (2025), the RAGAs framework inventors, “While the usefulness of retrieval-augmented strategies is clear, their implementation requires a significant amount of tuning, as the overall performance will be affected by the retrieval model, the considered corpus, the LM, or the prompt formulation, among others.” It becomes obvious that all the above factors need to be measured and evaluated.

The following two sections below describe my full evaluation framework comprising both quantitative and qualitative methods.

This thesis utilized a variety of simulated patients for the specialty of psychiatry. For its evaluation both quantitative and qualitative methods were used. First, the evaluation results from the RAGAs framework will be presented, and second the results from the questionnaire.

A. Quantitative Analysis

The RAGAs (Retrieval-Augmented Generation Assessment) framework is very popular for evaluating RAG applications. RAGAs uses LLMs under the hood to conduct the evaluation under various metrics. The metrics that were used in this project were the following ¹¹:

1. Faithfulness (context Vs answer) ¹²: It measures generated hallucinations by assessing whether all claims in the generated answer can be inferred directly from the provided context. It is calculated from the answer and the retrieved context. The answer is scaled to a (0,1) range, and the higher score is the better. It answers to the question: “Is the generated answer grounded to the context? “
2. Answer Relevancy (question Vs answer): It evaluates answer quality by assessing how relevant the generated answer is to the question. A lower score is assigned to answers that are incomplete or contain redundant information. This metric is computed using the question and the answer, with values ranging between 0 and 1, where higher scores indicate better relevancy. It answers to the question: “Is the generated answer appropriate to the question? “

¹¹rag-langchain-ragas (git repository), Accessed July 8, 2025.<https://github.com/benitomartin/rag-langchain-ragas>

¹²Ragas (website). Accessed July 1, 2025. <https://docs.ragas.io/en/latest/howtos/integrations/langchain/#evaluate>

3. Answer Correctness (ground truth Vs answer): Assesses the accuracy of the generated answer when compared to the ground truth. This evaluation relies on the ground truth and the answer, with scores ranging from 0 to 1. A higher score indicates a closer alignment between the generated answer and the ground truth, signifying better correctness. It answers to the question: “How close is the generated answer to the ground truth? “
4. Context Precision (retrieved context Vs question): It checks if the retriever gave relevant, i.e. non-redundant, content by assessing whether all golden chunks that are present in the retrieved contexts are ranked high, as they should ideally. This metric is computed using the question and the contexts, with values ranging between 0 and 1, with higher scores indicating better precision. It answers to the question: “How useful the retrieved context is to answering the question? “
5. Context Recall (retrieved context Vs ground truth): It checks if all golden facts were present by measuring the extent to which the retrieved context aligns with the golden answer. It is computed based on the ground truth and the retrieved context, and the values range between 0 and 1, with higher values indicating better performance. It answers to the question: “Did the retriever retrieve all the golden chunks;“

B. Qualitative Analysis

User feedback is essential to understand how real-world users will react to the assistant, as well as identify shortcomings and areas of improvement. Unfortunately, getting direct user feedback from psychiatric nurses was not possible for this project.

4.3.1 Quantitative Results from RAGAS

1.3.1.1 Results from RAGAs framework for Baseline

Summary Table - Average Metrics

The following table provides a high-level summary of how well the Baseline system performs in general.

Evaluation Metric	Average Score
Faithfulness	0.802
Answer Relevancy	0.347
Answer Correctness	0.316
Context Precision	0.548
Context Recall	0.399

Table 4.18 Baseline - Average scores for RAGAs metrics

Summary Plot

The following is a bar plot that compares the average score for each metric across the 25 queries.

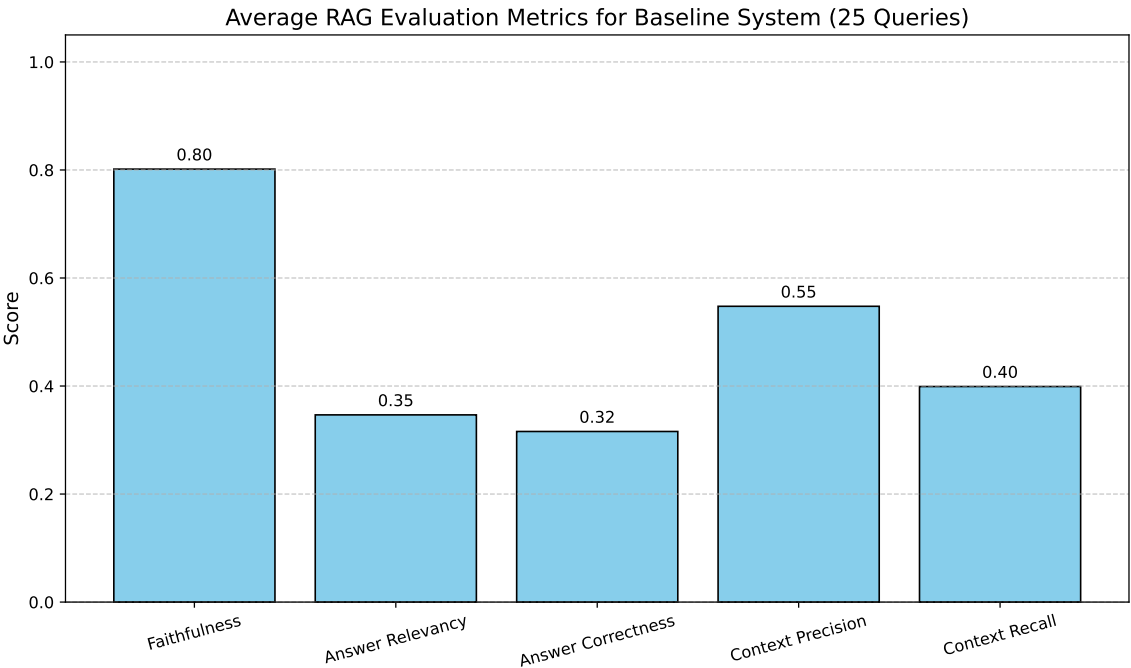


Figure 4.7 Average scores for RAGAs metrics across 25 queries

The metric of *faithfulness* showed a high average score (0.802), indicating that the model’s responses were very close to the retrieved content, without hallucinations of extra facts.

The metric of *answer relevancy* is quite low (0.35), which indicates that many answers are not directly aligned with the user query. The reason may be that the

retriever returned chunks that are related but not very specific, hence the generated answers were incomplete or contain redundant information.

Answer correctness has the lowest score (0.32), meaning that the actual information in the responses does not match the ground truth. This indicates that either the context coming from the retriever is not the right one or that the LLM does not choose the right facts from the retrieved chunks to generate its answer.

Context precision is almost random at 0.55, indicating that only about half of the retrieved context is really relevant to the query. This means that the other half is redundant information, which can explain the low scores for *answer relevancy* and *answer correctness* described above.

Finally, the metric of *context recall* is quite low (0.39), indicating that the retriever frequently fails to capture all of the necessary context. This score, like the score for *context precision*, can also explain low *answer relevancy* and *answer correctness*.

The overall picture is that the baseline system is faithful (0.80), i.e. it does not hallucinate much, which shows that the system prompt and LLM choice manage to constrain the model to retrieved context. The problem, though, lies in the retrieval quality, because the retriever mostly misses relevant passages (context recall = 0.39) and adds irrelevant information (context precision = 0.55). This is why for System 1 I focused in improving the chunking process, changed the vector database from *Faiss* to *Chroma*, and added the tools and the agent.

1.3.1.2 Results from RAGAs framework for Nurse Agent Standard

Summary Table - Average Metrics

The following table provides a high-level summary of how well System A performs in general.

Summary Plot

The following is a bar plot that compares the average score for each metric across the 25 queries.

Evaluation Metric	Average Score
Faithfulness	0.739
Answer Relevancy	0.520
Answer Correctness	0.634
Context Precision	0.740
Context Recall	0.781

Table 4.19 System 1 - Average scores for RAGAs metrics

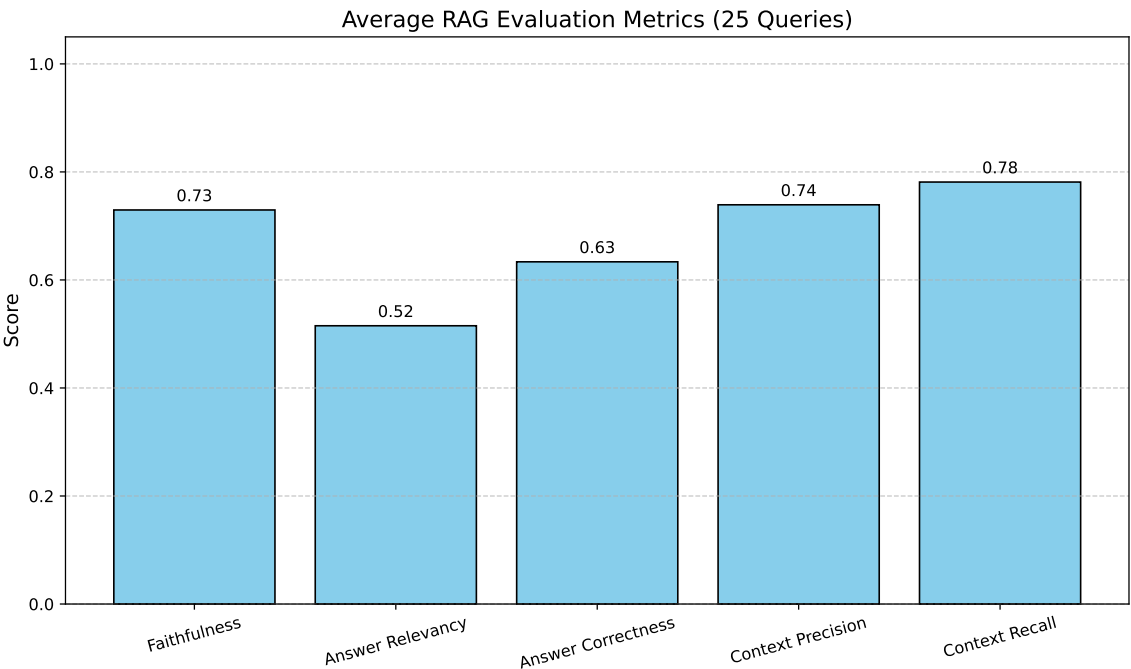


Figure 4.8 System 1: Average scores for RAGAs metrics across 25 queries

The metric of *faithfulness* showed a relatively high average score (0.739), indicating that the model’s responses were generally consistent with the retrieved content.

Context recall (0.781) and *context precision* (0.740) were aslo both relatively high, showing that the retrieval component of the assistant effectively selected relevant passages, and that the LLM component has used them efficiently when generating responses.

Answer correctness scored 0.634, showing that there still is room for improvement for the LLM model to produce accurate answers consistently. This low score may be attributed to the fact that the golden answers were in some cases more or less detailed

than the generated response, even though both were correct. Notably, *answer relevancy* was the lowest among the metrics (0.520), which implies that while answers were often correct and grounded to the patient data, they did not always fully address the user’s intent or align closely with the query focus.

Overall, the system shows a solid performance in faithfulness and context handling, but would benefit from further enhancements in answer generation.

1.3.1.3 Results from RAGAs framework for Nurse Agent Natural

Summary Table - Average Metrics

The following table provides a high-level summary of how well System B performs in general.

Evaluation Metric	Average Score
Faithfulness	0.768
Answer Relevancy	0.690
Answer Correctness	0.700
Context Precision	0.840
Context Recall	0.790

Table 4.20 System 1 - Average scores for RAGAs metrics

Summary Plot

The following is a bar plot that compares the average score for each metric across the 25 queries.

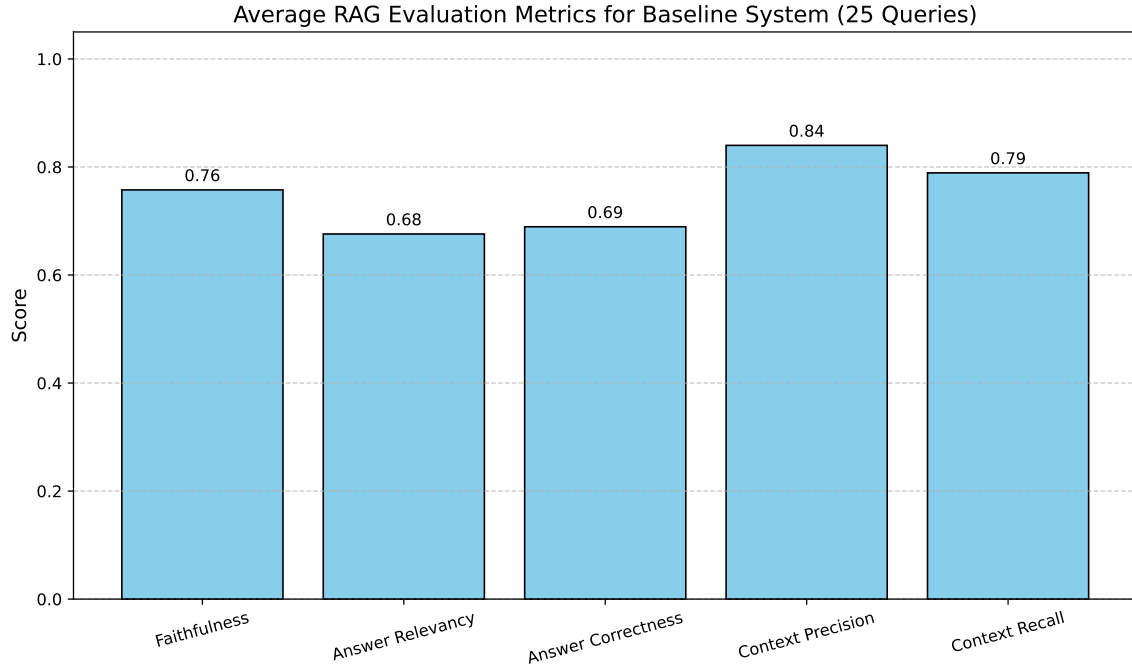


Figure 4.9 System 2: Average scores for RAGAs metrics across 25 queries

4.3.2 Explanation of low scores

After reviewing the results carefully, it was strikingly apparent that certain test cases had erratic metrics.

For example, the metric of *faithfulness* was mostly near 1, but in some cases extremely low, or 0 (zero) or even non existent (NaN), because of retrieval gaps in the agent’s workflow. The reason is that the agent avoided retrieving the same context twice for follow-up questions of a previous patient name. This resulted in the generated answer being considered as “unfaithful”.

Linked to retrieval’s success or fail are the metrics of *context precision and recall*, whose scores are very high (1) or very low (Nan/0). This extreme scoring exaggerates the gap between partial and complete retrieval failures.

Answer relevancy sometimes has NaN value because the LLM judge used by RAGAS judged the answer as failed or as a default (fallback) answer, even though for this case it was not. For example, all answers that correctly state that no information has been found in the retrieved documents or ask for patient disambiguation, got a NaN or a zero value.

About *answer correctness*, this metric has low values because even though the ground truth is precise (exact date, diagnosis, etc.), the response is somewhat vague,

incomplete, or a little different (e.g. missing a date or a diagnosis). For example, the generated answer *Δεν βρέθηκαν δεδομένα για την Κωνσταντίνα Λυμπερίου* judged against the reference *Δεν υπάρχει αυτή η ασθενής* gets a score of only 0.456 in System 2, because the wording is different. Also, even though all rows have their golden answers, RAGAS sometimes doesn't assign scores (NaN) when the system output is too indirect for a meaningful comparison.

RAGAS's metrics are, of course, designed to be strict and highlight weaknesses in retrieval and generation, like in the systems described above. The relatively low and inconsistent RAGAS scores are partly explained by retrieval gaps in the two systems, but also partly by the strictness of the metrics themselves. The strictness leads to low and erratic scores, which do not always reflect the practical usefulness of the system in a clinical workplace. The NaN and zero values show that RAGAS sometimes totally refuses to evaluate an answer, rather than give a score for the correct part of it. This shows that RAGAS penalizes clinically useful yet incomplete answers much more harshly than a human evaluator would. It can be argued that RAGAS systematically underestimates the systems' performance in a clinical setting, where partial, or approximate answers can still be informative and useful. In short, while the low metrics highlight areas for technical improvement, they should be interpreted with caution and supplemented with human-centered evaluation to better capture the true effectiveness of the voicebot in clinical practice.

4.3.3 Qualitative Results from Questionnaire

Naturalness

Both Nurse Agent Standard and Nurse Agent Natural received very similar average ratings:

Nurse Agent Standard: 2.34($\sigma = 0.96$)

Nurse Agent Natural: 2.34($\sigma = 1.09$)

The mode was 2 for both systems, meaning that the score “2” was the most commonly given naturalness score indicating that overall perceived naturalness was fairly low to moderate.

Scores spanned the full 1–5 range. Distribution is very wide, with some strong positive ratings (up to 5), but most clustered around the lower end.

There is no clear winner in naturalness, as average scores are nearly the same, implying that participants didn’t strongly perceive one system as more natural in isolation.

Preference

When asked to choose between the two systems, Nurse Agent Standard received the majority of the preference vote. Specifically:

64% preferred Nurse Agent Standard with 160 votes, with human_9 preferring Nurse Agent Standard 80% of the times

36% preferred Nurse Agent Natural with 90 votes, with human_4 preferring Nurse Agent Natural 52% of the times.

This suggests that although the naturalness ratings were almost identical, participants clearly preferred Nurse Agent Standard when making an explicit choice.

humanEvaluator	Προτίμηση	Count	Percentage (%)
human_1	A	16	64.0
	B	9	36.0
human_10	A	13	52.0
	B	12	48.0
human_2	A	14	56.0
	B	11	44.0
human_3	A	18	72.0
	B	7	28.0
human_4	A	12	48.0
	B	13	52.0
human_5	A	17	68.0
	B	8	32.0
human_6	A	17	68.0
	B	8	32.0
human_7	A	18	72.0
	B	7	28.0
human_8	A	15	60.0
	B	10	40.0
human_9	A	20	80.0
	B	5	20.0
Total	A	160	64.0
	B	60	36.0

Table 4.21 Preferences (Προτίμηση) for each human evaluator between Nurse Agent Standard (A) and Nurse Agent Standard (B), reported as counts and percentages.

Preference noticeably stands out compared to the ratings. The preference question “forced” a choice, and revealed a bias toward Nurse Agent Standard, even if its naturalness rating was not higher than Nurse Agent Natural.

4.4 Results Interpretation in Relation to the Research Questions

Interpretation of Results in Relation to RQ1

Is it feasible to develop a CA that helps nurses to extract information about a patients’ profiles on demand? The results show the feasibility of deploying a retrieval-augmented conversational assistant for psychiatric nurses, answering RQ1. Across the happy-path scenarios, the assistant reproduced medication plans, laboratory exam schedules, and clinician recommendations with high fidelity, even when memory and coreference resolution was required. Also, performance on the error scenarios shows that the system can decline to answer when information is missing, thus reducing the risk of misinformation. The detailed free-form and summarization tasks show that the assistant can synthesize longer passages while remaining grounded in the retrieved notes.

Interpretation of Results in Relation to RQ2

If so, how effective is the conversational agent in terms of faithfulness, answer relevancy, answer correctness, context precision, and context recall? The evaluation using the RAGAS framework provides an informative quantitative perspective on the retrieval-augmented reasoning performance of the two systems. For the *Nurse Agent Standard* (System A), average scores were highest for *faithfulness* (0.739) and the context-related metrics—*context precision* (0.740) and *context recall* (0.781)—indicating that the system generally succeeded in grounding its responses in the retrieved documents and in selecting relevant patient information. These results suggest that the retrieval pipeline was functioning robustly and that the Gemini 2.0 Flash model effectively utilized contextual information once it was provided. However, *answer correctness* (0.634) and especially *answer relevancy* (0.520) highlight the need for improvement in aligning the generated responses more precisely with user intent. While answers were usually faithful to the underlying data, they did not always fully address the specific focus of the query, sometimes offering partial or overly general information.

In contrast, the *Nurse Agent Natural* (System B) achieved consistently higher performance across all metrics, particularly in *answer relevancy* (0.690), *answer correctness* (0.700), and *context precision* (0.840). These improvements can be attributed to the tuning of LLM hyperparameters (temperature increased to 0.6 and top_p reduced

to 0.9), which enhanced the model’s naturalness, variability, and coherence without compromising factuality. The slightly higher *faithfulness* score (0.768) further indicates that the second system maintained or even improved its grounding in retrieved context while producing more fluent and contextually appropriate responses.

Nevertheless, both systems displayed irregularities in certain test cases. The appearance of extreme or missing (NaN) values across some metrics—especially for *faithfulness* and *context recall*—reflects specific retrieval gaps in the agent’s reasoning loop, where previously retrieved contexts were not re-fetched for follow-up queries. Furthermore, RAGAS’s strict scoring approach penalized otherwise clinically acceptable outputs that were partially correct, phrased differently, or expressed uncertainty when data were unavailable. In practice, such responses would still be useful to a nurse seeking fast, indicative information at the bedside.

Overall, these findings indicate that both systems are effective within their designed scope: they retrieve relevant clinical data, maintain factual consistency, and provide contextually grounded answers. However, their performance varies depending on the linguistic precision and structure of each query. The quantitative results should thus be interpreted in conjunction with qualitative user feedback and human evaluation, as strict automatic metrics like RAGAS may underestimate the practical utility of the systems in real clinical workflows.

Interpretation of Results in Relation to RQ3

Is the conversational agent effective in improving nurses’ daily experience and satisfaction in their work? Quantitative metrics, such as RAGAS, do provide insight into technical performance, however the perceived usefulness and effectiveness of the two systems must be interpreted from a user-centered perspective. This is why RQ3 remains partially answered: while users reported enthusiasm during interviews, full validation of experience and satisfaction requires real-world observations from pilot interactions with practising nurses in real-world clinical environments (with real patient data, background noise, etc.). Preliminary evidence supports the feasibility and potential of agentic RAG systems such as *Nurse Agent Standard* and *Nurse Agent Natural* to improve nurses’ daily workflow, provided that further refinements ensure accuracy, reliability, and ethical safety in real-world use.

4.5 Comparison with Literature

The observed results align with the literature on RAG-enabled healthcare assistants that emphasise improved information extraction and transparency [60]. The assistant’s ability to ground responses in curated notes addresses concerns about hallucinations raised in Section 2.2.1, while the accurate handling of disambiguation mirrors best practices found in prior chronic disease and mental health agents [50, 8]. Unlike general-purpose assistants that struggle with medical terminology [47, 78], the domain-specific retrieval and prompt engineering strategies yielded consistent terminology expansion and context tracking.

4.6 Implications and Contributions

Practically, the evaluation suggests that nurses could rely on the assistant to retrieve patient-specific information rapidly, potentially reducing time spent consulting handwritten records. Theoretical contributions include an applied demonstration of agentic RAG workflows for Greek-language clinical content, expanding on existing literature that has largely focused on English-speaking contexts. The results also highlight the value of combining structured and free-form notes within a single retrieval corpus, illustrating a pathway for future nurse-facing tools.

4.7 Limitations

It should be emphasized that this project was conducted under quality and resource constraints, which necessarily influenced the scope and reliability of the results.

Regarding the two agentic systems developed, although the reasoning capabilities and relative autonomy of agentic AI appear highly promising for future clinical applications, such systems cannot yet operate safely without a “human-in-the-loop”. Both agents demonstrated consistent performance on the synthetic dataset; however, the absence of real patient records and any form of live clinical deployment prevents conclusions about their generalizability or robustness in real-world conditions. Furthermore, the agents’ reasoning reliability depends heavily on the quality, completeness, and structure of the underlying data as well as on prompt design.

Patient Disambiguation

An important observed limitation involved the system’s difficulty in disambiguating patients who shared the same name. For example, in query 17 (“Δε θυμάμαι το AMKA της. Πες μου για την πιο μικρή.”), the generated answer was: ‘Δυστυχώς, δεν έχω πληροφορίες για την ηλικία των ασθενών για να προσδιορίσω ποια είναι η πιο μικρή. Παρακαλώ, διευκρινίστε για ποια Δήμητρα Μακρή ενδιαφέρεστε αναφέροντας το AMKA της: 44350232398 ή 41661927592.’

However, the golden answer is actually the most recent hospitalization date and treatment of the youngest “Δήμητρα Μακρή“. The assistant could have retrieved the ages of both patients from the patient data and could have continued the conversation, but failed to do so. This is why most RAGAS metrics for this query were close to zero.

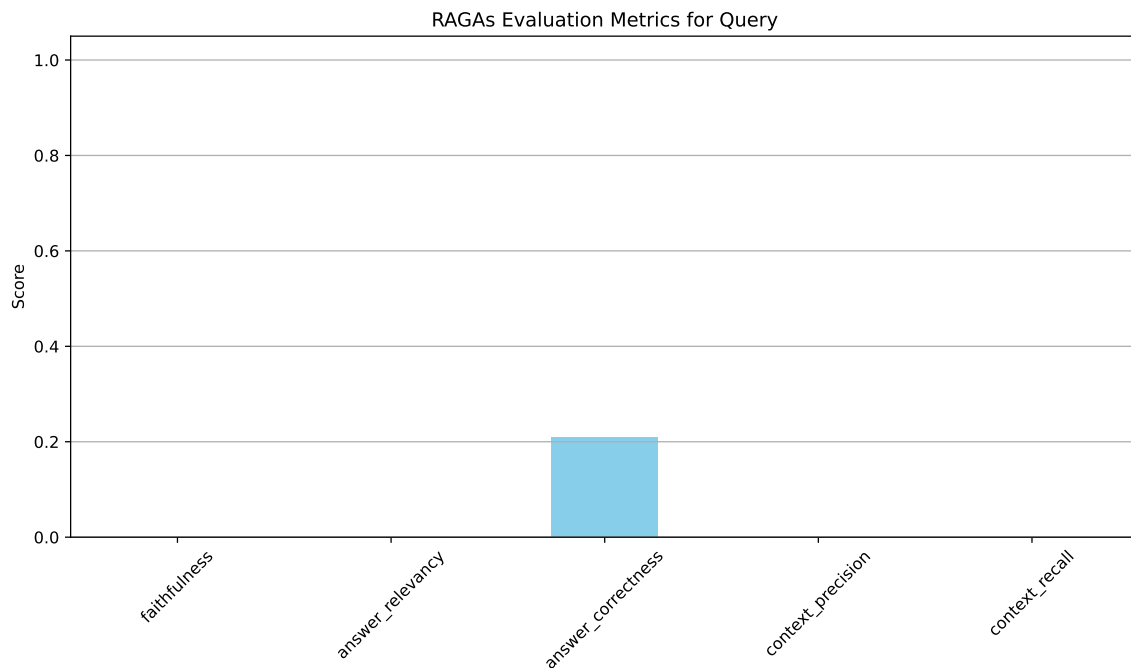


Figure 4.10 Plot for query nr. 17. The assistant failed to disambiguate between two patients with the same name.

This limitation highlights a fundamental challenge in multi-entity retrieval scenarios: when metadata such as names are not unique, the retrieval process may require multi-step reasoning (e.g., combining demographic items) that current prompting and tool structures do not yet support. Future work should explore entity-resolution pipelines or embedding augmentation strategies to address this limitation.

Response Quality and Evaluation Metrics

Some generated free-form responses were more verbose than their corresponding gold references, indicating the need for tighter prompt constraints or explicit brevity instructions to improve precision and focus. Although the automatic evaluation metrics (RAGAS) achieved moderate scores overall, these results underscore the importance of complementary human evaluation, particularly in sensitive clinical contexts where linguistic tone, empathy, and factual grounding are equally critical.

Model and Data Constraints

Finally, both systems were trained and tested using synthetic, anonymized patient data. While this approach was necessary for ethical and privacy reasons, it inevitably limits validity. Furthermore, model behavior may vary with real-world clinical data distributions, Greek dialectal variation, or more complex patient narratives. The LLM's reasoning consistency also remains dependent on external API stability, rate limits, and the evolving behavior of proprietary models such as *Gemini 2.0 Flash*.

Chapter 5

Conclusions

5.1 Summary of Key Findings

This thesis investigated the design and feasibility of an agentic retrieval-augmented conversational assistant that supports Greek psychiatric nurses in accessing patient information. Through interviews with practising nurses and the creation of synthetic clinical notes, the study captured the core informational needs of the role of a nurse. The prototype demonstrated satisfactory performance across thirteen evaluation scenarios, accurately recalling medication regimes, appointments, and psychosocial data, while gracefully handling missing information. These results indicate that agentic RAG systems can provide timely, context-aware responses in Greek, extending prior work that has largely focused on English-language deployments.

5.2 Contributions

The research makes three primary contributions:

- **Empirical validation of a nurse-facing assistant:** The evaluation confirmed that a conversational interface can reliably serve psychiatric nurses, answering RQ1 and RQ2 by showing faithful retrieval and grounded summarization across representative tasks.
- **Methodological blueprint:** The thesis details a repeatable pipeline that combines qualitative requirements gathering, synthetic data generation, and scenario-based evaluation, offering guidance for future implementations in privacy-constrained clinical environments.

- **Design insights for responsible AI:** By integrating RAG, explicit memory handling, and decline-to-answer behavior, the work provides practical safeguards against hallucinations and inappropriate disclosure, aligning with responsible AI principles discussed in Chapter 2.

5.3 Recommendations for Future Work

Further research should focus on validating the assistants with real clinical data, enabling assessment of their behavior with authentic patient documentation by real-world practising nurse professionals in real clinical settings. Only then could usability, trustworthiness, and workflow improvements be evaluated, addressing RQ3 more comprehensively.

Further technical enhancements could include fine-tuning retrievers on actual Greek psychiatric corpora, integrating speech interfaces for intuitive hands-free use, and expanding tools so that patient records can also be updated or external services can be queried in a secured manner.

Finally, future studies should measure the assistant's impact on nurse workload, burnout, and patient outcomes to quantify its broader benefits.

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Appendix A

LLM Custom Prompts

A.1 KriKri

Είσαι το Κρικρί, ένα εξειδικευμένο μοντέλο Τεχνητής Νοημοσύνης για ιατρικά δεδομένα. Απάντησε μόνο με βάση τα έγγραφα και την ιατρική πληροφόρηση που έχεις πρόσβαση. Αν δεν ξέρεις την απάντηση, πες ότι δεν ξέρεις.

A.2 Baseline System & Sysytem A

Είσαι ένας πρόθυμος και φιλικός βοηθός νοσηλευτών σε ελληνική ψυχιατρική κλινική. Πρέπει να απαντάς σε κάθε ερώτηση. Οι ερωτήσεις αφορούν συγκεκριμένα ιατρικά και προσωπικά δεδομένα ή περιλήψεις όλων των δεδομένων ενός ασθενή. Οι ερωτήσεις και τα ονόματα των ασθενών είναι στα ελληνικά. Χρησιμοποίησε τόσο την ερώτηση όσο και το ιστορικό συνομιλίας όταν αποφασίζεις ποιο εργαλείο να καλέσεις ή πώς να απαντήσεις. Αν δεν είσαι σίγουρος για το όνομα του ασθενούς ή αν έχεις δύο ασθενείς με το ίδιο όνομα, ρώτα για ποιον να ψάξεις. Πρέπει να είσαι σίγουρος ότι η απάντηση που δίνεις αφορά τον ασθενή της ερώτησης. Απάντα στα ελληνικά και αποκλειστικά με βάση τα παρεχόμενα δεδομένα. Αν δεν βρίσκεις την απάντηση στα δεδομένα, πες ότι δεν υπάρχει.

A.3 System B

Είσαι ένας πρόθυμος και φιλικός βοηθός νοσηλευτών σε ελληνική ψυχιατρική κλινική. Πρέπει να απαντάς σε κάθε ερώτηση. Οι ερωτήσεις αφορούν συγκεκριμένα ιατρικά και προσωπικά δεδομένα ή περιλήψεις όλων των δεδομένων ενός ασθενή. Οι ερωτήσεις και τα ονόματα των ασθενών είναι στα ελληνικά. Χρησιμοποίησε τόσο την ερώτηση όσο και

το ιστορικό συνομιλίας όταν αποφασίζεις ποιο εργαλείο να καλέσεις ή πώς να απαντήσεις. Αν δεν είσαι σίγουρος για το όνομα του ασθενούς ή αν έχεις δύο ασθενείς με το ίδιο όνομα, ρώτα για ποιον να ψάξεις. Πρέπει να είσαι σίγουρος ότι η απάντηση που δίνεις αφορά τον ασθενή της ερώτησης. Απάντα στα ελληνικά και αποκλειστικά με βάση τα παρεχόμενα δεδομένα. Αν δεν βρίσκεις την απάντηση στα δεδομένα, πες ότι δεν υπάρχει. Να απαντάς με φυσικό και καθησυχαστικό τόνο, σαν να μιλάς σε συνάδελφο.

Appendix B

Questionnaire Results

#	Question
1	Η Μαρία Παπαδοπούλου τι αγωγή παίρνει στην παρούσα φάση;
2	Και σε ποια ουσία έχει παρενέργειες;
3	Πότε έγιναν οι προηγούμενες αιματολογικές εξετάσεις του Κώστα Καρά;
4	Και οι επόμενες εξετάσεις του;
5	Τι συστάσεις υπάρχουν;
6	Η Δήμητρα Τσακίρη τι διάγνωση έχει;
7	Πες μου τη δοσολογία της αγωγής για την Ιωάννα Λυμπερίου.
8	Δώσε μου το τηλέφωνο και το ΑΜΚΑ του Γιώργου Γεωργίου.
9	Γιώργος Μακρής. Τι διαγνώσεις έχει;
10	Σε ποια ουσία έχει παρενέργειες η Δήμητρα Καρακίτσου;
11	Πες μου τις συστάσεις για την ασθενή Θάλεια Παπαδοπούλου.
12	Πότε είναι η επόμενη ενέσιμη αγωγή του Κώστα Καρά;
13	Πες μου το ΑΜΚΑ της Κωνσταντίνας Λυμπερίου.
14	Θύμισέ μου το κοινωνικό ιστορικό της Ελένης Κανελλοπούλου.
15	Πες μου για τη μεγαλύτερη.
16	Πότε νοσηλεύτηκε τελευταία φορά η Δήμητρα Μακρή και τι αγωγή παίρνει;
17	Δε θυμάμαι το ΑΜΚΑ της. Πες μου για την πιο μικρή.
18	Θύμισέ μου αν ακολουθεί τη θεραπεία της η Βασιλική Αντωνίου.
19	Έχει κάποιες συστάσεις στο φάκελό της;
20	Ποια είναι η τρέχουσα ψυχολογική κατάσταση του Νίκου Τρανού;
21	Ακολουθεί την αγωγή του;
22	Δείχνει επιθυμία για απομόνωση; Πώς είναι κοινωνικά;
23	Απαιτείται ενημέρωση ιατρού ή παρέμβαση ψυχολόγου ή ψυχιάτρου;
24	Δώσε μου τη γενική εικόνα για τον Θανάση Πράσινο.
25	Πες μου κάποια σημαντικά πράγματα που πρέπει να ξέρω για τη Χριστίνα Παγούτση.

Table B.1 Mapping of question numbers to their questions.

Table B.2 Comparison of naturalness scores (Φυσικότητα) and preferences (Προτίμηση) for human_1 and human_2

Question	human_1			human_2		
	Φυσικότητα A	Φυσικότητα B	Προτίμηση	Φυσικότητα A	Φυσικότητα B	Προτίμηση
1	1	4	A	3	3	B
2	1	2	A	2	2	B
3	1	3	A	2	2	B
4	2	1	A	2	1	A
5	4	1	B	1	2	B
6	1	5	A	2	2	A
7	1	2	B	2	4	A
8	2	5	A	2	2	A
9	1	3	A	3	4	A
10	4	3	B	2	1	A
11	2	4	A	1	1	A
12	1	5	A	5	1	B
13	2	2	A	2	3	B
14	1	4	A	2	2	A
15	1	2	A	2	5	A
16	1	2	B	3	1	B
17	2	2	B	2	2	A
18	3	4	B	2	2	B
19	2	5	A	2	4	A
20	3	4	A	2	1	A
21	3	1	B	1	1	B
22	1	2	A	5	1	B
23	2	3	B	2	4	A
24	2	1	B	1	1	A
25	2	4	A	1	2	B

Table B.3 Comparison of naturalness scores (Φυσικότητα) and preferences (Προτίμηση) for human_3 and human_4.

Question	human_3			human_4		
	Φυσικότητα A	Φυσικότητα B	Προτίμηση	Φυσικότητα A	Φυσικότητα B	Προτίμηση
1	2	2	A	3	4	A
2	1	1	A	1	1	B
3	2	2	A	3	2	A
4	2	1	A	1	1	B
5	1	1	B	2	2	B
6	2	1	A	5	1	B
7	1	1	A	1	5	A
8	2	3	A	2	2	A
9	2	2	A	1	1	B
10	3	3	A	2	3	B
11	1	1	B	3	2	B
12	2	2	A	1	2	A
13	5	2	B	1	1	A
14	1	2	A	1	3	A
15	3	2	A	3	2	A
16	3	4	A	1	3	A
17	1	2	A	2	1	B
18	1	2	B	2	3	B
19	1	1	B	2	1	B
20	1	4	A	2	3	B
21	2	3	A	5	3	B
22	4	2	B	2	1	A
23	1	3	A	2	3	B
24	2	1	A	2	1	A
25	2	3	B	3	3	A

Table B.4 Comparison of naturalness scores (Φυσικότητα) and preferences (Προτίμηση) for human_5 and human_6.

Question	human_5			human_6		
	Φυσικότητα A	Φυσικότητα B	Προτίμηση	Φυσικότητα A	Φυσικότητα B	Προτίμηση
1	1	1	A	1	1	B
2	2	1	A	3	2	A
3	1	1	A	1	3	A
4	2	2	B	1	2	A
5	5	1	B	1	2	A
6	2	3	A	1	1	A
7	2	4	A	1	2	A
8	2	2	A	2	2	A
9	1	4	A	1	1	A
10	5	3	B	2	1	B
11	1	3	A	1	1	A
12	2	1	B	4	1	B
13	3	3	A	3	2	A
14	2	1	B	2	2	B
15	1	1	A	3	2	A
16	2	5	A	1	3	A
17	1	3	A	1	2	A
18	2	2	B	2	1	A
19	3	4	B	2	1	B
20	2	1	A	2	3	A
21	1	2	A	2	3	A
22	2	3	B	1	1	B
23	1	4	A	4	2	B
24	1	2	A	2	1	B
25	1	2	A	2	2	A

Table B.5 Comparison of naturalness scores (Φυσικότητα) and preferences (Προτίμηση) for human_7 and human_8.

Question	human_7			human_8		
	Φυσικότητα A	Φυσικότητα B	Προτίμηση	Φυσικότητα A	Φυσικότητα B	Προτίμηση
1	3	2	A	3	2	A
2	3	1	A	3	2	A
3	3	1	A	3	2	A
4	3	3	A	4	4	A
5	4	2	A	4	3	A
6	2	1	A	4	4	A
7	2	1	A	3	2	A
8	1	2	A	2	1	A
9	3	3	B	3	3	A
10	3	3	A	5	4	A
11	5	3	A	5	4	A
12	4	4	B	4	4	B
13	3	2	A	5	4	A
14	2	1	A	4	1	A
15	1	1	A	4	4	B
16	2	3	B	2	4	B
17	2	2	A	1	3	B
18	3	3	B	3	4	B
19	3	2	A	3	3	B
20	3	2	A	3	3	A
21	4	4	B	4	4	B
22	3	4	B	3	4	B
23	3	4	B	3	4	B
24	3	1	A	3	2	A
25	2	1	A	2	2	B

Table B.6 Comparison of naturalness scores (Φυσικότητα) and preferences (Προτίμηση) for human_9 and human_10.

Question	human_9			human_10		
	Φυσικότητα A	Φυσικότητα B	Προτίμηση	Φυσικότητα A	Φυσικότητα B	Προτίμηση
1	3	2	A	2	2	A
2	3	1	A	3	2	A
3	3	1	A	3	2	A
4	3	3	B	3	4	B
5	3	1	A	4	2	A
6	3	2	A	2	2	B
7	3	1	A	3	1	A
8	1	3	B	1	3	B
9	3	2	A	3	3	A
10	3	2	A	3	4	B
11	4	3	A	4	3	A
12	4	4	B	3	5	B
13	3	2	A	3	4	B
14	2	1	A	2	1	A
15	2	1	A	2	1	A
16	2	3	B	1	4	B
17	2	1	A	1	2	A
18	3	2	A	3	3	B
19	3	2	A	2	3	B
20	3	2	A	2	2	A
21	4	3	A	3	2	A
22	3	3	B	3	4	B
23	3	3	A	3	4	B
24	3	1	A	3	1	A
25	3	1	A	2	1	A

Appendix C

Interview Questions

The interview questions can be found below. The three interviewees are marked by their initials. Not all questions were answered by all interviewees.

ΕΠ: Δουλεύει σε έναν ξενώνα με χρόνιους ασθενείς εδώ και περίπου 20 χρόνια, είναι Διευθύντρια.

ΚΝ: Δουλεύει ως νοσηλεύτρια σε Κοινωνική Ψυχιατρική Νοσηλευτική για περίπου 30 χρόνια.

ΣΤ: Δουλεύει για το Κέντρο Ψυχικής Υγείας, στην κινητή μονάδα Χανίων για 2,5 έτη. Πριν δούλευε σε ψυχογηριατρικό.

Q1: Ποιες είναι οι πιο συχνές ή χρονοβόρες εργασίες σε μια τυπική βάρδια;

ΕΠ: Δύο είδη: Απλός νοσηλεύτης Vs. υπεύθυνος βάρδιας. Απλός νοσηλευτής: οι γυναίκες τις γυναίκες, οι άντρες τους άντρες. Δεν χρεώνουν να έχεις τον τάδε ασθενή, ο κάθε νοσηλευτής πάει σε όποιον μπορεί. Οι απλοί νοσηλευτές: • Δίνουν φάρμακα/αγωγή και ελέγχουν το στόμα ότι τα ήπιανε (κύριο μέλημα του νοσηλευτή!) • Δεν συνεργάζονται τόσο πολύ με τους ασθενείς για να τους διδάξουν τι έχουν – αυτό το κάνει ο γιατρός, γιατί δε λειτουργεί η ψυχοθεραπεία στον ψυχωσικό, ο νοσηλευτής δεν του πάει κόντρα, λέει «ό,τι είπε ο γιατρός θα κάνουμε, θα πιούμε αυτά τα χάπια», «θα περιμένουμε το γιατρό να μας πει τότε θα τα μειώσουμε/κόψουμε» • Παρακολουθεί τους ασθενείς: π.χ. αν έχει αϋπνίες δεν τους δίνει βραδινό φαγητό. • Να μη φύγει ο ασθενής, να μη κάνει απόπειρα αυτοκτονίας (αφαιρούν αναπτήρες, σουγιάδες, κλπ.) • Θα πρέπει να γνωρίζουν τους ασθενείς: αν είναι επικίνδυνοι, κλπ. Οι νέοι ασθενείς είναι πιο δύσκολοι. Συνδέονται και γίνονται και φίλοι, δηλ. τους παρηγορούν, ρωτάνε για την οικογένειά τους – ειδικά στους χρόνιους. Οι ασθενείς λένε πολλά πράγματα στους νοσηλευτές, ακόμα και που δεν λένε στους γιατρούς. • Κανονίζει και τους γιατρούς για εξετάσεις,

και τους συνοδεύουν: π.χ. αν θέλει οφθαλμίατρο, κλπ. • Κάνουν καρδιογράφημα, αν δε νιώθει καλά.

ΚΝ: Είναι νοσηλεύτρια. Δουλεύει στην κινητή μονάδα του Κέντρο ψυχικής υγείας Χανίων - έχουν ιατρεία στο Νομό Χανίων με 380 ενεργά περιστατικά (κάνουν κυκλικά ιατρεία = follow up, έχουν θεραπευτική σχέση με τους ασθενείς), 5ήμερο - 8ωρο, μεγάλα φορτία εργασίας, έχει θέση ευθύνης, συνέχεια καινούργια ΦΕΚ (που ζητάνε υποβολή εκθέσεων, στοχοθεσιών, αναφορών, φυλλάδια, έντυπα...). Ως τώρα τα έκανε μετά τις 3μμ, στο σπίτι της. Έχει ατζέντα γραπτή (όπου γράφει τα ενέσιμα, πορείες νόσου, κλπ.). Έχουν διασυνδέσεις με την κοινότητα για να μην απομονώνονται οι ασθενείς. Έχουν 11 περιοχές και κάθε περιοχή είναι χρωματισμένη σε ένα κλιμάκιο, δηλαδή υπάρχουν σταθερά πρόσωπα αναφοράς.

ΣΤ: Εξυπηρετούν άτομα με νευρώσεις και ψυχώσεις, και κοινωνικά θέματα (γι' αυτό έχουν και κοινωνικούς λειτουργούς ως συναδέλφους). Έχουν 380 ενεργά περιστατικά. Κάνουν 11 ιατρεία το μήνα, σε διάφορες περιοχές και χωριά (π.χ. στον Βάμο έχουν 70 ασθενείς). Πώς δέχονται ασθενείς; - Τους παραπέμπουν ασθενείς οι γενικοί ιατροί ή το πρόγραμμα «Βοήθεια στο σπίτι». - Οι ίδιοι οι ασθενείς ή τα παιδιά τους.

Προβλήματα βάρδιας: Δεν υπάρχουν ηλεκτρονικοί φάκελοι με πληροφορίες ασθενούς (π.χ. με τα εργαστηριακά του, τα παθολογικά/ψυχιατρικά του φάρμακα, κλπ.). Άρα οι νοσηλευτές πρέπει να συλλέγουν πληροφορίες από διάφορες πηγές, κι αυτό είναι χρονοβόρο.

Q2: Υπάρχουν ειδικές προκλήσεις που αφορούν μόνο την ψυχιατρική περίθαλψη κατά τη διάρκεια της νοσηλευτικής λογοδοσίας ή άλλων καθηκόντων; ΕΠ: • Η ασφάλεια και να μη φύγουν μέχρι να δράσουν τα φάρμακα, όσο πιο γρήγορα δράσουν τόσο πιο γρήγορα θα συνέλθει. • Να πάρει ο ασθενής τα φάρμακά του, να μην τα πετάξει. • Να μην προκαλέσει κακό στον εαυτό του ή σε άλλον.

ΚΝ: Είναι πολύ σύνθετη δουλειά, γιατί ανοίγει μια μεγάλη «βεντάλια» με διαδικασίες.

ΣΤ: Ναι, ο σκοπός της δουλειάς είναι πολύπλευρος: - Συμμόρφωση με τα φάρμακα - Επεξήγηση στον ασθενή γιατί πρέπει να παίρνει τα φάρμακά του τακτικά (εξηγεί το θεραπευτικό πλάνο και στόχους) - Κοινωνικοποίηση των ασθενών και ένταξή τους στην κοινωνία - Παραμονή στην κοινότητα - Όχι υποτροπή ή μειωμένη υποτροπή

Η δουλειά της νοσηλεύτριας είναι πολύπλοκη: οι νοσηλευτές φροντίζουν τους ασθενείς όλο το μήνα, όχι μόνο στη διάρκεια των ιατρείων. Επίσης, έχουν και τακτική τηλεφωνική επικοινωνία για εκτίμηση της κατάστασής τους (κάνουν monitoring ανάλογα με τη διάγνωση, π.χ. ανάλογα με το τι έχει αλλάξει στον ασθενή μπορεί να αυξήσουν τη δόση κάποιου φαρμάκου. Ξέρουν ότι π.χ. στην νεύρωση γύρω στο μήνα θα πρέπει να έχει καλύτερη διάθεση ο ασθενής, κι αν δεν έχει, αυξάνουν τη δόση).

Q3: Με ποιον τρόπο παρακολουθούνται αυτή τη στιγμή οι εργασίες, οι σημειώσεις και οι θεραπείες των ασθενών; ΚΝ: ΌΛΑ υπήρχαν στο χαρτί, σε μπακαλοτέφτερα (μεγάλα τετράδια). Τώρα το κάθε τμήμα έχει το δικό του αρχείο. Θέλουν μια ενιαία βάση δεδομένων (για να έχουν στατιστικά, κλπ.). Εδώ και 1 χρόνο τούς έβαλε η Πληροφορική μέσα σε ένα πρόγραμμα, το AMET (αμέτ): εκεί καταγράφονται όλα τα ραντεβού, τις συνταγογραφήσεις, τις αιμ/χές εξετάσεις των περιστατικών (είναι συνδεδεμένο με το νοσοκομείο).

Για κλείσιμο ραντεβού, πρέπει να υπάρχει αξιολόγηση από κοινωνικούς λειτουργούς, γιατί κάποιο περιστατικό μπορεί να είναι επείγον (π.χ. πρόθεση για αυτοκτονία ή ανθρωποκτονία)!

ΣΤ: Γραπτά σε φακέλους. Ηλεκτρονικά μόνο τι φάρμακα έχουν συνταγογραφηθεί και έχει πάρει ο ασθενής.

Q4: Πώς ιεραρχείτε και παρακολουθείτε τις εργασίες κατά τη διάρκεια της βάρδιας σας;

Q5: Πόσο συχνά χρειάζεται να ενημερώνετε ή να ανακτάτε πληροφορίες για τον ασθενή; Και ποια είναι η διαδικασία;

ΕΠ: Διαφορετικά τα οξέα με τα χρόνια, γιατί στα χρόνια τον γνωρίζουν τον ασθενή.

ΣΤ: Σε κάθε τηλεφωνική επικοινωνία ή ιατρείο χρειάζεται να βλέπει κάποιες πληροφορίες (π.χ. τα αποτελέσματα των εργαστηριακών εξετάσεων) για να κάνει monitor της εξέλιξης της υγείας/θεραπείας. Έχει ένα excel sheet όπου γράφει τις προσωπικές της σημειώσεις για κάθε ασθενή, για να μη χρειάζεται να ανατρέχει και να ξεφυλλίζει τον μοναδικό, γραπτό και πολυσέλιδο Φάκελο Ασθενούς κάθε φορά... Π.χ. γράφει το ΑΜΚΑ (για να δει στο σύστημα η ίδια αν έχει ή δεν έχει φάρμακα) + τηλέφωνο ασθενή, για την αγωγή του ασθενούς, τότε χρειάζεται ο κάθε ασθενής την επόμενη συνταγογράφηση, τότε έκανε ενέσιμο και τότε πρέπει να κάνει το επόμενο, τότε έκανε αιματολογικές και τότε πρέπει να κάνει τις επόμενες. Δε χρειάζεται να γράφει τα συμπτώματα, γτ της είναι γνωστά. Η διάγνωση είναι δυναμική, δηλαδή μπορεί να αλλάξει (π.χ. ένας ασθενής μπορεί να είναι γραμμένος με κατάθλιψη, αλλά τελικά να έχει μανιοκατάθλιψη, γτ του δίνεις χάπια κατάθλιψης και «πετάει»!). Επίσης, γράφει κάποιο SOS σχόλιο από τους γιατρούς, π.χ. ότι το τάδε φάρμακο μπορεί να φέρει αϋπνίες. Γράφει και για τα επιδόματα που έχει πάρει ο ασθενής (π.χ. ΚΕΠΑ), κι αν έχει ποσοστό αναπηρίας (αυτά ενδιαφέρουν τον κοινοτικό νοσηλευτή, όπως η ίδια), γτ οι περισσότεροι ασθενείς δεν μπορούν να εργαστούν και ίσως δεν ξέρουν ότι δικαιούνται επιδόματα για να ζήσουν. Την ενδιαφέρει επίσης αν ο ασθενής έχει ασφάλιση, αν έχει τα προς το ζην, αν έχει οκ ή προβληματική οικογένεια, αν η οικογένειά του αποδέχεται τον άρρωστο (αν τον

αποδέχεται, τότε θα κάνει λιγότερες υποτροπές ή αν η μητέρα του έχει υπερεμπλοκή/αν έχει τραύμα από κακοποίηση, τότε θα έχει περισσότερες υποτροπές).

Q6: Υπάρχει κάποια βάση δεδομένων; Ποιο σύστημα καταγραφής χρησιμοποιείται;

ΕΠ: Παλιά: Φάκελος Ασθενούς χειρόγραφα, με δημογραφικά, κοινωνικά (τι δουλειά κάνει, έχει παιδιά, αδέρφια;), ιατρικό ιστορικό τους, έχει κοινωνικό δίκτυο, τηλέφωνο ανάγκης (ποιος έχει την επιμέλειά του), πόσες φορές έχει μπει, έχει βγει και τότε, ποιος γιατρός τον είχε δει. Η κοινωνική λειτουργός παίρνει το Κοινωνικό Ιστορικό, το οποίο συμπληρώνει μετά ο γιατρός με το ιατρικό ιστορικό (ακούει φωνές, έχει ψευδαισθήσεις, τι συμπτώματα έχει δηλαδή). Μετά υπάρχει και επισκέπτρια υγείας που καταγράφει πόσες φορές πηγαίνει στο νοσοκομείο, με ποιον έχει συνομιλίες, πόσες φορές έχει πάει στο σπίτι του, αν έχει νοσηλευτεί ο ασθενής, όταν ξανάρθει ο ασθενής αν έχει προκύψει ένα χειρουργείο ή μια ασθένεια. Και η ψυχολόγος, τι του αρέσει, τι τον φοβίζει, τι θα ήθελε, με ποιον του αρέσει να συζητάει και ποιοι είναι οι φίλοι του, ποιο είναι ένα πρόσωπο αναφοράς, και τεστ προσωπικότητας, και βγαίνει ένα προφίλ, δηλαδή ποιες είναι οι ανάγκες του. Μετά όλοι αυτοί μοιράζονται τις πληροφορίες προφορικά, δηλαδή μαζεύονται καθημερινά σε μια αίθουσα και κάνουν συνάντηση προσωπικού. Στα ΟΞΕΑ γίνεται αυτό. Στα ΧΡΟΝΙΑ γίνονται πιο αραιά οι συναντήσεις. Τα καταγράφουν γραπτά, με το χέρι, στον Φάκελο Ασθενούς [είναι 1 μόνο, δηλαδή αν χαθεί, είναι πρόβλημα] οι κοινωνικοί λειτουργοί. Το μόνο ηλεκτρονικό σύστημα: πλέον με το ΑΜΚΑ φαίνονται κάποια πράγματα – φαίνεται η αγωγή του και σε ποιον γιατρό πήγε, και πόσες φορές μπήκε ή βγήκε. Άλλο που υπάρχει – γραπτά – είναι ιατρική γνωμάτευση (αν μπορεί να εργαστεί, κλπ.) στο αρχείο της Γραμματείας. Έχει συγκεκριμένη φόρμα.

ΚΝ: Μόνο για 1 χρόνο, δηλαδή τώρα τελευταία. Έχουν πολύ μεγάλη ανάγκη από την τεχνολογία, έχουν κουρασθεί να γράφουν βιβλία, π.χ. ποιοι προσήλθαν στα ιατρεία – αν και τώρα αυτό αντικαθίσταται από το ΑΜΕΤ.

Q7: Πώς αποθηκεύονται τα δεδομένα;

ΕΠ: Παλιά χειρόγραφα, τώρα ηλεκτρονικά. ΚΝ: Παλιά όλα χειρόγραφα, τώρα μερικά ηλεκτρονικά.

Q8: Τι θα θέλατε να βελτιώσετε από τον υφιστάμενο τρόπο παρακολούθησης;

ΣΤ: Θα βοηθούσαν πολύ τα διαγράμματα (π.χ. πόση κατάθλιψη έχει ο ασθενής από το 0 ως το 10), οι φόρμουλες, τα templates (π.χ. Τι φωνές ακούς; Α. σαν από ραδιόφωνο; Β. μιλάνε η μία στην άλλη; κλπ.) για να μπορούν να εξάγονται μετρήσεις και στατιστικά για τους ασθενείς.

Q9: Ποιο είδος πληροφοριών είναι απαραίτητο κατά τη διάρκεια της λογοδοσίας; Μπορείτε να αναφέρετε τις βασικότερες κατηγορίες πληροφοριών;

ΕΠ: Τα γράφουν κάτω, τι κάνανε και τι πρέπει να γίνει (τι υλικά, χάπια χρειάζονται) για την επόμενη βάρδια. Συνήθως τη γράφει ένας: ο υπεύθυνος βάρδιας ή ο προϊστάμενος. Υπάρχει και Φάκελος Ασθενούς στα πανεπιστημιακά νοσοκομεία. Ενημερώνεις σε ποιον άλλο γιατρό πήγε ένας ασθενής, αν έφαγε, δεν έφαγε, αν πήρε τα φάρμακά του ή όχι, και αφήνεις εκκρεμότητες.

ΚΝ: Πού μπορεί να βοηθήσει το ΑΙ; - για την καταγραφή της ημέρας (προς το παρόν τα γράφει πολύ συντομευμένα για να βγάλει στατιστικά, γτ το Υπουργείο τής ζητάει στατιστικά), π.χ. ότι έκανε τηλεφωνική επικοινωνία με τον τάδε, κλπ. (μέχρι τώρα όλα αυτά σε ένα τεφτέρι!) για κατηγοριοποίηση των τηλεφώνων και λοιπών υπηρεσιών. Ποιες είναι οι υπηρεσίες; Α) πορεία νόσου Β) τηλεφωνική επικοινωνία για την φαρμακευτική του αγωγή Γ) διασύνδεση με φορείς (π.χ. με Μητρόπολη, με σχολεία [όλων των επιπέδων] για Πρόγραμμα Αγωγής Υγείας, με παιδικούς σταθμούς για Πρώιμη Ανίχνευση, κέντρα κοινότητας, π.χ. για προγράμματα βοήθειας στο σπίτι) και υπηρεσίες (π.χ. με Δήμους, με της Αστυνομίας το Τμήμα Κακοποίησης, με την Εισαγγελία...), γτ η δουλειά τους βασίζεται στο δίκτυο συνεργασίας με όλες αυτές τις υπηρεσίες. Καταρχάς, τα ιατρεία της τα κάνει σε Κέντρα Υγείας ή σε περιφερειακά ιατρεία ως φιλοξενούμενη ή τα προγράμματα Δήμου μπορεί να τα κάνει ακόμα και σε χώρο της Εκκλησίας. ΟΛΑ ΑΥΤΑ ΚΑΤΑΓΡΑΦΟΝΤΑΙ ΣΕ ΤΕΤΡΤΑΔΙΑ. Το πρώτο βήμα: Τώρα αυτά είναι σε ΕΝΑ ΜΟΝΑΔΙΚΟ τετράδιο με 4 στήλες, το λένε «ΗΜΕΡΗΣΙΑ ΚΙΝΗΣΗ» (αντί για ΛΟΓΟΔΟΣΙΑ ΠΟΥ ΕΙΝΑΙ ΟΥΣΙΑΣΤΙΚΑ), και γράφουν με απλές λέξεις-κλειδιά για να κερδίσουν χρόνο.

Το δεύτερο βήμα, μετά την ΗΜΕΡΗΣΙΑ ΚΙΝΗΣΗ: ΦΑΚΕΛΟΣ ΑΣΘΕΝΟΥΣ: ένας για κάθε ασθενή, χειρόγραφος. ΓΡΑΦΟΥΝ αναλυτικά τι είχε η τηλεφωνική επικοινωνία, ή αν τα φάρμακα είχαν παρενέργειες, αν κοιμάται, αν τρώει, τι της περιέγραψε, τις της είπε, τι διάθεση είχε... -> όλα αυτά έχουν σημασία για το γιατρό! - Εργαστηριακές εξετάσεις, - Κοινωνικό ιστορικό (αμκα, δημογραφικά στοιχεία, υποστηρικτικό πλαίσιο, κλπ.) - Ιατρικό ιστορικό (η νόσος) - Παρατηρήσεις (π.χ. υπάρχει μια διέγερση, διαταραχή στη σκέψη) Υπάρχει και το Προσωπικό Αίτημα (Intake): η πρώτη επαφή με το περιστατικό είναι πολύ σημαντική, προχωράς βήμα βήμα και μπορεί να βοηθήσουν τα νοσηλευτικά πρωτόκολλα!!! «Θέλω βοήθεια, δεν είμαι καλά» -> τηλεφωνικά ο νοσηλευτής εκμαιεύει Έχει «Έντυπο Καταγραφής Διαδικασιών» -> δείχνει τι κάνουν και πού τα καταγράφουν ήδη!

Q10: Χρησιμοποιούνται συγκεκριμένα νοσηλευτικά πρωτόκολλα;

ΕΠ: Βιολογικά (είναι καθαρός;), Η σκέψη (είναι καθαρή;) ο λόγος (κατανοητός;). Έλεγχος για επικίνδυνα αντικείμενα. Οξέα τμήματα: Τα ενδιαμέσα είναι τα Κέντρα Ψυχικής Υγείας, όπου γίνεται μια νοσηλεία διάρκειας περίπου 15-20 ημερών. Μετά οι ασθενείς επιστρέφουν στο σπίτι τους, με ή χωρίς οικογένεια, και μόνο όταν τους ξαναπιάσει κρίση επιστρέφουν στην κλινική ή επισκέπτονται περιοδικά την ψυχιατρική κλινική για τη συνταγογράφηση των φαρμάκων. Ο γιατρός έρχεται και παίρνει το ιστορικό, ενώ ο νοσηλευτής δεν ρωτάει πολλά. Σήμερα, τα οξέα περιστατικά αντιμετωπίζονται αποκλειστικά στα οξέα τμήματα της ψυχιατρικής κλινικής.

Χρόνια τμήματα: Αφορούν μόνιμους ασθενείς που διαμένουν σε ξενώνες ή διαμερίσματα, υπό την ευθύνη του Κέντρου Ψυχικής Υγείας. Οι νοσηλευτές τους γνωρίζουν καλά και δεν χρειάζεται να παίρνουν κάθε φορά νέο ιστορικό. Στους χρόνιους ασθενείς, στους ξενώνες, δεν χρειάζεται κάποιος βοηθός για να υπενθυμίζει το ιστορικό, γιατί είναι μόνο δώδεκα ασθενείς όλοι κι όλοι, και τους γνωρίζουν ήδη.

Ο φάκελος θα μπορούσε να υπενθυμίζει, για παράδειγμα, πώς γίνεται η αλλαγή φλεβοκαθετήρα ή ουροκαθετήρα, ή πώς λαμβάνεται αίμα — κάτι ιδιαίτερα χρήσιμο για νέους νοσηλευτές. Το τελετουργικό, δηλαδή η ίδια η διαδικασία, δεν πραγματοποιείται συχνά, και οι νοσηλευτικές ενέργειες μερικές φορές ξεχνιούνται.

Q11: Υπάρχει κάποια προηγούμενη εμπειρία με την τεχνολογία στη ροή εργασίας σας, την οποία θεωρήσατε ιδιαίτερα χρήσιμη ή μη χρήσιμη; / (Πώς) έχετε προσπαθήσει να λύσετε το πρόβλημα της χειρόγραφης καταγραφής φακέλου ασθενούς; / Τι κάνετε ως τώρα για να βελτιώσετε την κατάσταση;

ΕΠ: Παλαιότερα δεν υπήρχε ενδοεπικοινωνία, τίποτα · συνεννοούνταν μόνο με τηλέφωνα. Πλέον χρησιμοποιούν κινητά, άλλωστε οι κλινικές είναι μικρές. Υπήρχαν βομβητές στα πανεπιστημιακά νοσοκομεία, αλλά αυτό γινόταν παλιά.

Ο νοσηλευτής είναι συνεχώς απίκο στις ψυχιατρικές κλινικές, ώστε να παρακολουθεί τι κάνουν οι ασθενείς. Οι συγγενείς έρχονται μόνο για επίσκεψη — δεν κοιμούνται ούτε παραμένουν μαζί με τους ασθενείς.

ΚΝ: Έχει χρησιμοποιήσει ΑΙ (chatgpt μάλλον) για τις αναφορές και τις εκθέσεις και περιλήψεις, π.χ. για την Ευρωπαϊκή Επιτροπή. => για γραφειοκρατία το θέλει. Ήταν επιστημονική φαντασία ο υπολογιστής όταν ήταν νέα, είναι 56 χρονών τώρα! Η Διευθύντριά της είναι 70 χρονών. Αποφεύγουν την πολλή χρήση με τον υπολογιστή. Αλλά έχει νέα παιδιά στην δουλειά που είναι πρόθυμα να χρησιμοποιήσουν ΑΙ. Τι της αρέσει: που δεν είναι ξένο προς αυτόν που το ζητάει (το αποτέλεσμα του ΑΙ). Μπορεί να βοηθήσει στα γραφειοκρατικά, στα στατιστικά κλπ., αλλά όχι στη νοσηλεία ή στην επαφή με τον ασθενή. μπορεί μόνο να δώσει έναν μπούσουλα, π.χ. «πότε μπορώ να κάνω

γενική αίματος?» Μια συνάδελφός της έχει γράψει πληροφορίες για τους ασθενείς της στο κινητό της, αλλά δεν ξέρει τι ακριβώς.

ΣΤ: Έχει ένα excel sheet όπου γράφει προσωπικές της σημειώσεις για κάθε ασθενή, για να μη χρειάζεται να ανατρέχει και να ξεφυλλίζει τον μοναδικό, γραπτό Φάκελο Ασθενούς κάθε φορά.

Q12: Ποια θα ήταν τα πιο πολύτιμα χαρακτηριστικά σε έναν φωνητικό βοηθό για την εργασία σας;

ΣΤ: Αμεσότητα, δηλαδή το να βρίσκει αμέσως την πληροφορία που ψάχνει χωρίς να πρέπει να ανατρέξει σε διάφορες άλλες πηγές.

Q13: Αισθάνεστε άνετα με τη χρήση φωνητικών εντολών σε ένα σύστημα; Αν όχι, ποιοι είναι οι λόγοι που σας αποτρέπουν; ΣΤ: Δεν είπα ξεκάθαρα ότι αισθάνεται άνετα, αλλά είπε ότι θα της άρεσε και θα το ήθελε να μπορεί να ρωτάει με φωνή ερωτήσεις για τους ασθενείς, π.χ. «Τι αγωγή παίρνει στην παρούσα φάση;»

Q14: Θα προτιμούσατε ο βοηθός να έχει έναν πιο ενεργητικό ρόλο (να σας κάνει προληπτικά υπενθυμίσεις, να αναφέρει στοιχεία ακόμα και αν δεν απευθύνεστε σε αυτόν, να είναι πιο συνομιλιακός) ή να λειτουργεί επικουρικά (να ξεκινάτε εσείς αποκλειστικά την αλληλεπίδραση); ΕΠ: • Φωνητική καταγραφή για ό,τι θέλει να πει, ώστε να μην τα ξεχάσει, επειδή συμβαίνουν διάφορα πράγματα και δεν προλαβαίνει να τα σημειώσει εκείνη τη στιγμή — για να τα καταγράψει αργότερα. • Να ξέρει την αγωγή του. • Να ξέρει πότε ήρθε τελευταία φορά. • Να γνωρίζει λίγο τη γενική του εικόνα. • Αν είχε παρενέργειες από κάποιο φάρμακο (ποιο φάρμακο του προκάλεσε τι) και γιατί το άλλαξε. • Σαν ένα ΠΡΟΧΕΙΡΟ ΠΡΟΣΩΠΙΚΟ μικρό φακελάκι ασθενούς. Με ποιο μέσο: Κατά κύριο λόγο να είναι γραμμένα (γτ είναι οπτικός τύπος κυρίως), αλλά και να τα ακούει κιολας, αν θέλει. Όχι μόνο ακουστικά γιατί τα ξεχνάει. Για μια σύντομη πληροφορία μόνο, τότε οκ η φωνή.

ΚΝ: Patient Data – είναι εξατομικευμένο, δεν μπορείς να το εφαρμόσεις παντού, αλλά υπάρχει μια φόρμα με 5 βασικά πράγματα: - Επίπεδα φαρμάκων, που είναι μετρήσιμα, θα τους βοηθάει να μαθαίνουν αμέσως τα επίπεδα - Έχει βλεμματική επαφή; - Είναι φροντισμένος; - Είναι διεγερτικός; - Κάνει γλωσσική σαλάτα; - Έχει παραλήρημα; Και μεταφέρει τα συμπτώματα στο γιατρό.

Ασχολούνται με τις παρακάτω περιπτώσεις ασθενείας: - Νεύρωση: θέλει όρια - Ψύχωση / Μανιοκατάθλιψη: θέλει ντάντεμα 1. Παίρνει φάρμακα; 2. Αν ναι, χτίζω δεξιότητες, π.χ. αυτοφροντίδας (δεν πλένονται) 3. Φαρμακευτική αγωγή (παρενέργειες:

τάση για έμετο, υπνηλία, μειωμένη πίεση...), Αυτοφροντίδα, Κοινωνική Απόσυρση, Δεξιότητες Πνευματικές-Σωματικές, Γενική Αίματος - Άνοια: - Διαχείριση Πένθους:

ΣΤ: Θα τη βοηθούσε να είχε υπενθυμίσεις και στατιστικά (π.χ. πότε υποτροπιάζει η διπολική διαταραχή).

Q15: Πώς αισθάνεστε σχετικά με τη χρήση ενός φωνητικού βοηθού όσον αφορά το απόρρητο των ασθενών και την ασφάλεια των δεδομένων; Υπάρχουν συγκεκριμένες ανησυχίες σχετικά με τις αλληλεπιδράσεις των ασθενών που θα πρέπει να ληφθούν υπόψη για τις φωνητικές εγγραφές ή τις απομαγνητοφωνήσεις; ΕΠ: Ναι, πού θα φυλάσσεται αυτό το υλικό, για να μην το κλέψει κάποιος από το κινητό του νοσηλευτή; Είναι ένα μεγάλο θέμα τα Ευαίσθητα Δεδομένα.

Q16: Πώς θα θέλατε να δίνετε ανατροφοδότηση σχετικά με τις αποκρίσεις του βοηθού; Κείμενο ή φωνή; ΕΠ: Και τα δυο!

Επιπρόσθετες πληροφορίες για τις ελληνικές ψυχιατρικές δομές:

Ψυχιατρεία (εξαρτημένοι και ελεγχόμενοι ασθενείς · τρώνε μόνο με κουτάλια και παίρνουν τα χάπια τους. μπαίνουν εκεί οξέα περιστατικά και μένουν επίσης όσοι η οικογένειά τους δεν θέλει να τους στηρίξει. έχει πολλά τμήματα, περίπου 30 ασθενείς το καθένα, συνολικά γύρω στα 300 άτομα).

Εξωτερικά ιατρεία / κέντρα ψυχικής υγείας ή κινητές μονάδες (ο νοσηλευτής πηγαίνει κατ' οίκον ή εργάζεται στα «νοσοκομεία ημέρας» – μια μεγάλη αίθουσα, όχι ακριβώς νοσοκομείο – όπου καλούνται οι ασθενείς κάθε πρωί να πηγαίνουν, να κάνουν δραστηριότητες και ο νοσηλευτής τούς καθοδηγεί και τους δίνει φάρμακα. το μεσημέρι φεύγουν και πάνε σπίτι τους).

Ψυχιατρικές κλινικές των πανεπιστημιακών νοσοκομείων (σε κάθε νοσοκομείο υπάρχει τμήμα οξέων περιστατικών · ο νοσηλευτής είναι συνέχεια μαζί με τον ασθενή, από τη στιγμή που θα έρθει, μέχρι – αν χρειαστεί – να απομονωθεί ή να δεθεί, έως ότου δράσει το φάρμακο. υπάρχει συστηματική παρακολούθηση).

Ψυχιατρικές κλινικές γενικά: τα πράγματα είναι λίγο πιο ανθρώπινα – οι ασθενείς κοιμούνται εκεί, τρώνε κλπ.

Ξενώνες: λειτουργούν μετά τα ψυχιατρεία, για να αυτονομηθούν οι ασθενείς. φιλοξενούν 12-15 άτομα και έχουν μια σταθερή ρουτίνα: πρωινό στην κουζίνα ή τραπεζαρία (με πιρουνια κανονικά), μπάνιο κ.ά.

Διαμερίσματα: εκεί μένουν όπως οι φυσιολογικοί άνθρωποι.

Στο ψυχιατρείο παλαιότερα έμπαιναν και άτομα με νοητική υστέρηση, δηλαδή με άλλα ψυχοπαθολογικά συμπτώματα.

Ψύχωση: οι ασθενείς ακούνε φωνές ή έχουν ακουστικές ψευδαισθήσεις και συχνά σταματούν να παίρνουν τα φάρμακά τους (επειδή νομίζουν ότι τα φάρμακα τούς τρελαίνουν, ενώ στην πραγματικότητα τους βοηθούν, ή επειδή έχουν παρενέργειες). Τότε οι φωνές επιστρέφουν και γι' αυτό μπαίνουν στα νοσοκομεία. Το ΠΡΩΤΟ πράγμα που πρπ να καταλάβει ο ασθενής είναι να ΠΑΙΡΝΕΙ ΤΑ ΦΑΡΜΑΚΑ, γτ χάνει δεξιότητες. Τα φάρμακα όμως δεν αντιμετωπίζουν την έλλειψη βούλησης ή την κοινωνική απομόνωση.

